Two Rankings from One Posterior: A Bayesian Power Rating and a Wins-Above-Bubble Resume for College Basketball

Thomas Reinke

2026-11-03

Abstract

A college basketball ranking is asked two questions: which team is better, and which has earned a place in the tournament. The first is predictive and should use margin of victory; the second is retrodictive and should reward only results. We answer both from one Bayesian model. Team strength, in points per 100 possessions against an average Division I team on a neutral floor, is observed through the regulation margin and a luck-adjusted efficiency margin built from a shot-quality model and shooters' free-throw rates, modelled jointly so that one performance is not counted twice and weighted by the number of possessions. Strength is pooled within conferences, carried over between seasons through a preseason regression on last season's rating, a public preseason rating and roster turnover, and allowed to drift week to week as a random walk. The power rating is the posterior of strength. The resume, Wins Above Bubble, counts the wins a team has beyond those the last at-large team in the field would expect against the same schedule, where in each posterior draw the automatic bids leave the ranking before the bubble is found. We evaluate the system on every week of every season since 2014–15 with convergence gates, posterior predictive and prior-sensitivity checks, simulation-based calibration, ablations, and a rolling out-of-sample backtest against results, closing betting lines and totals, a public rating system and a least-squares baseline, and compare the resume with the selection committee's choices.

Table of contents

1 Introduction **4**
- 1.1 Notation . . . 6

2 Data **6**

3 The Power Model **10**
- 3.1 Likelihood . . . 10
- 3.2 Hierarchy, Preseason Information and the Weekly Walk . . . 12
- 3.3 Tempo, Totals, Offence and Defence . . . 14
- 3.4 Priors . . . 15

4 What the Efficiency Signal Adds **16**

5 The Resume **18**
- 5.1 Wins Above Bubble . . . 18
- 5.2 The Rest of the Season . . . 20
- 5.3 Strength of Record, Quadrants and Ranks . . . 20
- 5.4 Record-only Reference: the Colley Matrix . . . 22
- 5.5 Choosing the Benchmark . . . 22
- 5.6 How Margin Reaches the Resume . . . 23

6 Computation and Pipeline **24**
- 6.1 Sampling and Gates . . . 24
- 6.2 The Pipeline . . . 24

7 Model Checking **26**
- 7.1 Posterior Predictive Checks . . . 26
- 7.2 Prior Sensitivity . . . 28
- 7.3 Leave-one-out Cross-validation . . . 28
- 7.4 Choosing the Margin Family . . . 29

8 Out-of-sample Evaluation **30**
- 8.1 Design . . . 30
- 8.2 Accuracy . . . 30
- 8.3 Calibration . . . 32
- 8.4 Ablations and Baselines . . . 34
- 8.5 Priors and Preseason Information Out of Sample . . . 35
- 8.6 Tuned Choices . . . 36

9 The 2025–26 Season **36**
- 9.1 The 2025–26 Field . . . 43
- 9.2 The Resume Against the Committee . . . 44

10 Discussion **44**

11 Appendix: Supplementary Posterior Checks **46**
11.1 Sampler Behaviour Across All Snapshots 46
11.2 Sampler Diagnostics 47
11.3 Convergence 48
11.4 Posterior Geometry 53
11.5 Prior and Posterior 53
11.6 Justification of the Informative Priors 55
11.7 Induced Priors 56
11.8 Posterior Predictive Checks 57
11.9 Influence and PSIS Diagnostics 60
11.10 Power-scaling Sensitivity 61
11.11 Sensitivity to the Prior: Perturbation Refits 61
11.12 Simulation-based Calibration 63

12 Appendix: The Stan Program **64**

13 Appendix: Reproducibility **70**

References **70**

1 Introduction

Every winter some 360 Division I men's basketball teams play about thirty games each, and in March a committee chooses the 68 teams of the national tournament: one automatic bid for each conference champion, the rest "at large". The schedule graph is denser than in football — a team plays fewer than a tenth of the other teams, many of them twice within its conference, and November and December link the conferences through hundreds of non-conference games — but the information in a single game is small. A basketball game is a sequence of some seventy possessions each way, and its margin is the difference of two noisy sums. Any ranking has to propagate strength through the graph and decide how much of each result is noise. The history of the problem is largely the history of ways to do so: least squares on scoreboard margin [1, 2], paired-comparison models for wins and losses [3], linear-algebraic record-only schemes such as the Colley matrix [4, 5], and the tempo-free efficiency ratings that now dominate public analysis of the college game [6, 7], built on the possession accounting of Oliver [8] and Kubatko et al. [9].

These methods disagree partly because they answer different questions. A *predictive* rating asks which team would win a game played tonight, and by how much. Margin is informative about that, so a predictive rating should use it, together with any other signal of how well a team played. A *retrodictive* ranking asks what a team has earned by its results, and is the question a selection committee is asked. Here margin is usually excluded on purpose, so that a team is neither rewarded for running up a score nor penalised for resting its starters, and a win over a strong opponent is worth more than a win over a weak one. The committee's own team sheets combine both kinds of number — a predictive rating, quadrant records, and since 2024–25 "wins above bubble" [10] — and leave the weighing to its members.

This paper keeps the two questions apart but derives both answers from one posterior distribution. The contributions are four.

1. A **power model** (Section 3) for team strength in points per 100 possessions against an average Division I team on a neutral floor, observed through the regulation scoreboard margin and through a *luck-adjusted* efficiency margin: actual two-point baskets, expected three-point baskets from a shot-quality model, and expected free throws from each shooter's prior free-throw rate. The margin is modelled *conditionally* on the efficiency residual, so the two signals share one latent strength without being treated as independent evidence, and their relative weight is estimated rather than set. Each game is weighted by its number of possessions. Strength is pooled within conferences, carried over from the previous season through a preseason regression on last season's rating, a public preseason rating and the roster's turnover, and moves through the season as a random walk whose step size is estimated. A ridge model for possessions and points per possession turns the per-100 ratings into spreads and totals in points, and splits each rating into offence and defence (Section 3.3).
2. A result on how much the efficiency signal adds (Section 4). Under a Gaussian version of the model the gain in per-game Fisher information about strength has a closed form, and at the fitted parameters it is 17%–34% — about 9 extra games' worth over a season. An ablation backtest measures the same thing out of sample (Section 8.4).
3. A **resume** measure, Wins Above Bubble (Section 5): for each posterior draw, the number of wins a team has beyond those the draw's last at-large team would be expected to earn

against the same opponents at the same venues. The bubble is found as the committee would find it: in each draw the conference champions — the automatic bids, known or simulated — and the teams not yet eligible for the tournament leave the power ranking, and the bubble team is the last of the at-large places among the rest. The measure is additive over games, never rewards a loss, is centred at zero for a team exactly as strong as the bubble, and inherits uncertainty in the opponents, the bubble and the ranks directly from the posterior.

4. A **validation** of the whole system (Section 6, Section 7, Section 8) on every week of every season from 2014–15 to 2025–26: 277 posteriors fitted by the No-U-Turn sampler, of which 274 pass the convergence gates; posterior predictive checks, prior-sensitivity diagnostics, simulation-based calibration, ablations and baselines; and a rolling backtest of 65,268 games, each predicted from the posterior fitted before it, scored against the result, the closing betting line and the ratings of a public system at the same date. The resume is compared with the committee's at-large selections in every season with a tournament (Section 9.2).

Figure 1 shows how these pieces fit together. Section 2 describes the data, Section 3 and Section 4 the power model, and Section 5 the resume and the reference rankings. Section 6 describes fitting and the pipeline, Section 7 model checking, Section 8 the out-of-sample evaluation, Section 9 the 2025–26 season as a worked example, and Section 10 the limitations.

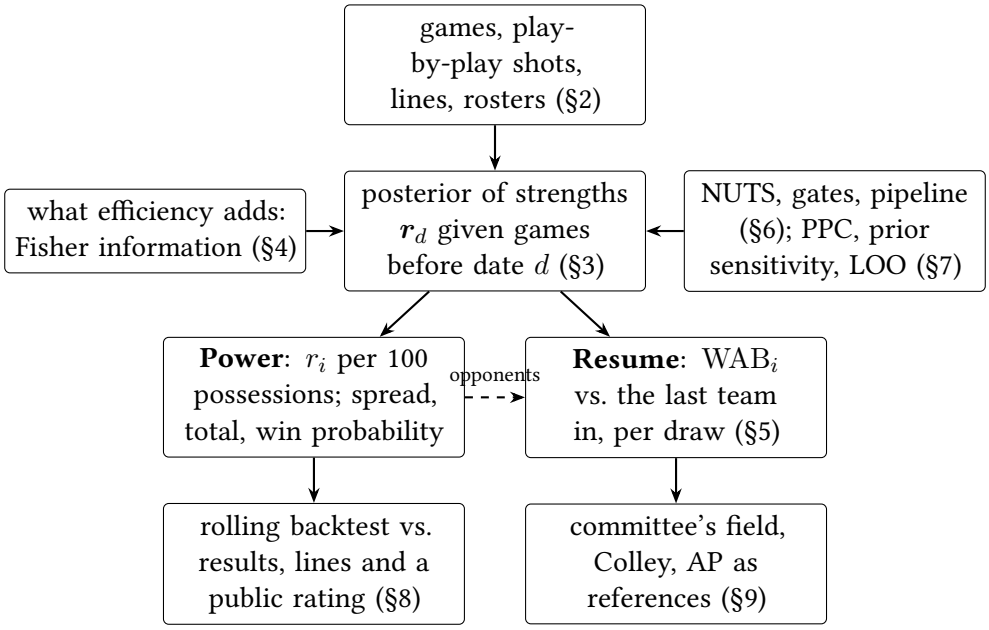


Figure 1: Structure of the paper. One posterior per snapshot date feeds both products; the dashed arrow is the only link between them: the Resume values results using opponent and bubble strengths drawn from the Power posterior, but never uses margin. Each box names the section where it is developed.

Related work. The margin model descends from Harville [1] and Harville [11], who fit mixed linear models to scoreboard margin with random team effects, and from Stern [12], who showed that margins are close to normal around the betting line, so that a spread determines a win

probability. For college basketball specifically, Harville and Smith [13] estimated the home-court advantage and its variation between teams, Kvam and Sokol [14] and Lopez and Matthews [15] built tournament prediction models from regular-season results, and Coleman and Lynch [16] modelled the committee's at-large choices from the team-sheet criteria. Glickman and Stern [17] made team strengths evolve over time in a Bayesian state-space model; the weekly random walk here is a version of theirs, and dynamic paired-comparison models [18, 19] let strengths drift between games in the same way. The tempo-free ratings of Pomeroy [6] and Torvik [7] measure offence and defence per possession and adjust for opponents and venue by iterative least squares; Torvik's ratings, archived by date, serve here both as an input to the preseason rating and as a benchmark in the backtest, and Torvik also publishes a wins-above-bubble measure, against which ours is compared (Section 9.2). Wins Above Bubble differs from the strength of record of the team sheets in its reference team and in how uncertainty enters; Lopez et al. [20] quantify how much of a season's outcome is noise, which is the reason a resume should report its uncertainty. The workflow follows Gelman et al. [21].

1.1 Notation

Table 1 collects the notation. Games are indexed by g, teams by i and j, conferences by c, and the model's time periods (weeks of seven days from the first game of the season) by w. Game g is played in period $w(g)$. Posterior draws are indexed by $s = 1, \ldots, S$, and a superscript (s) marks a quantity evaluated at draw s. A team's rating without a period index, r_i, is its rating at the date of the snapshot being published. Ratings and margins in the model are per 100 possessions; spreads and totals are in points.

2 Data

Games, box scores and play-by-play come from ESPN through the bulk files of the SportsDataverse project [22]: every game of every Division I team from 2013–14 to 2026–27, with team box scores and, for most games, play-by-play with shot locations. The CollegeBasketballData.com API [23] supplies the scores by half, conferences, closing betting lines (spread and over/under), NCAA tournament seeds, polls and recruiting ranks, and maps its games to ESPN's. Bart Torvik's site [7] supplies preseason ratings, the site's ratings as they stood on any past date (its "time machine"), and player tables with minutes and box plus-minus. Division I membership by season comes from ESPN; teams moving up from Division II, which are ineligible for the tournament for several seasons, and teams under a postseason ban are kept in hand-maintained tables. Teams are keyed by ESPN's identifiers throughout. The following derived quantities enter the model.

Possessions. The number of possessions in a game is estimated from the box score as the average over the two teams of field-goal attempts minus offensive rebounds plus turnovers plus 0.475 times free-throw attempts [6, 8]. For an overtime game the regulation possessions n_g are the total scaled by the share of playing time in regulation, $40/(40 + 5k)$ for k overtimes.

Regulation margin. $y_g = 100\,(\text{home} - \text{away})/n_g$, the home team's margin after 40 minutes per 100 regulation possessions. A game decided in overtime (6.1% of Division I games) contributes a margin of zero: the teams were level after regulation, and the overtime is a short and different

Table 1: Notation. The last column names the corresponding variable in `stan/power.stan` or the R code.

Symbol	Meaning	Code
$h(g), a(g)$	home and away team of game g	`home`, `away`
δ_g	site: $+1$ on the home team's floor, -1 on the away team's, 0 neutral	`site`
y_g	regulation margin per 100 regulation possessions, home minus away	`margin100`
e_g	luck-adjusted efficiency margin per 100 live possessions	`lux100`
n_g, n_g^{live}	regulation and live possessions; weights $\omega_g = n_g/70$, $\omega_g^e = n_g^{\text{live}}/70$	`poss_reg`, `w_m`, `w_e`
$r_{i,w}$	strength of team i in period w, points per 100 possessions	`r_state`, `r`
r_i^0	preseason level of team i	`r0`
ϕ	random-walk step sd per period	`tau`
η	home-court advantage per 100 possessions	`hca`
μ_g	$r_{h(g),w(g)} - r_{a(g),w(g)} + \eta\delta_g$, expected margin per 100	`mu`
λ, σ_e	scale and noise of the efficiency margin	`lambda`, `sigma_eff`
κ, σ_m	loading and scale of the margin	`kappa`, `sigma_m`
$\bar{\sigma}$	$(\sigma_m^2 + \kappa^2\sigma_e^2)^{1/2}$, marginal margin scale at 70 possessions	`sigma_game`
α_c, ω	mean strength of conference c and its spread	`mu_conf`, `sigma_conf`
τ	within-conference spread of the preseason level	`sigma_team`
d_i, s_i, β	preseason prediction of the rating, its uncertainty, and its loading	`prev_dev`, `prev_sd`, `b_prev`
$d_i^{(2)}, \beta_2$	final rating two seasons back and its loading	`prev2_dev`, `b_prev2`
$m_{\text{nd}}, \sigma_{\text{nd}}$	level and extra spread of non-Division I opponents	`r_nd`, `sigma_nd`
$B^{(s)}$	strength of the bubble team in draw s	`rb`
π_g	probability that the bubble team wins game g's fixture	`p_bench`
$R_i^{(s)}$	rank of team i in draw s	`rk`

Table 2: Out-of-sample calibration of the shot model: predicted against actual make rate by kind of shot, all field-goal attempts of the complete seasons, each season predicted by the model trained on the two before it. The largest season gap is over seasons with at least 1,000 attempts of the kind.

Kind	Attempts	Predicted	Actual	Largest season gap
three	2,916,203	34.2%	34.1%	1.3 pts
layup	2,431,394	55.7%	55.9%	3.4 pts
jumper	2,068,348	37.1%	37.5%	1.7 pts
dunk	261,230	88.2%	88.3%	1.6 pts
tip	70,447	65.9%	66.2%	6.3 pts
hook	8,951	40.7%	47.6%	6.1 pts

game. The win still counts as a win in the resume.

Efficiency margin. For each side, the *luck-adjusted* points are the points actually scored on two-point field goals, plus the *expected* points of its three-point attempts, plus the *expected* points of its free throws, all in live play: regulation time outside garbage time. A three-point attempt is worth three times its probability of being made under a shot model (below), and a free throw its shooter's free-throw rate in all earlier games, this season and last, shrunk towards the league rate as if the shooter had 40 more attempts at it. The efficiency margin is

$$e_g = 100\,\frac{\mathrm{lux}_{h(g)} - \mathrm{lux}_{a(g)}}{n_g^{\mathrm{live}}},$$

per 100 live possessions, with n_g^{live} the regulation possessions scaled by the share of regulation before garbage time began. The idea is to keep what a team controls — the shots it takes and allows, its two-point finishing — and to replace by their expectations the two quantities that are most variable from game to game and least repeatable: whether the threes go in, and how the free throws fall. A defence decides which threes it allows far more than whether they go in. A pure shot-quality margin, which values *every* field goal at its expected value, was built as well; it is compared in the ablations (Section 8.4). The efficiency margin is used when both teams' shots are in the play-by-play and the game has at least 20 live possessions, and is missing otherwise.

Shot model. The make probability of a field-goal attempt comes from two logistic regressions, one for two-point and one for three-point attempts, on a natural spline in shot distance (four and three degrees of freedom), the kind of shot (jumper, layup, dunk, tip, hook; for twos), whether it is a putback within four seconds of an offensive rebound and whether it comes within eight seconds of a change of possession. Distance comes from the shot coordinates or the play text, capped at 30 feet and imputed by the median of its kind when missing. The model for a season is trained on the two previous seasons' shots only, so the efficiency margin of a game never uses its own result. Table 2 shows its calibration out of sample. Only the three-point predictions enter the published efficiency margin; the two-point ones are used only by the shot-quality variant of Section 8.4. ESPN labels hook shots as such only from 2025–26, so that season's model learned them from a few hundred earlier labels, nearly all of made shots; the hook row's gap is that season's.

Garbage time. Play in the second half counts as garbage time from the moment one team leads by 20 or more with ten minutes or fewer left, or by 15 or more with five minutes or fewer. The efficiency margin excludes it; the scoreboard margin cannot.

Venue. The site δ_g is $+1$ when the game is on the home team's floor, -1 when it is on the away team's, and 0 at a neutral site. ESPN's neutral-site flag is the starting point, with one correction: a "neutral" game played at a venue where one of the teams played at least two non-neutral home games that season or the previous one is that team's home game (124 games over the complete seasons, mostly early-season events and conference-tournament games played on one team's own floor).

Opponents outside Division I. Games against Division II, III and NAIA teams (5,587 over the complete seasons) enter through their regulation margin only, against one pooled level for all such opponents with an extra spread for the mixture of divisions (Section 3.2). They have no efficiency margin, since the play-by-play of these games is rarely complete.

Preseason information. Three quantities from before the season enter each team's preseason level (Section 3.2): our own final rating of the team last season; Torvik's preseason rating, from the site's archive the day before opening night; and roster turnover from its player tables — the value of returning players (box plus-minus times minutes share last season), of incoming transfers at their previous teams, of departing players, the minutes shares of returners and transfers, and the number of highly rated freshmen. They are combined by a regression of each past season's final rating on these quantities, fitted on earlier seasons only (Section 3.2), and our final rating two seasons back enters the model separately.

Snapshots. A posterior is fitted on opening night (prior only), on every Monday from the first after the opening slate, and on the day after the championship game. The Monday before the tournament is Selection Monday. Each snapshot uses exactly the games played before its date and predicts the Division I games until the next snapshot.

Unusual seasons. The 2019–20 season stopped during the conference tournaments, and no national tournament was held. In 2020–21 most arenas were empty or nearly so, many games were cancelled or postponed, and the season had about a quarter fewer games than usual; its home-court advantage has its own prior (Section 3.4).

Missing efficiency. The efficiency margin is missing for 5% of Division I games over the complete seasons, from 0.2% in the best-covered season to 11% in the worst, mostly the earliest. Whether ESPN recorded a game's shots depends on the broadcaster and the season, which are unrelated to how the game went, so the data are missing at random in the sense of Rubin [24] and the likelihood of the observed data (Equation 2 for these games) is the right one to use without modelling why the efficiency margin is missing.

Table 3 summarises the data.

* In progress. The first season is fitted once, at its end, to supply the next season's carry-over.

Table 3: Data by season (named by the year of its tournament). *Teams*: Division I teams; *ineligible*: of these, those not eligible for the tournament (moving up to Division I, or banned). *Games*: completed Division I games and games against non-Division I opponents. *Eff.*, *Line*: shares of Division I games with an efficiency margin and with a closing line. *Snap.*: posteriors fitted.

Season	Teams	Ineligible	D-I games	Non-D-I	Eff.	Line	Neutral	OT	Snap.
2013–14	351	6	5,491	433	87%	71%	636	363	1
2014–15	351	8	5,492	435	89%	73%	607	355	23
2015–16	351	8	5,506	433	91%	72%	641	343	23
2016–17	351	4	5,485	441	93%	72%	630	331	23
2017–18	351	0	5,500	467	94%	73%	635	394	23
2018–19	353	2	5,595	452	92%	99%	652	345	24
2019–20	353	3	5,324	433	94%	100%	446	300	20
2020–21	347	7	3,943	340	95%	100%	477	218	21
2021–22	358	8	5,475	490	98%	100%	685	319	23
2022–23	363	11	5,714	506	99%	100%	751	360	23
2023–24	362	11	5,723	517	99%	93%	733	343	24
2024–25	364	9	5,759	526	98%	94%	731	325	24
2025–26	365	4	5,752	547	100%	100%	693	325	24
2026–27*	367	4	121	62	100%	100%	14	3	2

3 The Power Model

3.1 Likelihood

For game g let

$$\mu_g = r_{h(g),w(g)} - r_{a(g),w(g)} + \eta\,\delta_g$$

be the expected regulation margin per 100 possessions: the difference between the two teams' strengths in the period of the game, plus the home-court advantage on the floor of whichever team is at home. Both observed signals depend on the teams only through μ_g. How strengths move through the season is the subject of Section 3.2.

A rate per 100 possessions is noisier when it is computed from fewer possessions: the margin of a 60-possession game varies more, per possession, than that of an 80-possession game. Each signal's variance is therefore divided by its game's weight, $\omega_g = n_g/70$ for the margin and $\omega_g^e = n_g^{\text{live}}/70$ for the efficiency margin, so that the scales σ_e and σ_m refer to a game of 70 possessions.

Power model likelihood. For a game with an efficiency margin,

$$e_g \mid \mu_g \sim \mathcal{N}\big(\lambda\mu_g,\ \sigma_e^2/\omega_g^e\big), \qquad y_g \mid e_g, \mu_g \sim \mathcal{D}\big(\mu_g + \kappa(e_g - \lambda\mu_g),\ \sigma_m^2/\omega_g\big), \tag{1}$$

where $\mathcal{D}(\xi, \sigma^2)$ is a law with location ξ and scale σ from one of the families of Table 4. For a game without one,

$$y_g \mid \mu_g \sim \mathcal{D}\big(\mu_g,\ \sigma_m^2/\omega_g + \kappa^2\sigma_e^2/\omega_g^e\big), \tag{2}$$

Table 4: The margin families. $t_\nu(\xi, \sigma)$ is the Student-t law with ν degrees of freedom. The published family, chosen on out-of-sample predictions (Section 7.4), is the normal.

Family	$\mathcal{D}(\xi, \sigma^2)$	Extra parameters
Normal (published)	$\mathcal{N}(\xi, \sigma^2)$	none
Student-t	$t_\nu(\xi, \sigma)$	ν
Contaminated normal	$(1-p_o)\,\mathcal{N}(\xi, \sigma^2) + p_o\,\mathcal{N}(\xi, k_o^2\sigma^2)$, $k_o > 1$	p_o, k_o

with $\omega_g^e = 1$ when the live possessions are unknown. Games are conditionally independent given the parameters.

Three features of Equation 1 matter. First, both signals load on the same μ_g, so strength is measured on the scale of the scoreboard margin: λ converts it to the efficiency scale, and it is estimated. Second, the margin is modelled *given* the efficiency residual $e_g - \lambda\mu_g$. A team that shoots better than its expected efficiency usually outscores its expected margin too, and the loading κ absorbs that shared component. Were the two modelled as conditionally independent given μ_g, one good performance would be counted twice. Third, the noise is heteroscedastic in a known way, through the possession weights; beyond that, the family is a modelling choice made once, on out-of-sample predictions (Section 7.4), and the published one is the normal: a margin is the sum of some seventy possessions each way, and heavier tails did not predict better.

The variance in Equation 2 is chosen so that the two cases agree for the normal family.

Marginal Margin Lemma. Let $\mathcal{D}$ in Equation 1 be normal. Then, marginally over e_g, $y_g \mid \mu_g \sim \mathcal{N}\big(\mu_g, \sigma_m^2/\omega_g + \kappa^2\sigma_e^2/\omega_g^e\big)$, and (e_g, y_g) is bivariate normal with mean $(\lambda\mu_g, \mu_g)$ and covariance

$$\Sigma_g = \begin{pmatrix} \sigma_e^2/\omega_g^e & \kappa\sigma_e^2/\omega_g^e \\ \kappa\sigma_e^2/\omega_g^e & \sigma_m^2/\omega_g + \kappa^2\sigma_e^2/\omega_g^e \end{pmatrix}.$$

Proof. Write $e_g = \lambda\mu_g + \varepsilon_e$ and $y_g = \mu_g + \kappa\varepsilon_e + \varepsilon_m$ with $\varepsilon_e \sim \mathcal{N}(0, \sigma_e^2/\omega_g^e)$ and $\varepsilon_m \sim \mathcal{N}(0, \sigma_m^2/\omega_g)$ independent. Then (e_g, y_g) is a linear image of a Gaussian vector, with the stated mean, $\mathrm{Var}(y_g) = \kappa^2\sigma_e^2/\omega_g^e + \sigma_m^2/\omega_g$ and $\mathrm{Cov}(e_g, y_g) = \kappa\sigma_e^2/\omega_g^e$. □

For the contaminated normal the same argument applies to each component of the mixture with the common scale inflated, and for the t the marginal law of Equation 2 is an approximation that matches the location and the scale; both matter only for the alternative families. At 70 possessions the marginal scale of the margin is $\bar\sigma = (\sigma_m^2 + \kappa^2\sigma_e^2)^{1/2}$, 15.0 points per 100 in a typical complete season, or about 10.5 points per game. The same law defines the win probability.

Spread and win probability. For a game between teams i and j at site $v \in \{-1, 0, 1\}$ from i's side, with $\hat n$ expected possessions (Section 3.3) and $\hat\omega = \hat n/70$, the predicted spread is the posterior mean of $\mu_{ij}(v)\,\hat n/100$, $\mu_{ij}^{(s)}(v) = r_i^{(s)} - r_j^{(s)} + \eta^{(s)}v$, and the win probability is

$$p_{ij}(v) = \frac{1}{S}\sum_{s=1}^{S} \Phi\big(\mu_{ij}^{(s)}(v)\,\hat\omega^{1/2}/\bar\sigma^{(s)}\big).$$

This is the rule of Stern [12] averaged over the posterior, which widens the predictive distribution by the uncertainty in the ratings, with one basketball-specific feature: a faster game is more predictable. The same difference in strength per 100 possessions translates into a larger expected margin over more possessions, while the noise of the margin grows only with their square root, so p_{ij} moves away from one half as $\hat{n}$ grows. The ratings are those at the snapshot date; a game later in the week adds no drift.

3.2 Hierarchy, Preseason Information and the Weekly Walk

A team's strength starts the season at a preseason level, pooled within its conference, and then moves from period to period.

Strength hierarchy. For team i in conference $c(i)$, the preseason level is

$$r_i^0 = \alpha_{c(i)} + \beta\, d_i + \beta_2\, d_i^{(2)} + \left(\tau^2 + \beta^2 s_i^2\right)^{1/2} z_i, \qquad \alpha_c = \omega\, u_c, \qquad z_i, u_c \overset{\text{iid}}{\sim} \mathcal{N}(0,1). \tag{3}$$

Here d_i is a preseason prediction of team i's final rating on the scale of the model, s_i the posterior standard deviation of its final rating last season (3 points when it has none), and $d_i^{(2)}$ its final rating two seasons back (zero when unknown).

The prediction d_i is the fitted value of a linear regression, fitted on every earlier season, of a team's final rating on its final rating the previous season, Torvik's preseason rating and the roster quantities of Section 2. Where the regression has fewer than 300 team-seasons to learn from, in the first two seasons, d_i is the even average of last season's rating and Torvik's preseason rating. The loading β then measures how much of that prediction the season's games confirm, and is estimated jointly with everything else; with a well-calibrated prediction it should be near one. Because d_i is a prediction of the rating itself, not of the deviation from the conference mean, it carries conference strength over as well, and the conference effects α_c absorb only what the prediction misses about a conference as a whole — which is why their spread ω is small (Section 3.4).

Weekly random walk. Let $w_1 < w_2 < \cdots$ be the periods in which team i plays, and $w_0 = 1/2$, the preseason level sitting half a period before the first. Its strength in those periods is

$$r_{i,w_k} = r_{i,w_{k-1}} + \phi\,(w_k - w_{k-1})^{1/2}\,\varepsilon_{i,k}, \qquad r_{i,w_0} = r_i^0, \qquad \varepsilon_{i,k} \overset{\text{iid}}{\sim} \mathcal{N}(0,1), \tag{4}$$

so a team that does not play in a period accumulates that period's variance at its next game. Its rating at a snapshot date in period w is its last state plus the drift since, $r_i = r_{i,w_K} + \phi\,(w - \frac{1}{2} - w_K)_+^{1/2}\varepsilon_i$.

Only the periods in which a team plays carry a state, so a season adds about twenty parameters per team. The step ϕ is shared by all teams and estimated. It is the model's answer to how quickly a rating should forget: with $\phi = 0$ every game of the season counts equally, as in a static model, and the larger ϕ is relative to the noise of a game, the more the latest games dominate. Figure 2 shows the estimate by season: about 0.56 points per 100 per week, so over a twenty-week season a team's strength drifts by about 2.5 points (one sd), small next to the spread of Division I strengths

but not next to the margin of a close game. Whether the drift improves predictions over static ratings is tested directly in Section 8.4.

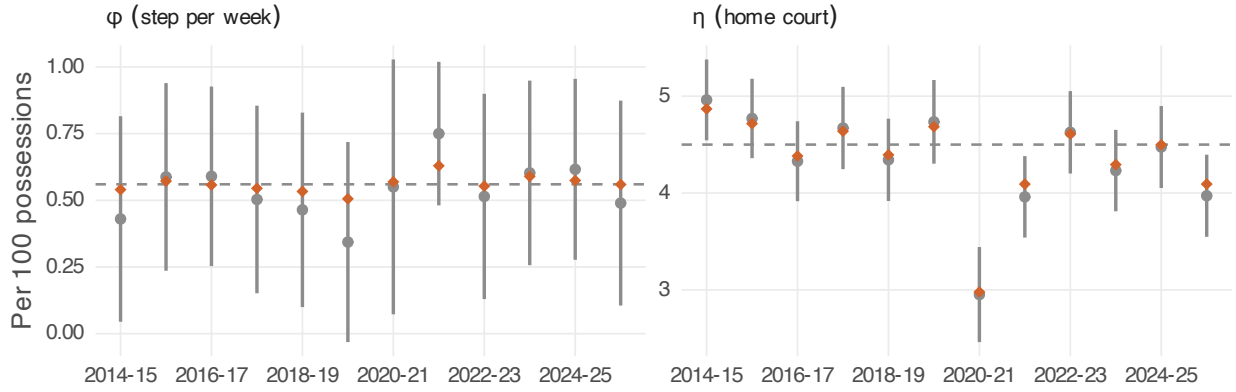


Figure 2: Random-walk step ϕ (left) and home-court advantage η (right) by season, per 100 possessions: posterior mean ± 2 sd of the full-season refit under vague priors (grey) and the published final posterior mean (orange). The dashed lines are the published prior means; the home-court prior of 2020–21 differs.

The parameterisation is non-centred [25] throughout, in z_i, u_c and the steps $\varepsilon_{i,k}$: in the first weeks each team has one or two games and the data barely constrain z_i, the regime in which a centred hierarchy produces the funnel geometry that defeats Hamiltonian samplers [26]. The random walk has a funnel of its own, between ϕ and the steps, and one season identifies ϕ only weakly, which is why its prior is informative and keeps it away from zero (Section 3.4).

Uncertainty in last season's rating. The term $\beta^2 s_i^2$ in Equation 3 integrates out the uncertainty of last season's rating, treated as $\mathcal{N}(d_i, s_i^2)$: a team whose rating was poorly determined is shrunk further towards its conference. Using a posterior mean as the covariate is regression calibration [27], and since the posterior mean is already the best predictor of the true rating, the regression on it is not attenuated as a regression on a noisy measurement would be.

One home-court advantage. η is one number per season for every home game. Arenas, crowds and travel surely differ [13], and a team-specific home edge shrunk to the season's was tried and made no difference out of sample; the data identify one average well and many small effects poorly. Figure 2 shows the estimate by season: stable at about 4.5 points per 100 possessions (3.1 points in a 70-possession game), except in 2020–21, played in empty or near-empty arenas, when it falls to 3.0.

Opponents outside Division I. In a game against a team below Division I the Division I team's margin follows Equation 2 with $\mu_g = r_{i,w(g)} - m_{\mathrm{nd}} + \eta\,\delta_g$ and an extra variance σ^2_{nd}, one level m_{nd} for all such opponents. The spread σ_{nd} absorbs the difference between a good Division II team and an NAIA one. These games inform the Division I teams' strengths little — they are mostly lopsided — but they fill the early weeks, when a team may have played only one Division I game.

Remark (identification). The likelihood of Division I games depends on the strengths only through differences within a connected component of the schedule graph. Every prior in the model is

proper, so the posterior is proper regardless, and the published ratings are centred, $\tilde{r}_i = r_i - \bar{r}$ over Division I teams, draw by draw: a rating of $+10$ means ten points per 100 possessions better than an average Division I team on a neutral floor. Between components the relative strengths come from the prior alone — from the preseason prediction and the conference structure. In basketball that matters only in the first days of a season: at the first Monday the graph of teams that have played has 37–122 components across seasons, and from the second Monday in a typical season it is connected.

Figure 3 draws the full model as a directed acyclic graph.

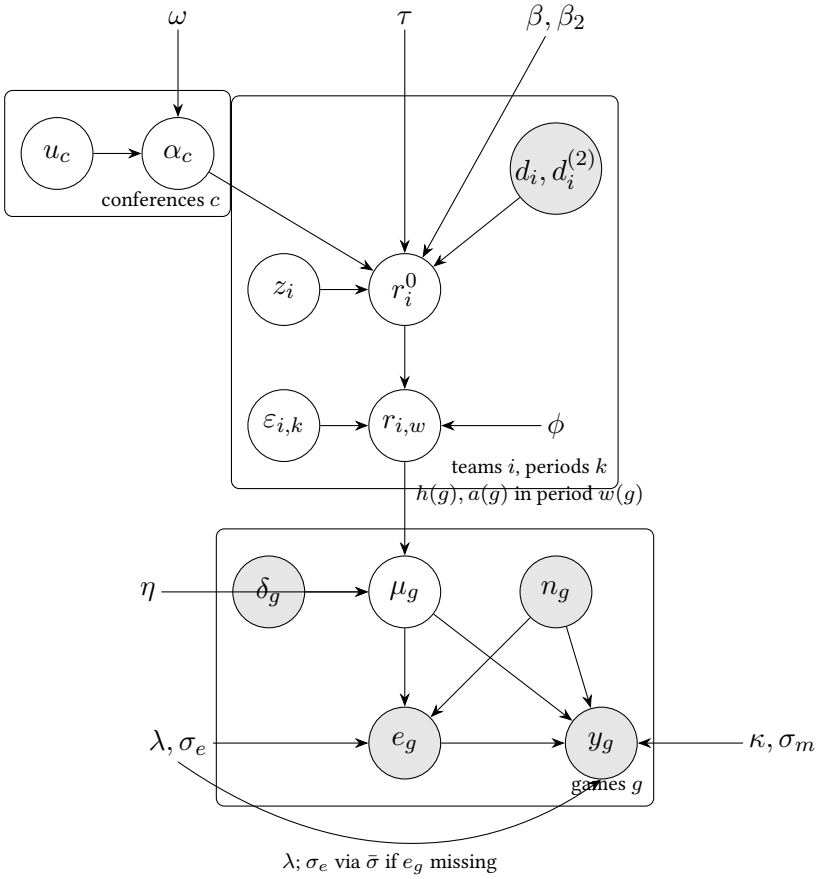


Figure 3: The power model. Shaded nodes are observed; plates repeat over conferences, teams (and, for $\varepsilon_{i,k}$ and $r_{i,w}$, the periods in which a team plays) and games; s_i, the uncertainty of d_i, enters r_i^0 alongside it and is not drawn, nor are the games against non-Division I opponents. The possessions n_g set the weights of both signals. The efficiency margin e_g is a parent of the scoreboard margin y_g, which is how the conditional likelihood of Equation 1 avoids counting one performance twice.

3.3 Tempo, Totals, Offence and Defence

The power rating is per possession; a spread in points needs the number of possessions, and a total needs the points per possession as well. Both come from a model outside the Bayesian

fit. Regulation possessions are regressed on a league mean and an additive effect for each team, and total points per 100 possessions (both teams) likewise, each a ridge regression — a Gaussian random-effects model with the variance ratios fixed — whose team effects shrink towards a fraction of last season's (0.58 for tempo and 0.38 for scoring, the season-to-season regression slopes over the data), with games weighted by recency (half-life 60 days) and points per 100 weighted by possessions. A game's expected possessions $\hat{n}$ are the sum of the mean and the two teams' tempo effects, and its expected total is $\hat{n}/100$ times the expected points per 100 plus a small allowance for overtime. With t_i a team's scoring effect and $\bar{x}$ the league's points per 100, its **offence** and **defence** — points scored and allowed per 100 possessions against an average team — are

$$\mathrm{Off}_i = \tfrac{1}{2}\bar{x} + \tfrac{1}{2}(\tilde{r}_i + t_i), \qquad \mathrm{Def}_i = \tfrac{1}{2}\bar{x} + \tfrac{1}{2}(t_i - \tilde{r}_i),$$

so that $\mathrm{Off}_i - \mathrm{Def}_i = \tilde{r}_i$ by construction. The tempo model shares nothing with the power model, so it changes neither the power rating nor the resume, only the spreads, totals and the split into offence and defence. Its variance ratios and half-life were set on a weekly backtest of totals; it is scored against the closing over/under in Section 8.2.

3.4 Priors

The hyperparameters of the hierarchy — the spreads ω and τ, the step ϕ and the home-court advantage η — describe the sport rather than any team. In the first weeks of a season, with a game or two per team, the data cannot estimate them, and with vague priors the sampler fails the convergence gates of Section 6.1. They are therefore given informative priors, set from history. Each complete season's final snapshot (about 5,500 games) is refitted under deliberately vague priors on every structural parameter (Section 11.6); these refits are the `prior_refits` target of the pipeline. The season-by-season posterior means $\hat{\theta}_s$, with posterior standard deviations v_s, are then pooled by a random-effects meta-analysis [28]. With the DerSimonian–Laird estimate $\hat{\sigma}_b^2$ of the between-season variance and weights $w_s = (v_s^2 + \hat{\sigma}_b^2)^{-1}$, each prior is centred at $\sum_s w_s\hat{\theta}_s / \sum_s w_s$ and its standard deviation is the predictive standard deviation for a new season,

$$\sigma^2_{\text{prior}} = \hat{\sigma}_b^2 + \Big(\sum_s w_s\Big)^{-1}, \tag{5}$$

the season-to-season variation in excess of estimation noise plus the uncertainty of the centre, rounded up to the next 0.05 and to at least 0.1. Home court leaves out 2020–21, which has its own prior, $\mathcal{N}(3, 0.5^2)$.

The other structural parameters — the loadings β and β_2, the scales and loadings of the two signals, and the non-Division I level — are informed by thousands of games even in November, and get weakly informative priors on the scale of the sport. Table 5 lists every prior with the range of the full-season posterior means under the vague priors and as published.

The priors in use were pooled from an earlier run of the refits, under which only the four pooled scales had vague priors; the current refits, with every prior vague, pool to $\omega \sim \mathcal{N}(1.05, 0.16^2)$, $\tau \sim \mathcal{N}(4.01, 0.10^2)$, $\phi \sim \mathcal{N}(0.56, 0.05^2)$ and $\eta \sim \mathcal{N}(4.46, 0.27^2)$ before rounding. The published prior means of ω and τ are more than half a prior standard deviation from the current pooled mean; the priors lag the refits by one rebuild, and Section 11.11 shows how little the published ratings

Table 5: Priors (normal, truncated to the support where a parameter is constrained) and the range over complete seasons (2014–15 to 2025–26) of the full-season posterior mean, refitted under vague priors and as published. The last column is the median posterior sd of the vague-prior refits. The first four are pooled from the refits (Equation 5).

Parameter	Prior	Vague priors	Published	Post. sd
ω conference spread	$\mathcal{N}(1.48, 0.20^2)$	0.62 to 1.91	1.37 to 1.53	0.57
τ team spread	$\mathcal{N}(4.27, 0.35^2)$	3.56 to 4.50	3.94 to 4.39	0.34
ϕ weekly step	$\mathcal{N}(0.56, 0.10^2)$	0.34 to 0.75	0.51 to 0.63	0.18
η home court (per 100)	$\mathcal{N}(4.50, 0.40^2)$	2.95 to 4.96	2.98 to 4.87	0.21
β loading of the preseason prediction	$\mathcal{N}(0.70, 0.20^2)$	0.77 to 1.07	0.76 to 1.04	0.05
β_2 loading two seasons back	$\mathcal{N}(0.20, 0.20^2)$	-0.04 to 0.23	-0.00 to 0.23	0.05
λ efficiency scale	$\mathcal{N}(0.70, 0.30^2)$	0.90 to 0.98	0.90 to 0.98	0.01
σ_e efficiency noise	$\mathcal{N}(12.00, 5.00^2)$	13.52 to 16.42	13.51 to 16.40	0.14
κ loading of the margin on it	$\mathcal{N}(0.50, 0.50^2)$	0.45 to 0.62	0.44 to 0.62	0.01
σ_m margin noise	$\mathcal{N}(14.00, 5.00^2)$	12.12 to 13.32	12.12 to 13.31	0.12
m_{nd} non-Division I level	$\mathcal{N}(-25.00, 10.00^2)$	-55.92 to -38.62	–	1.14
σ_{nd} its extra spread	$\mathcal{N}(10.00, 5.00^2)$	13.84 to 20.83	–	0.97

depend on them. The most consequential of the four is the conference spread: once the preseason prediction carries conference strength over, conference means move only about 1.1 points per 100 between seasons. For 6 of the 12 parameters (β_2, η, λ, ω, τ and ϕ) the between-season standard deviation of the posterior means is less than 1.3 times the typical within-season posterior standard deviation, about what sampling noise alone would produce: the history gives no evidence that these parameters change from season to season.

The priors are empirical Bayes at the level of the sport, and two consequences need checking. First, a season's own data help set its prior, so saying that its posterior lies inside that prior would be circular. Section 11.6 therefore also builds, for each season, the prior pooled from the *other* seasons only, and asks whether the season's own estimate lies inside it. Second, the early-season backtest of Section 8 for, say, 2015–16 uses priors informed by later seasons. Section 8.5 refits the first snapshots of every season under leave-one-season-out priors and measures the difference. Section 11.7 gives the priors they induce on team strengths, win probabilities and margins, and Section 11.11 refits the model under widened and shifted priors.

4 What the Efficiency Signal Adds

The efficiency margin is expensive — it needs play-by-play data, a shot model and every shooter's free-throw history — and a natural question is how much it sharpens the estimate of strength once the scoreboard is known. The Gaussian version of the model answers it in closed form. Take a game of 70 possessions, $\omega_g = \omega_g^e = 1$; other lengths scale both informations by the same factor.

Information Proposition. In the Gaussian model of the Marginal Margin Lemma at unit weights, the Fisher information about μ_g carried by one game is

$$\mathcal{I}_y = \frac{1}{\Sigma_{22}} = \frac{1}{\sigma_m^2 + \kappa^2 \sigma_e^2}$$

from the margin alone, and

$$\mathcal{I}_{e,y} = \frac{\lambda^2}{\sigma_e^2} + \frac{(1-\kappa\lambda)^2}{\sigma_m^2} = \mathcal{I}_y + \frac{(\lambda - \lambda^*)^2}{\Sigma_{11} - \Sigma_{12}^2/\Sigma_{22}}, \qquad \lambda^* = \frac{\Sigma_{12}}{\Sigma_{22}} = \frac{\kappa\sigma_e^2}{\sigma_m^2 + \kappa^2\sigma_e^2}, \tag{6}$$

from both signals. Hence $\mathcal{I}_{e,y} \geq \mathcal{I}_y$, with equality exactly when $\lambda = \lambda^*$.

Proof. For a Gaussian vector with mean $\boldsymbol{a}\mu$ and known covariance Σ the information about μ is $\boldsymbol{a}^\top \Sigma^{-1} \boldsymbol{a}$. With $\boldsymbol{a} = (\lambda, 1)^\top$, factor the joint density as $p(y)\, p(e \mid y)$: the margin contributes $1/\Sigma_{22}$, and $e \mid y$ is normal with mean $\lambda\mu + (\Sigma_{12}/\Sigma_{22})(y - \mu)$ and variance $\Sigma_{11} - \Sigma_{12}^2/\Sigma_{22}$, whose derivative in μ is $\lambda - \lambda^*$. Adding the two informations gives the second form of Equation 6. The first form follows by factoring instead as $p(e)\, p(y \mid e)$: e contributes λ^2/σ_e^2 and $y \mid e$ has mean $(1-\kappa\lambda)\mu + \kappa e$ and variance σ_m^2. □

The proposition says that the efficiency margin helps exactly insofar as its scale λ differs from λ^*, the slope of the regression of e on y. If efficiency were nothing but a noisy copy of the scoreboard, its mean would move with μ exactly as the regression predicts from y, and it would add nothing. A luck-adjusted margin is not a noisy copy: by replacing the made threes and free throws by their expectations it removes the largest source of game-to-game noise from the scoreboard, so its residual variance is smaller, and its slope on strength is not the slope implied by its correlation with the margin.

Example. At the end of 2025–26 the posterior means are $\lambda = 0.927$, $\kappa = 0.587$, $\sigma_e = 13.94$ and $\sigma_m = 12.98$. The margin alone then has scale $\bar{\sigma} = 15.35$ points per 100 and carries $\mathcal{I}_y = 1/\bar{\sigma}^2 = 4.25 \times 10^{-3}$ per game. The slope at which efficiency would add nothing is $\lambda^* = \kappa\sigma_e^2/\bar{\sigma}^2 = 0.484$, 0.442 from the fitted λ, and both signals together carry $\mathcal{I}_{e,y} = 5.65 \times 10^{-3}$: a gain of 33%, or about 10 extra games' worth of information over a thirty-game season.

Table 6 evaluates Equation 6 at the posterior means of each complete season's final fit.

The efficiency residual explains 23%–32% of the margin's variance, and the fitted λ lies 0.38–0.45 from λ^*. The per-game information gain is 17% to 34%, about 5–10 extra games' worth over a thirty-game season. Two caveats limit how far this should be read. The calculation holds λ, κ and the scales fixed, whereas in the fit they are estimated jointly with strength; and it is per game, at the posterior mean. The backtest is the arbiter. Section 8.4 refits the model without the efficiency likelihood at every snapshot of 2022–23 to 2025–26: on the same 22,948 games, the margin-only model's mean absolute error is 0.04 points (standard error 0.01) higher and its log loss 2.2 $\times 10^{-3}$ (se 0.6) higher. At each season's final posterior the expected log predictive density of the margin itself (Section 7.3) favours the full model by 61 in a typical season (full minus margin-only: 44 to 82 across seasons, standard errors about 8).

Table 6: Per-game Fisher information about the expected margin per 100 possessions under the Gaussian model, at 70 possessions and each season's final posterior means. R^2 is the share of the margin's variance explained by the efficiency residual, $\kappa^2\sigma_e^2/\bar{\sigma}^2$.

Season	λ	λ^*	R^2	$10^3\mathcal{I}_y$	$10^3\mathcal{I}_{e,y}$	Gain
2014–15	0.978	0.542	0.30	4.65	5.94	28%
2015–16	0.953	0.525	0.30	4.73	6.11	29%
2016–17	0.901	0.519	0.30	4.61	5.70	24%
2017–18	0.926	0.480	0.28	4.49	6.01	34%
2018–19	0.912	0.493	0.29	4.43	5.72	29%
2019–20	0.913	0.519	0.23	4.34	5.09	17%
2020–21	0.934	0.508	0.31	4.19	5.54	32%
2021–22	0.931	0.513	0.30	4.52	5.83	29%
2022–23	0.928	0.523	0.32	4.40	5.63	28%
2023–24	0.925	0.512	0.32	4.37	5.71	30%
2024–25	0.910	0.483	0.28	4.28	5.57	30%
2025–26	0.927	0.484	0.28	4.25	5.65	33%

5 The Resume

5.1 Wins Above Bubble

A resume should value results, not performances, and should value them relative to a standard that matters for selection. The tournament takes 68 teams, but not the 68 best: each conference's champion qualifies automatically, however weak, and the committee chooses the rest. The natural standard for an at-large resume is therefore the last team to get in.

Bubble. In posterior draw s, let $A^{(s)}$ be the automatic bids — each conference's tournament champion, or its regular-season leader where it holds no tournament; known once decided, and otherwise simulated in the draw (Section 5.2) — and E the teams eligible for the tournament. With a field of F teams, the bubble strength $B^{(s)}$ is the $(F - |A^{(s)}|)$-th largest of $\{r_i^{(s)} : i \in E \setminus A^{(s)}\}$.

The bubble is a strength, not a team: in different draws it is a different team, its uncertainty is part of the posterior, and so is the uncertainty in which conference champions remove themselves from the at-large pool. Since a weak champion takes a place without taking an at-large one, the bubble sits near the 46th-best team by power in a typical season (the posterior mean rank of the bubble at Selection Monday), not the 68th.

Wins Above Bubble. Let team i have played Division I games $g \in G_i$, with result $W_g \in \{0, 1\}$ (overtime wins count), opponent $o(g)$, site $v_g \in \{-1, 0, 1\}$ from i's side and possessions n_g. The probability that the bubble team would have won the same fixture is

$$\pi_g^{(s)} = \Phi\Big(\big(B^{(s)} - r_{o(g)}^{(s)} + \eta^{(s)} v_g\big)\, \omega_g^{1/2}/\bar{\sigma}^{(s)}\Big),$$

the win probability of Section 3.1 with the bubble in team i's place, and

$$\mathrm{WAB}_i^{(s)} = \sum_{g \in G_i} \left(W_g - \pi_g^{(s)} \right). \tag{7}$$

The published value is the posterior mean $\mathrm{WAB}_i = S^{-1} \sum_s \mathrm{WAB}_i^{(s)}$ with an 80% interval.

Equation 7 uses the scoreboard only through W_g: margin enters the resume solely by having informed the strengths of the opponents and the bubble. Both are taken at the date of the snapshot, not at the date of the game: an opponent is valued at what the posterior now believes it to be, so a November win over a team that improves gains value as it does. The following properties are elementary but they are the reason for the choice.

WAB Properties. For every draw s:

(i) *Additivity.* $\mathrm{WAB}_i^{(s)}$ is a sum of per-game contributions $W_g - \pi_g^{(s)}$, each in $(-1, 1)$.

(ii) *Monotonicity.* A win contributes $1 - \pi_g^{(s)} > 0$ and a loss $-\pi_g^{(s)} < 0$; with the draw held fixed, changing any loss to a win raises $\mathrm{WAB}_i^{(s)}$ by exactly one, whatever the opponent.

(iii) *Schedule response.* The contribution of a win is increasing in the opponent's strength and that of a loss is increasing too (a loss to a strong team costs less), and both are decreasing in the site v_g: road wins count for more than home wins.

(iv) *Calibration.* If the model is correct and team i has strength $r_i^{(s)} = B^{(s)}$, then $\mathbb{E}[\mathrm{WAB}_i^{(s)} \mid \text{parameters}] = 0$ for every schedule.

Proof. Parts (i) and (ii) are immediate from Equation 7, since $0 < \pi_g^{(s)} < 1$. For (iii), $\pi_g^{(s)}$ is decreasing in $r_{o(g)}$ and increasing in v_g, and both contributions have the form $\text{const} - \pi_g^{(s)}$. For (iv), under the model $\Pr(W_g = 1) = \Phi\big((r_i - r_{o(g)} + \eta v_g)\, \omega_g^{1/2} / \bar{\sigma}\big) = \pi_g^{(s)}$ when $r_i = B^{(s)}$; an overtime game, a regulation tie, is decided by a coin flip of equal teams in the model's terms, which leaves the expectation unchanged when the teams are equal and moves it little otherwise. □

Property (ii) holds draw by draw. The *published* WAB_i changes by a slightly different amount when a result flips, because the result is also data for the posterior: a win moves the opponent's strength down, and the bubble with it. Section 5.6 measures a related effect of the margin. Property (ii) excludes what a strength-of-schedule bonus does, rewarding a team for *playing* strong opponents regardless of result; (iii) is the precise sense in which the schedule is counted once; and (iv) gives the scale its meaning: a positive WAB is evidence of having done better than a bubble team would have with the same games. Summing over games makes WAB a count, not a rate, so a team that wins an extra game in a conference tournament gains from it. That is intended: every game is a chance to earn or lose standing, and the committee sees the same games.

The possession weight in π_g is the only place where the margin's scale enters the resume, and it enters as the model's estimate of how decisive a game of that length is, not through the game's own result.

Example. Table 7 works through WAB for Texas, 17–14 at Selection Monday of 2025–26 and the last of the 37 at-large teams by WAB, given the actual automatic bids; Seton Hall was the first

team out. For each game, $\bar{\pi}_g$ is the posterior mean probability that the bubble team would have won the same fixture at the same site; the contribution is $W_g - \bar{\pi}_g$, and WAB is their sum, 0.00 in this refit (the published value, from a separate fit, is 0.00). Texas was in the committee's field, as an 11 seed.

5.2 The Rest of the Season

Before the automatic bids are decided, the bubble depends on who will win them. In each posterior draw the rest of the season is therefore played out: the remaining regular-season games from the draw's ratings and home court, then each conference tournament as a single-elimination bracket of the size the conference used the previous season, seeded by the simulated standings, with byes for the top seeds; a champion already crowned is fixed, as is the standings leader of a conference without a tournament. A champion still moving up to Division I is ineligible and the bid passes to the best eligible team in the standings, as those conferences did. The simulated champions are the draw's automatic bids, which set its bubble, and the draw's projected $\mathrm{WAB} - \mathrm{WAB}$ so far plus the simulated games, each scored against the same bubble — picks the at-large teams among the rest. The published probabilities of an automatic bid, an at-large bid and a place in the field come out jointly, and sum correctly over teams in every draw. The simulation uses the same draws as the resume, so its uncertainty is the posterior's; it does not simulate the selection committee, only a committee that selects by projected WAB.

5.3 Strength of Record, Quadrants and Ranks

Three further quantities are reported with the resume, as on the committee's team sheets. The **strength of record** is the probability that a top-25 team would do at least as well as team i did against its schedule: $\mathrm{SOR}_i = \Pr\big(\sum_g T_g \geq \sum_g W_g\big)$ with independent $T_g \sim \mathrm{Bernoulli}(\bar{\pi}_g^{25})$, where $\bar{\pi}_g^{25}$ is the posterior mean probability that the draw's 25th-best team wins game g's fixture. The sum of independent Bernoulli variables with unequal probabilities is Poisson-binomial, and its distribution is computed exactly by the recursion

$$P_0^{(0)} = 1, \qquad P_k^{(t)} = (1 - \bar{\pi}_t) P_k^{(t-1)} + \bar{\pi}_t P_{k-1}^{(t-1)}, \quad k = 0, \ldots, t,$$

over a team's games [29]. Smaller is more impressive. **Quadrant records** count wins and losses against opponents grouped by their power rank and the site, with the cut-offs the committee uses for its own ranking (top 30 at home, 50 neutral, 75 away for the first quadrant, and so on), applied to our ranks. **Schedule strength** is the bubble team's expected loss rate against the schedule, $1 - |G_i|^{-1} \sum_g \bar{\pi}_g$.

SOR plugs in the posterior mean $\bar{\pi}_g^{25}$ rather than averaging the per-draw probabilities as WAB does, and the two differ because the tail probability is non-linear in π_g. Refitting each Selection Monday with every draw kept, the largest difference over all teams and seasons is 0.017 and the rank correlation of the two versions is at least 0.963, so the cheaper version loses nothing.

Uncertainty in ranks is reported by ranking within each draw.

Table 7: Wins Above Bubble, game by game, for Texas at Selection Monday of 2025–26. H, A, N: home, away, neutral. $\bar{\pi}_g$: probability that the bubble team would win the same fixture.

Date	Opponent	Site	Score	W_g	$\bar{\pi}_g$	$W_g - \bar{\pi}_g$	Running WAB
Nov 04	Duke	N	60–75	0	0.088	-0.088	-0.09
Nov 08	Lafayette	H	97–60	1	0.992	0.008	-0.08
Nov 12	Fairleigh Dickinson	H	93–58	1	0.993	0.007	-0.07
Nov 15	Kansas City	H	71–55	1	0.998	0.002	-0.07
Nov 18	Rider	H	99–65	1	0.998	0.002	-0.07
Nov 24	Arizona State	N	86–87	0	0.661	-0.661	-0.73
Nov 26	NC State	N	102–97	1	0.450	0.550	-0.18
Dec 03	Virginia	H	69–88	0	0.373	-0.373	-0.55
Dec 08	Southern	H	95–69	1	0.982	0.018	-0.54
Dec 12	UConn	A	63–71	0	0.160	-0.160	-0.69
Dec 16	Le Moyne	H	95–53	1	0.988	0.012	-0.68
Dec 22	Maryland-Eastern Shore	H	94–71	1	0.997	0.003	-0.68
Jan 03	Mississippi State	H	98–101	0	0.754	-0.754	-1.43
Jan 06	Tennessee	A	71–85	0	0.177	-0.177	-1.61
Jan 10	Alabama	A	92–88	1	0.203	0.797	-0.81
Jan 14	Vanderbilt	H	80–64	1	0.342	0.658	-0.16
Jan 17	Texas A&M	H	70–74	0	0.568	-0.568	-0.72
Jan 21	Kentucky	A	80–85	0	0.239	-0.239	-0.96
Jan 24	Georgia	H	87–67	1	0.492	0.508	-0.46
Jan 28	Auburn	A	82–88	0	0.296	-0.296	-0.75
Jan 31	Oklahoma	A	79–69	1	0.388	0.612	-0.14
Feb 03	South Carolina	H	84–75	1	0.820	0.180	0.04
Feb 07	Ole Miss	H	79–68	1	0.694	0.306	0.35
Feb 14	Missouri	A	85–68	1	0.375	0.625	0.97
Feb 17	LSU	H	88–85	1	0.726	0.274	1.25
Feb 21	Georgia	A	80–91	0	0.303	-0.303	0.94
Feb 25	Florida	H	71–84	0	0.161	-0.161	0.78
Feb 28	Texas A&M	A	76–70	1	0.367	0.633	1.41
Mar 04	Arkansas	A	85–105	0	0.215	-0.215	1.20
Mar 07	Oklahoma	H	85–88	0	0.593	-0.593	0.61
Mar 11	Ole Miss	N	66–76	0	0.606	-0.606	0.00

Posterior rank summaries. Let $R_i^{(s)}$ be the rank of $\mathrm{WAB}_i^{(s)}$ among Division I teams in draw s (1 is best). The published rank orders the posterior means WAB_i, and the 80% rank interval is the 10% and 90% quantiles of $\{R_i^{(s)}\}_s$. The power rank is defined in the same way from $\tilde{r}_i^{(s)}$.

The published rank orders posterior means rather than taking the median rank, because the median ranks of different teams need not form a permutation.

5.4 Record-only Reference: the Colley Matrix

The model's resume uses opponent strengths from a model that sees margins. As a reference that sees only wins and losses we compute the Colley rating [4, 5].

Colley rating. Let n_i be team i's games, w_i and l_i its wins and losses, and n_{ij} the number of games between i and j. The Colley ratings $\boldsymbol{c}$ solve $C\boldsymbol{c} = \boldsymbol{f}$ with

$$C_{ii} = 2 + n_i, \qquad C_{ij} = -n_{ij}\ (i \neq j), \qquad f_i = 1 + \tfrac{1}{2}(w_i - l_i).$$

Colley Lemma. C is symmetric positive definite, so $\boldsymbol{c}$ exists and is unique, and $\frac{1}{n}\sum_i c_i = \frac{1}{2}$.

Proof. $C = 2I + L$, where L is the Laplacian of the schedule multigraph, which is positive semidefinite; so C is positive definite. Every column of L sums to zero, so $\mathbf{1}^\top C = 2\mathbf{1}^\top$ and $2\,\mathbf{1}^\top\boldsymbol{c} = \mathbf{1}^\top C\boldsymbol{c} = \mathbf{1}^\top\boldsymbol{f} = n + \frac{1}{2}\sum_i(w_i - l_i) = n$, since every game produces one win and one loss. □

The rating is Laplace's rule of succession, $(1 + w_i)/(2 + n_i)$, adjusted for the average rating of the opponents. We compute it on the same Division I games as the posterior; it ignores site, so a home win counts as much as a road win.

5.5 Choosing the Benchmark

The bubble is a choice about what the resume is for, not a parameter that data could estimate: a resume is retrodictive, and there is no future outcome against which a backtest could score it. It answers the question the committee answers for at-large teams — would a team on the edge of the field have done this well? — and two features of its construction follow from that question rather than from convenience: it removes the automatic bids before counting, and it uses the field's size, not a fixed rank. Table 8 shows how much the choice matters, by recomputing WAB at Selection Monday of every complete season with tournament under four alternatives: the bubble moved four places up or down the at-large order, and two benchmarks that ignore the bids, the 45th and the 25th-best team by power. The last column counts how many of the committee's at-large teams each benchmark's top WAB would have selected, given the actual automatic bids.

Over the whole of Division I the order barely depends on the benchmark (median rank correlation at least 0.997), because a benchmark changes the price of each result but not which results are good. The edge of the field does move: a stronger benchmark makes a loss cheap and a win over a weak team nearly worthless, which favours difficult schedules, while a weaker one makes

Table 8: Sensitivity of the Selection Monday resume to the benchmark, over complete seasons with a tournament. Rank correlation of WAB with the published bubble, among all Division I teams; the number of at-large selections by WAB that change; the mean posterior power rank of the benchmark; and the at-large teams the committee also chose (mean over seasons, of the number of at-large places). Median over seasons, with the range.

Benchmark	ρ with published	At-large picks changed	Benchmark rank	Committee's at-la
Last team in (published)	–	–	45	
Last four byes	1.000 (1.000–1.000)	0 (0–1)	41	
First four out	1.000 (1.000–1.000)	0 (0–1)	50	
Power rank 45	1.000 (1.000–1.000)	0 (0–0)	45	
Power rank 25	0.997 (0.995–0.998)	1 (0–3)	25	

Table 9: Effect of a blowout margin on the resume, 2025–26 final posterior. For each game: the change in each team's WAB and power rating (per 100 possessions) when the result is refitted as a one-point win for the same team, and the mean absolute change in WAB of every other team.

Game	Margin	ΔWAB winner	ΔWAB loser	Δ power winner	Δ pow
UConn over UMass Lowell	63	-0.141	0.025	-1.6	
Saint Louis over Bethune-Cookman	59	-0.034	0.000	-1.3	
Iowa State over Alcorn State	64	-0.058	0.012	-1.4	
Georgia over Maryland Eastern Shore	65	-0.029	-0.008	-1.5	
Gonzaga over Southern Utah	72	-0.050	0.091	-1.6	
Florida over Prairie View A&M	59	-0.053	0.014	-2.4	

every win count and every loss costly. Moving the bubble four places changes at most 1 of the at-large selections in any season, and none in a typical one. The 45th-best team by power, which ignores the bids, sits close to the published bubble in rank and selects the same at-large teams in every season; only the 25th-best team, a much stronger benchmark, moves the edge of the field. Which benchmark agrees best with the committee is a question about the committee, not about the measure; Section 9.2 compares the published one with its choices.

5.6 How Margin Reaches the Resume

WAB does not use margin directly, but margin informs the posterior strengths of the opponents and the bubble, so it reaches the resume indirectly. A blowout win makes the loser look weaker, which lowers the winner's credit for beating it. Table 9 measures this on the 6 most lopsided Division I games of 2025–26: each is refitted at the final snapshot with its regulation margin replaced by a one-point win for the same team (and its efficiency margin dropped), and the two teams' WAB and power ratings recomputed.

Turning a blowout of 59–72 points into a one-point win moves the winner's power rating by 1.3–2.4 points per 100, but its WAB by -0.141 to -0.029 wins and the loser's by -0.008 to 0.091; every other team's moves by at most 0.029 on average. For comparison, the Monte Carlo standard

Table 10: Monte Carlo standard errors of published quantities at Selection Monday, all Division I teams of every complete season with a tournament: median and largest over teams and seasons, with every post-warm-up draw.

Power rating	WAB	$\Pr(\text{WAB} > 0)$	Rank, 10% quantile	Rank, 90% quantile
0.032 / 0.077	0.006 / 0.011	0.003 / 0.008	0.000 / 1.000	0.000 / 0.500

error of a published WAB is about 0.006 (Section 6.1). The margin's influence on the resume runs through the opponents' strengths, as designed, and a team gains little by running up a score. Most of these games are against the weakest teams in Division I, whose strength barely matters to anyone's resume.

6 Computation and Pipeline

6.1 Sampling and Gates

Each posterior is sampled with the No-U-Turn sampler [30] in Stan [31] via CmdStan: 4 chains of 1,000 warm-up and 1,000 sampling iterations, target acceptance 0.9. A fit is checked against four gates on every team's centred rating at the snapshot date and every structural parameter: rank-normalised $\widehat{R} \leq 1.01$ and bulk effective sample size of at least 400 [32], no divergent transitions, and E-BFMI above 0.2 in every chain [26]. A snapshot that fails is still published, with the failure recorded; the informative priors of Section 3.4 were adopted precisely because they make failures rare. The resume, the team sheet and the season simulation use all 4,000 draws. Table 10 measures what that buys, from the Selection Monday refits of every complete season: the Monte Carlo standard errors of the published quantities.

The largest Monte Carlo error of a published WAB is 0.011 wins, and of a rating 0.077 points per 100: small next to the posterior uncertainty they summarise, so the second decimal of a published WAB is meaningful and the third is not.

Of the 277 posteriors, 274 pass every gate. Table 11 summarises them by season.

The 3 failures are all of one kind: no divergences, the smallest effective sample size or largest $\widehat{R}$ belongs to ϕ, with $\widehat{R}$ at most 1.014 and effective sample size at least 644: slow mixing of a weakly identified scale, not a failure of the sampler to explore, and too small to move a published rating.

6.2 The Pipeline

The computation is a `targets` pipeline in R [33, 34], drawn in Figure 4. Its structure enforces the two properties the evaluation depends on. First, the input to a snapshot contains only games played before its date, and the schedule of later games without their results, and is hashed, so a cached posterior is reused exactly when its input is unchanged, and new results or betting lines never invalidate it. Second, each season's preseason information is the final posterior of the previous season, and the roster regression is fitted on the seasons before; the pipeline tracks both

Table 11: Convergence gates by season, over all snapshots. Worst values are taken over snapshots and over every monitored parameter; *Minutes* is total sampling time, four chains in parallel.

Season	Snapshots	Passed	Max $\widehat{R}$	Min ESS	Divergent	Min E-BFMI	Minutes
2014–15	23	23	1.008	565	0	0.84	25
2015–16	23	23	1.008	774	0	0.83	25
2016–17	23	23	1.009	649	0	0.88	25
2017–18	23	22	1.014	644	0	0.86	25
2018–19	24	24	1.009	685	0	0.88	27
2019–20	20	19	1.011	733	0	0.86	18
2020–21	21	21	1.008	999	0	0.88	15
2021–22	23	23	1.010	610	0	0.85	25
2022–23	23	23	1.009	615	0	0.86	27
2023–24	24	24	1.008	558	0	0.86	29
2024–25	24	23	1.010	637	0	0.85	29
2025–26	24	24	1.009	624	0	0.77	29
2026–27	2	2	1.005	1,841	0	0.89	0

dependencies, so refitting one season refits every later one. The shot model of a season is trained on the two seasons before it, for the same reason. The analyses of this paper are targets too, built on request; they read the published inputs and never write to them.

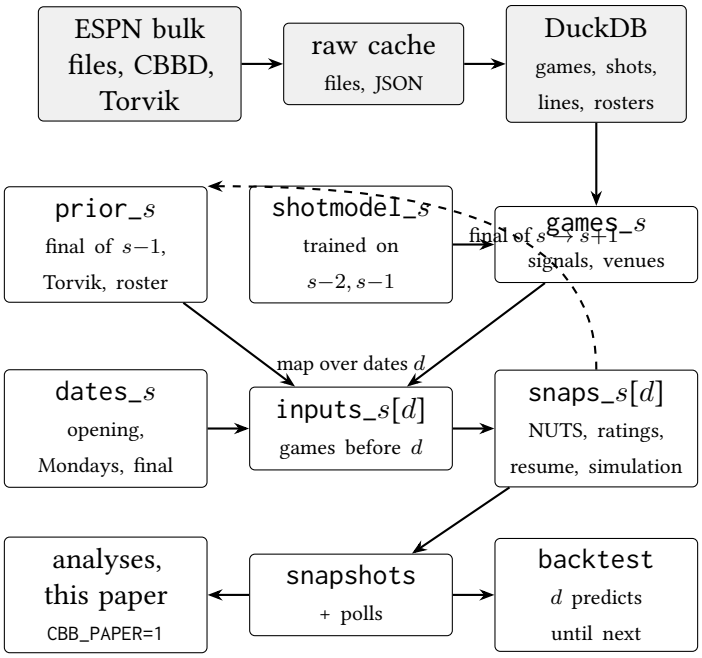


Figure 4: The computation as a `targets` pipeline. Grey boxes are external stores; white boxes are targets, with $[d]$ marking a dynamic branch per snapshot date and s a season. The dashed edge is the only cross-season dependency of the published fits: season s's final ratings become season $s+1$'s preseason information. In season, a daily build fetches the new results and refits only the latest snapshot.

7 Model Checking

Each complete season's final posterior is refitted with the pointwise log-likelihood stored, and three checks are run on it [21, 35]. This section summarises them across seasons; Section 11 examines two posteriors in full detail — sampler diagnostics, convergence, posterior geometry, prior-to-posterior learning, predictive distributions and influence.

7.1 Posterior Predictive Checks

Replicated regulation margins per 100 possessions are drawn from Equation 2, each game at its expected margin in the period it was played, turned into points with the game's regulation possessions and rounded to whole points, so that they can land exactly on the thresholds of the statistics. They are compared with six observed statistics: the standard deviation of the margin in points, the shares of games within three and within ten points, the share decided by 20 or more, the share tied after regulation, and the home team's regulation win rate on its own floor; the efficiency margin is compared on its standard deviation. Figure 5 shows the result for every season.

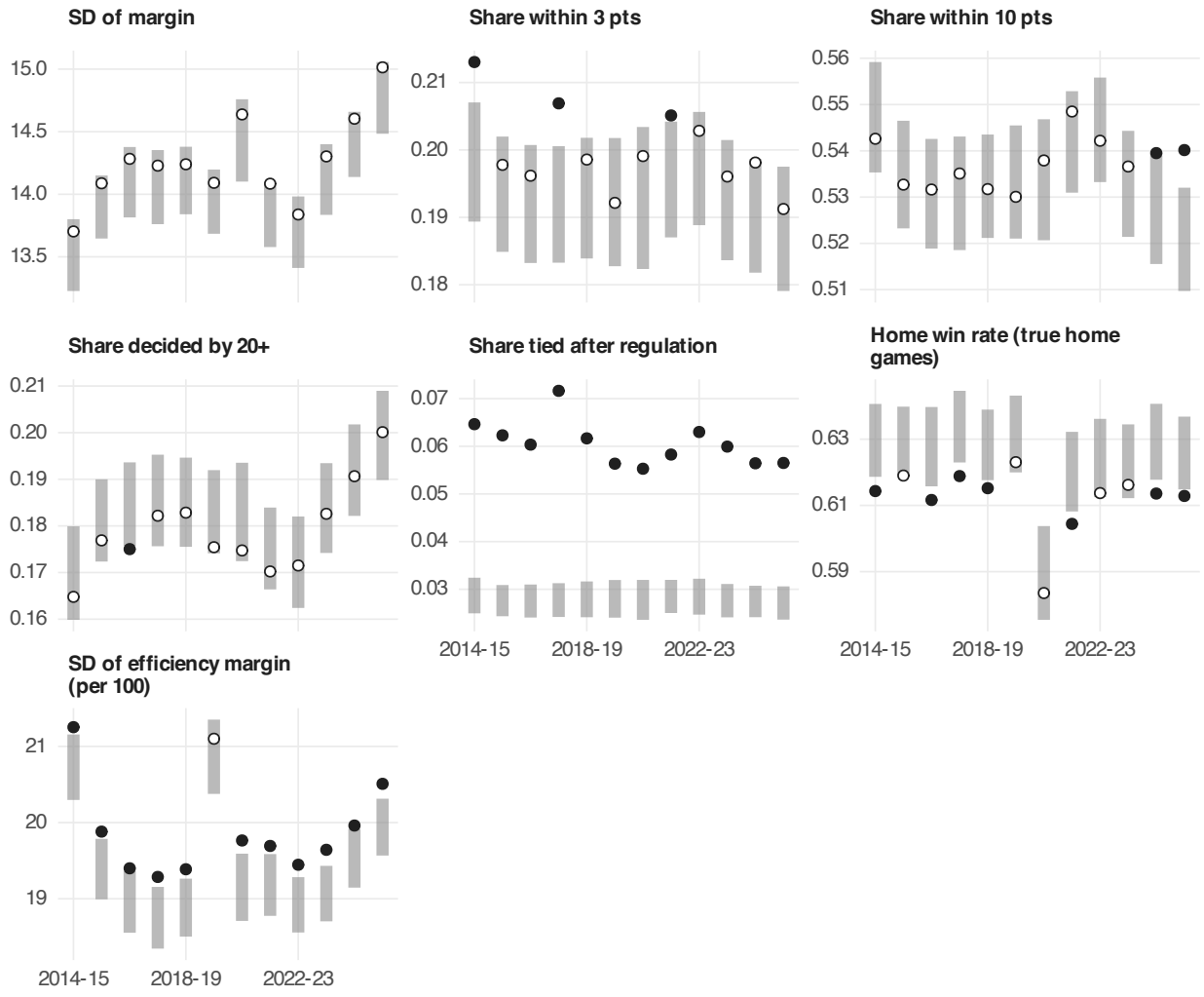


Figure 5: Posterior predictive checks at each complete season's final posterior. Bars: central 90% interval of the statistic over 400 replicated seasons; points: the observed value, filled when it falls outside the interval.

The model reproduces sd of margin and share decided by 20+ (outside the interval in at most one season each). It misses home win rate (true home games) in 7 of 12 seasons (observed 0.584–0.623, replicated mean 0.589–0.634), sd of efficiency margin (per 100) in 11 of 12 seasons (observed 19.286–21.253, replicated mean 18.755–20.850), share tied after regulation in 12 of 12 seasons (observed 0.055–0.072, replicated mean 0.027–0.029), share within 10 pts in 2 of 12 seasons (observed 0.530–0.548, replicated mean 0.521–0.546), and share within 3 pts in 3 of 12 seasons (observed 0.191–0.213, replicated mean 0.188–0.198). The share tied after regulation is the clearest case: a continuous law around the rating difference gives a tie only when a replicate rounds to zero, while real games are tied at the end of regulation more often, because the end of a close game is played to the score — the trailing team fouls and shoots threes to tie, the leading team plays for the last shot. These ties are the overtime games, whose regulation margin is zero by construction. The misfit is in the *shape* of the margin distribution near zero; the win probabilities,

Table 12: Power-scaling sensitivity by parameter, at each complete season's final posterior. Sensitivities are the cumulative Jensen–Shannon distance between the base and the power-scaled posteriors; the conventional threshold is 0.05.

Parameter	Prior (median)	Likelihood (median)	Seasons flagged: conflict	weak likelihood
η	0.047	0.158	5/12	0/12
β	0.059	0.113	10/12	0/12
β_2	0.051	0.110	7/12	0/12
λ	0.006	0.151	0/12	0/12
κ	0.008	0.544	0/12	0/12
σ_e	0.007	0.680	0/12	0/12
σ_m	0.007	0.279	0/12	0/12
ω	0.097	0.089	12/12	0/12
τ	0.081	0.154	9/12	0/12
ϕ	0.076	1.190	12/12	0/12

which depend on the location and the scale, are well calibrated (Section 8).

7.2 Prior Sensitivity

Power-scaling sensitivity [36] perturbs the prior and the likelihood by raising each to a power near one, estimates the resulting change in each marginal posterior by importance sampling, and flags a parameter whose posterior moves under both (prior–data conflict) or under the prior alone (weak likelihood). The likelihood here is that of the Division I games, efficiency and margin together. Table 12 counts the flags across seasons.

None of the 10 parameters are flagged for a weak likelihood in any season, and 6 for prior–data conflict in at least one (β, β_2, η, ω, τ and ϕ). The game-level parameters — the scales and loadings of the two signals — are set by thousands of games and their priors do not matter. The flags fall on the parameters that one season identifies only weakly: the hierarchy's spreads, the step, home court and the preseason loadings. For the four pooled ones that is what the pooling of Section 3.4 produces, and not a sign that the seasons disagree with their history: each pooled prior carries the information of a dozen seasons (Equation 5), so its standard deviation is comparable to or smaller than that of a single season's full-data posterior, and power-scaling either the prior or the likelihood then moves the posterior; priorsense labels a parameter that is sensitive to both a conflict. Whether a season actually departs from its prior is answered directly by the leave-one-season-out column of Table 24. How much the priors move the published products themselves is the subject of Section 11.11.

7.3 Leave-one-out Cross-validation

Pareto-smoothed importance-sampling leave-one-out cross-validation [37, 38] estimates the expected log predictive density of a new game, with the Pareto $\hat{k}$ diagnostic flagging games whose removal would change the posterior too much for the approximation.

Table 13: PSIS-LOO at each complete season's final posterior, on each Division I game's joint density of efficiency and margin. p_{loo} is the effective number of parameters; the last column counts games with $\hat{k} > 0.7$.

Season	Games	Teams	$\widehat{\text{elpd}}_{\text{loo}}$	SE	p_{loo}	$\hat{k} > 0.7$
2014–15	5,492	351	-42,552	134	316	0
2015–16	5,506	351	-42,418	113	315	0
2016–17	5,485	351	-42,661	108	312	0
2017–18	5,500	351	-43,154	103	312	0
2018–19	5,595	353	-43,420	116	314	0
2019–20	5,324	353	-42,955	830	371	1
2020–21	3,943	347	-31,167	86	266	0
2021–22	5,475	358	-43,821	89	329	0
2022–23	5,714	363	-45,951	87	322	0
2023–24	5,723	362	-45,894	87	333	0
2024–25	5,759	364	-46,435	91	330	0
2025–26	5,752	365	-46,747	83	332	0

The effective number of parameters is 266–371, 0.8–1.1 per team: a team's strength is a level and a walk of some twenty states, which the random walk and the hierarchy shrink heavily. At most 1 game per season has $\hat{k} > 0.7$: with thirty games per team no single game determines a team's strength. Because `log_lik` is the joint density of both signals where efficiency is observed, its $\widehat{\text{elpd}}$ cannot be compared with that of a model without efficiency; the program therefore also stores the marginal density of the margin alone, on which Section 4 compares the two models.

7.4 Choosing the Margin Family

The family $\mathcal{D}$ of Equation 1 was chosen on out-of-sample predictions: in a backtest of model variants on a fixed set of snapshots during development, the normal beat the Student-t on mean absolute error and log loss, and the contaminated normal tied it on log loss without improving the spreads and fitted more slowly. Table 14 repeats the comparison with the evidence of this paper: the ablation backtest of Section 8.4 on every snapshot of 2022–23 to 2025–26, and PSIS-LOO at each complete season's final posterior under the published priors, pooled over seasons (the sum of the per-season differences, with the root sum of squared standard errors).

By LOO the student-t fits the final posterior better than the normal in 12 of 12 seasons, by 13 in a typical season (392 pooled, standard error 175; 263 of it from 2019–20 alone, standard error 174), while the backtest favours the normal. The two answer different questions: LOO scores the density of every observed margin given the season's other games, including the tails that a heavier-tailed law fits better, while the backtest scores the spread and the win probability of the next week's games, which depend on the location and the scale. The published ratings and the resume use the win probability, so the backtest decides. None of the families reproduces the share of games tied after regulation (Section 7.1): the spike at zero is a property of how close games end, which a continuous location-scale law of any of these shapes does not reproduce (Section 10).

Table 14: Margin families. Backtest: difference from the normal in mean absolute error (points) and log loss on the same games, with standard errors clustered by snapshot (positive is worse). LOO: pooled difference in $\widehat{\mathrm{elpd}}$ from the normal over complete seasons (positive is better), and the seasons in which it is positive. Tied: seasons whose observed share of games tied after regulation lies outside the 90% predictive interval.

Family	Δ MAE	Δ log loss	$\Delta\widehat{\mathrm{elpd}}$	Seasons better	Tied
Normal (published)	–	–	–	–	12/12
Student-t	-0.001 (0.001)	+0.0002 (0.0001)	392 (175)	12/12	12/12
Contaminated normal	-0.002 (0.001)	-0.0003 (0.0002)	379 (186)	12/12	12/12

8 Out-of-sample Evaluation

8.1 Design

Every Division I game is predicted by the latest snapshot before it: the preseason snapshot predicts the first days of the season, each Monday's posterior the games of the following week, and so on. By construction (Section 6.2) a snapshot contains no information from its own date or later. This is a rolling-origin evaluation [39] with one origin per week, covering every game of every complete season from 2014–15 to 2025–26, the conference and national tournaments included. Predictions are scored by the absolute error of the spread, the continuous ranked probability score of the predictive distribution of the margin, and the Brier score [40] and log loss of the home win probability, all strictly proper except the absolute error [41]. Two references are scored on the same games. The closing betting line is the sharpest public forecast available, since it aggregates information — injuries, rest, motivation — that no ratings model sees [42]; beating it is not the aim, and coming close to it with public results alone is evidence that the ratings are sound. Torvik's ratings as they stood the day before each snapshot, from the site's archive, give a second reference built from the same kind of data: Torvik's adjusted offensive and defensive efficiencies and tempos turn into a spread by the usual multiplicative formula, with a home-court advantage and a margin scale estimated from earlier seasons only.

8.2 Accuracy

Table 15 scores every predicted game of the complete seasons, overall and by part of the season, and Table 16 compares the model with the closing line and with Torvik on the games that have both.

Over all 65,268 predicted games the mean absolute error is 8.80 points (95% interval 8.73–8.87), the Brier score 0.1763 (0.1732–0.1791) and the log loss 0.5230 (0.5152–0.5304), and the favourite wins 73.0% of the time. Games predicted by the same posterior share its errors, so these intervals come from a bootstrap that resamples whole snapshots, and the comparisons below use standard errors clustered the same way. On the 58,416 games with a closing line the model's error is 8.81 points against the line's 8.75. The difference, 0.062 points per game with standard error 0.014 ($z = 4.4$ in a Diebold–Mariano comparison of absolute errors [43]), is distinguishable from zero: the market

Table 15: Rolling out-of-sample accuracy, all predicted Division I games of the complete seasons. MAE and RMSE are errors of the predicted margin in points; CRPS scores the predictive distribution of the margin, in points; Brier score, log loss and accuracy score the home win probability.

Games	N	MAE	RMSE	CRPS	Brier	Log loss	Accuracy
All games	65,268	8.80	11.19	6.26	0.1763	0.5230	73.0%
January–April	40,805	8.61	10.91	6.12	0.1901	0.5589	70.5%
November–December	24,463	9.12	11.63	6.51	0.1532	0.4632	77.1%

Table 16: The model against the closing line and against Torvik's ratings at the same date, on the games with a line (and, for Torvik, with Torvik ratings for both teams). The line's win probability is normal around the line with the standard deviation of earlier seasons' line errors. ATS is the share of games in which the model's side of the line covered, pushes excluded; the break-even rate against a standard -110 price is 52.4%.

Comparison	Games	MAE, model	MAE, other	Log loss, model	Log loss, other	ATS
Model vs. closing line	58,416	8.81	8.75	0.5262	0.5236	51.8%
Model vs. Torvik	54,428	8.85	8.98	0.5261	0.5324	–

is better, though by little. Its win probabilities are worse than the line's (Brier difference 0.0009, $z = 2.8$). Against the line the model picks the covering side in 51.8% of 57,729 games (95% interval 51.4%–52.2%), above the 50% of a forecast with no edge on the market, but short of the 52.4% needed to profit at standard prices. Against Torvik's ratings at the same dates the model's absolute error is 0.134 points smaller ($z = -10.9$) and its log loss 6.3 $\times 10^{-3}$ smaller ($z = -8.8$), on the same 54,428 games.

Figure 6 shows how accuracy evolves through the season. The error is largest in the first week, predicted from the preseason information alone, and falls through November and December as the games accumulate. The gap to the line is smaller early than late (0.05 points in November and December, 0.07 from January): the preseason regression keeps up with the market's knowledge of rosters, and the market's edge lies in what it learns during the season and the ratings do not see. The log loss *rises* from January, and the right panel shows why: conference play pairs more evenly matched teams, so the predicted probabilities move towards one half and every forecast, the line's included, is less certain.

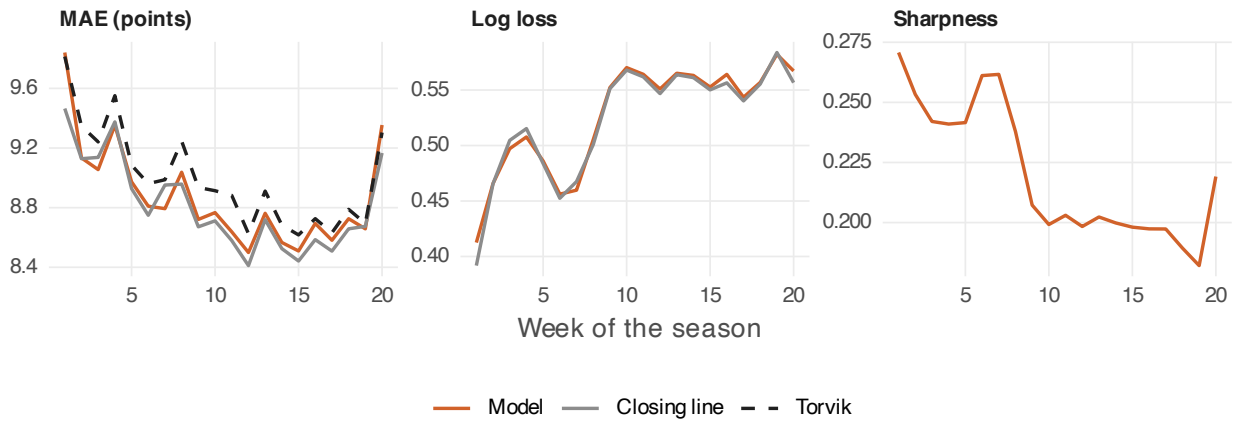


Figure 6: Rolling out-of-sample accuracy by week of the season (week 1 is the first seven days, predicted from the preseason). Left: mean absolute error of the model's spread, of the closing line and of Torvik's ratings, on the games with all three. Centre: log loss of the model's and the line's win probabilities. Right: sharpness, the mean of $|p - 1/2|$ over the model's predicted probabilities. Weeks with fewer than 200 games are omitted.

The predictive distribution of the margin. A spread is a point forecast; the model also implies a predictive distribution of the margin, normal with the standard deviation implied by its win probability, which combines the uncertainty in the ratings at the snapshot with the game noise. Its central 50%, 80% and 95% intervals contain 51.1%, 80.3%, and 94.5% of observed margins, and its continuous ranked probability score is 6.26 points. The slope of the observed margin on the predicted one is 1.030 (the line's: 1.007), so the spreads are neither too bold nor too timid.

Totals. The tempo model of Section 3.3 predicts the final combined score, overtime included, as the over/under is settled. On the 57,772 games with an over/under its mean absolute error is 13.36 points against the over/under's 13.13. The gap, 0.23 points, is larger than the gap on the spread: pace and shooting depend on who plays, and the market knows the injury report.

8.3 Calibration

A win probability is useful to the resume only if it is calibrated, because WAB subtracts π_g from an observed result. Figure 7 groups all predicted games into ten fixed-width bins of their win probability and compares the mean prediction in each bin with the observed frequency, overall and by part of the season.

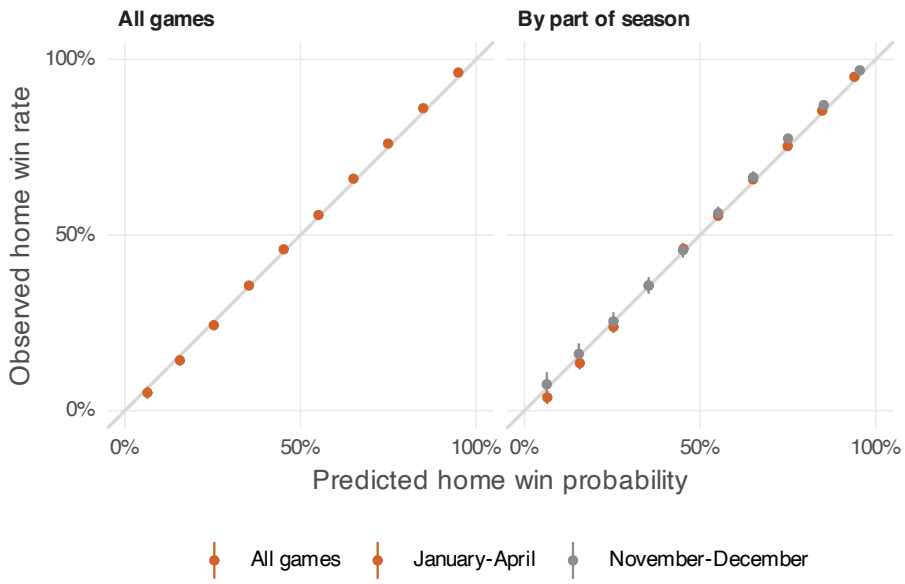


Figure 7: Calibration of out-of-sample home win probabilities. Games are grouped into fixed-width bins of predicted probability; each point is a bin's mean prediction and observed home win rate, with a 95% binomial interval. The diagonal is perfect calibration. Left: all games. Right: November–December and January–April.

Every bin lies within 1.3 percentage points of the diagonal, but with 65,268 games a gap of a point can be real. 4 of the 10 bins are more than two standard errors from it: (0.6,0.7], predicted 0.651 against observed 0.661 ($z = 2.1$); (0.7,0.8], predicted 0.749 against observed 0.760 ($z = 2.6$); (0.8,0.9], predicted 0.849 against observed 0.861 ($z = 3.3$); (0.9,1], predicted 0.949 against observed 0.963 ($z = 6.6$). The expected calibration error, the game-weighted mean gap between prediction and outcome within bins, is 0.010, and Spiegelhalter's test [44] gives $z = -7.0$: the predictions are slightly too cautious, the observed outcomes a little further from one half than predicted. The caution is the posterior's: averaging the win probability over the uncertainty in the ratings pulls it towards one half, and early in the season, when that uncertainty is largest, the spreads are slightly over-shrunk as well (the slope of the margin on the spread is above one in November and December). A recalibration of the log-odds fitted on earlier seasons only, which sharpens them by about five percent, improved the log loss by less than 10^{-3} in development, too little to be worth the extra step, and is not applied.

The calibration of predicted games bears on the resume only indirectly. The bubble probabilities π_g of Equation 7 use the same predictive law, but they describe a hypothetical fixture — the bubble team against the opponent — and are computed in sample, from the posterior that includes the game itself; no outcome can check them directly. That the model's out-of-sample probabilities for real fixtures are close to calibrated, in conference play especially, is supporting evidence that the scale of π_g is right, not proof.

Figure 8 repeats the comparison with the line and with Torvik season by season. The gaps are stable, which suggests they reflect information the market has and the model does not rather than

Table 17: Rolling out-of-sample accuracy of model variants, every Division I game of 2022–23 to 2025–26, each predicted by the snapshot before it. Each variant changes one thing. Δ columns: difference from the published model on the same games, with a standard error clustered by snapshot; positive is worse.

Variant	MAE	Log loss	Δ MAE	Δ Brier $\times 10^3$	Δ log loss $\times 10^3$
Published model	8.934	0.5273			
Margin only	8.975	0.5295	+0.041 (0.008)	+0.95 (0.25)	+2.19 (0.61)
Shot-quality efficiency	8.942	0.5282	+0.008 (0.008)	+0.36 (0.19)	+0.93 (0.44)
No preseason information	9.413	0.5488	+0.478 (0.142)	+8.55 (2.32)	+21.48 (5.87)
No two-season carry-over	8.974	0.5280	+0.040 (0.019)	+0.15 (0.11)	+0.71 (0.40)
No conference level	8.936	0.5274	+0.002 (0.001)	+0.03 (0.04)	+0.11 (0.11)
Static ratings	8.937	0.5277	+0.003 (0.002)	+0.14 (0.05)	+0.38 (0.13)
No non-D1 games	8.939	0.5277	+0.005 (0.003)	+0.17 (0.09)	+0.41 (0.23)
Student-t margin	8.934	0.5275	-0.001 (0.001)	+0.02 (0.04)	+0.20 (0.11)
Contaminated-normal margin	8.932	0.5270	-0.002 (0.001)	-0.08 (0.05)	-0.32 (0.16)

a drifting model.

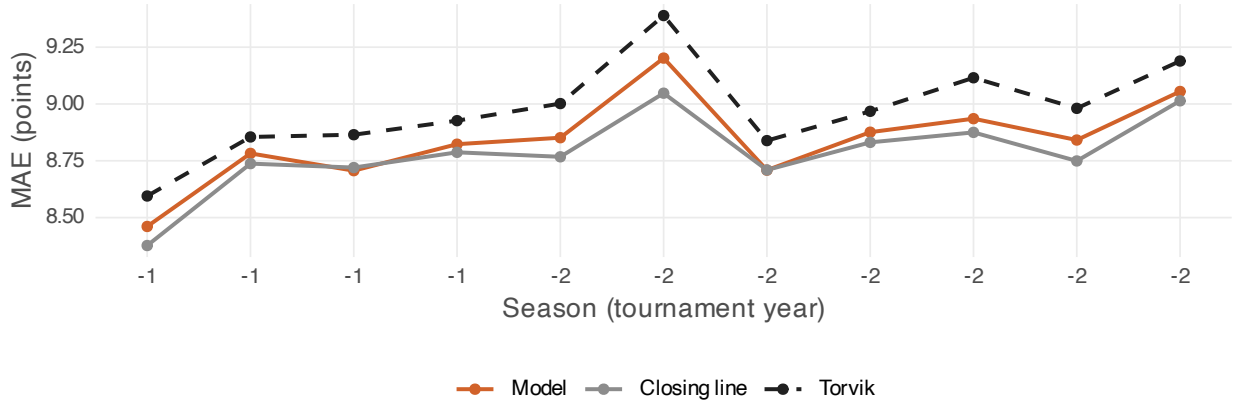


Figure 8: Out-of-sample mean absolute error by season on the games with a closing line and Torvik's ratings: the model, the closing line and Torvik on the same games.

8.4 Ablations and Baselines

Each component of the model is tested by removing it and repeating the rolling backtest on the same games. Refitting every snapshot of every season for each variant is expensive, so the ablations cover the 4 most recent complete seasons, 2022–23 to 2025–26 (22,948 games), with shorter chains than the published fits (two chains of 400 draws; the predictions are posterior means, for which that is ample). The published configuration is refitted the same way, so every row of Table 17 is comparable.

Removing a component makes the spreads measurably worse (by more than two standard errors)

for: no two-season carry-over, margin only, and no preseason information. No variant beats it by more than two standard errors. The preseason information matters most, as expected in a season whose first month is played between teams that have barely played anyone: without it, 0.48 points of absolute error per game over the whole season. The margin-only model settles the question of Section 4 empirically: the efficiency likelihood changes the mean absolute error by -0.041 points and the log loss by -2.19 $\times 10^{-3}$ (published minus margin-only). Against the pure shot-quality margin the luck-adjusted one is indistinguishable on the spread (+0.008 points, se 0.008) and better on the log loss (+0.93, se 0.44, $\times 10^{-3}$, shot-quality minus published): actual two-point finishing carries information about strength that a shot model's average make rate for the same shot does not. Freezing ratings within the season (*static ratings*, $\phi = 0$) changes the mean absolute error by +0.003 points (se 0.002) and the log loss by +0.38 $\times 10^{-3}$ (se 0.13): the spreads barely change, but letting ratings drift measurably sharpens the win probabilities, which is what the resume uses.

The model can also be compared with a rating system built differently. Least-squares ratings on the margin per 100 possessions, refitted at every snapshot with a weak ridge penalty and a home-court term (the Massey method, unpooled and static) and turned into spreads with the same expected possessions, predict the same games from the first Monday on: mean absolute error 9.32 against the model's 8.78, log loss 0.5566 against 0.5297, on 56,916 games (the least-squares win probabilities use a normal law with the error of earlier seasons, so the first season is not scored). Torvik's ratings, the stronger baseline, were compared above. At the end of each complete season the final power rating's rank correlation with Torvik's final ratings is 0.979–0.988.

8.5 Priors and Preseason Information Out of Sample

The leakage of pooling the priors over every season can be measured directly. The first snapshots of every complete season — opening night and the first three Mondays — refitted with priors pooled from the *other* seasons only predict the games until the next snapshot with a mean absolute error of 9.292 points against 9.288 for the published predictions of the same 14,616 games, and a log loss of 0.4644 against 0.4644: the leakage is negligible in the weeks where the priors matter most, and none at all later, when the data dominate them.

The preseason prediction d_i is the largest single input of the first weeks (Table 17), and its sources can be compared without refitting. Table 18 predicts every Division I game before each season's second Monday from the preseason information alone — the difference of the two teams' preseason values plus the season's home-court advantage, in points with the game's possessions — for each source, and correlates each with the final ratings.

Our own final rating from last season is the weakest single source (10.06 points), since college rosters turn over heavily from one year to the next; Torvik's preseason rating, which uses the incoming roster, is better (9.66), and the regression that combines both with the roster's turnover is the best (9.51). The transfer portal has made turnover larger and more uneven since 2020–21, which is what the roster terms — the value of incoming transfers and of departing players — are there to capture.

Table 18: Preseason sources, every Division I game before the second Monday of each season, predicted from the preseason value alone, and its correlation with the season's final rating; 2015–16 to 2025–26. The published prior is the roster regression, fitted on earlier seasons only, where it has enough history, and the even blend before.

Source	Games	MAE	Correlation with final
Our final rating last season	6,371	10.06	0.866
Torvik's preseason rating	6,371	9.66	0.907
Even blend of the two	6,371	9.59	0.909
Published (roster regression)	6,371	9.51	0.921

8.6 Tuned Choices

Every choice that was tuned or set from data is listed in Table 19, with the data it used. Several used the same seasons that the backtest reports, which makes the reported backtest slightly optimistic for those choices; Section 8.5 bounds the effect for the pooled priors, and the preseason regression is fitted on earlier seasons only.

9 The 2025–26 Season

This section uses the 2025–26 season to show what the two products look like and how they relate. Most of it uses Selection Monday, when the resume matters, and the final posterior, after the tournament.

Example. The 2025–26 national championship game, Michigan against UConn at a neutral site, was predicted by the posterior of April 6. There the posterior mean ratings were 40.7 and 32.0 points per 100 possessions, so the expected margin is 8.7 per 100 for Michigan; with 68.0 expected possessions from the tempo model, the predicted spread is 5.9 points. With $\bar{\sigma} = 15.35$ at the posterior mean, the plug-in win probability is $\Phi(8.72 \times 0.985/15.35) = 0.712$. The published probability averages over the posterior draws instead, 0.707, closer to one half because the uncertainty in the two ratings widens the predictive distribution. The predicted total was 149.1 against a closing over/under of 146.5. The closing line had Michigan favoured by 6.5, and the result was 69–63.

Table 20 lists the top of the resume at Selection Monday.

The rank intervals are the point of reporting a posterior. Figure 9 shows them for the top 25 in both products. Among the resume top 16 the median width of the 80% rank interval is 4 places for the resume and 12 for power. Resume ranks are the sharper, because once the results are fixed only the valuation of the opponents is uncertain, while power ranks describe strength itself.

Table 19: Every tuned or data-driven setting, how it was set, and whether the data overlap the reported backtest.

Setting	Value	How set	Overlaps backtest
Pooled priors ω, τ, ϕ, η	Table 5	Random-effects pooling of vague-prior refits, all complete seasons (Equation 5)	Yes (Section 8.5)
Other structural priors	Table 5	Weakly informative, on the scale of the sport	No
Margin family	normal	Development backtest of variants (Section 7.4)	Yes
Efficiency signal	luck-adjusted	Development backtest of variants	Yes
Preseason regression	own, Torvik, roster	Least squares on earlier seasons only	No
Two-season carry-over β_2	on	Development backtest of variants	Yes
Garbage-time rule	20 pts / 10 min, 15 / 5	Judgement; tighter, looser and none within noise	Yes
Free-throw shrinkage	40 attempts	Judgement	No
Tempo model ratios, half-life	Section 3.3	Weekly backtest of totals	Yes
Bubble	last team in	The committee's question (Section 5.5)	No
Sampler and gates	4 chains, 1000/1000	Conventional thresholds	No

Table 20: The 2025–26 resume at Selection Monday, top 25. WAB with its 80% interval; the resume rank with its 80% rank interval; the power rating (per 100 possessions above an average Division I team) with its rank; quadrant-1 record; strength of record; and the NCAA tournament seed.

	Team	Record	WAB	Rank 80%	Power	Q1	SOR	Seed
1	Michigan	31–3	14.01 (13.2, 14.8)	1–2	39.1 (3)	15–3	<0.001	1
2	Arizona	32–2	13.46 (12.7, 14.1)	1–3	38.0 (4)	17–2	<0.001	1
3	Duke	32–2	13.24 (12.5, 14.0)	2–3	40.6 (1)	16–2	<0.001	1
4	UConn	29–5	9.39 (8.4, 10.3)	4–7	30.8 (10)	10–4	0.0091	2
5	Florida	26–7	9.19 (8.4, 9.9)	4–7	39.1 (2)	14–7	0.019	1
6	Purdue	27–8	8.90 (8.1, 9.7)	4–8	34.4 (7)	12–8	0.035	2
7	Houston	28–6	8.81 (8.1, 9.5)	5–8	35.3 (5)	10–6	0.018	2
8	Arkansas	26–8	7.83 (7.1, 8.6)	7–12	26.5 (19)	12–8	0.055	4
9	Virginia	29–5	7.64 (6.9, 8.3)	8–14	28.3 (16)	9–4	0.036	3
10	St. John's	28–6	7.49 (6.5, 8.5)	8–15	29.9 (11)	9–5	0.085	5
11	Vanderbilt	26–8	7.35 (6.6, 8.1)	9–15	29.6 (13)	12–7	0.094	5
12	Nebraska	26–6	7.29 (6.6, 8.0)	9–15	24.9 (24)	10–6	0.044	4
13	Alabama	23–9	7.27 (6.5, 8.0)	10–15	27.7 (17)	11–7	0.096	4
14	Michigan State	25–7	7.25 (6.6, 7.9)	9–15	29.5 (14)	9–7	0.055	3
15	Iowa State	27–7	7.21 (6.5, 7.9)	9–15	35.0 (6)	8–7	0.068	2
16	Kansas	23–10	6.39 (5.7, 7.1)	14–18	24.7 (25)	11–9	0.16	4
17	Illinois	24–8	6.25 (5.6, 6.9)	15–18	33.7 (8)	9–8	0.14	3
18	Gonzaga	30–3	5.84 (5.2, 6.5)	16–19	31.6 (9)	5–2	0.074	3
19	Wisconsin	24–10	5.66 (4.9, 6.4)	17–20	25.2 (23)	10–8	0.23	5
20	Texas Tech	22–10	4.94 (4.3, 5.5)	19–22	23.5 (29)	8–9	0.27	5
21	North Carolina	24–8	4.66 (4.0, 5.3)	20–23	26.3 (20)	6–7	0.32	6
22	Tennessee	22–11	4.60 (3.9, 5.3)	20–24	29.6 (12)	8–10	0.39	6
23	Villanova	24–8	4.21 (3.3, 5.1)	20–28	21.2 (33)	6–6	0.42	8
24	Kentucky	21–13	3.88 (3.2, 4.6)	23–28	26.0 (21)	7–12	0.51	7
25	BYU	23–11	3.81 (3.1, 4.5)	23–29	26.8 (18)	8–10	0.54	6

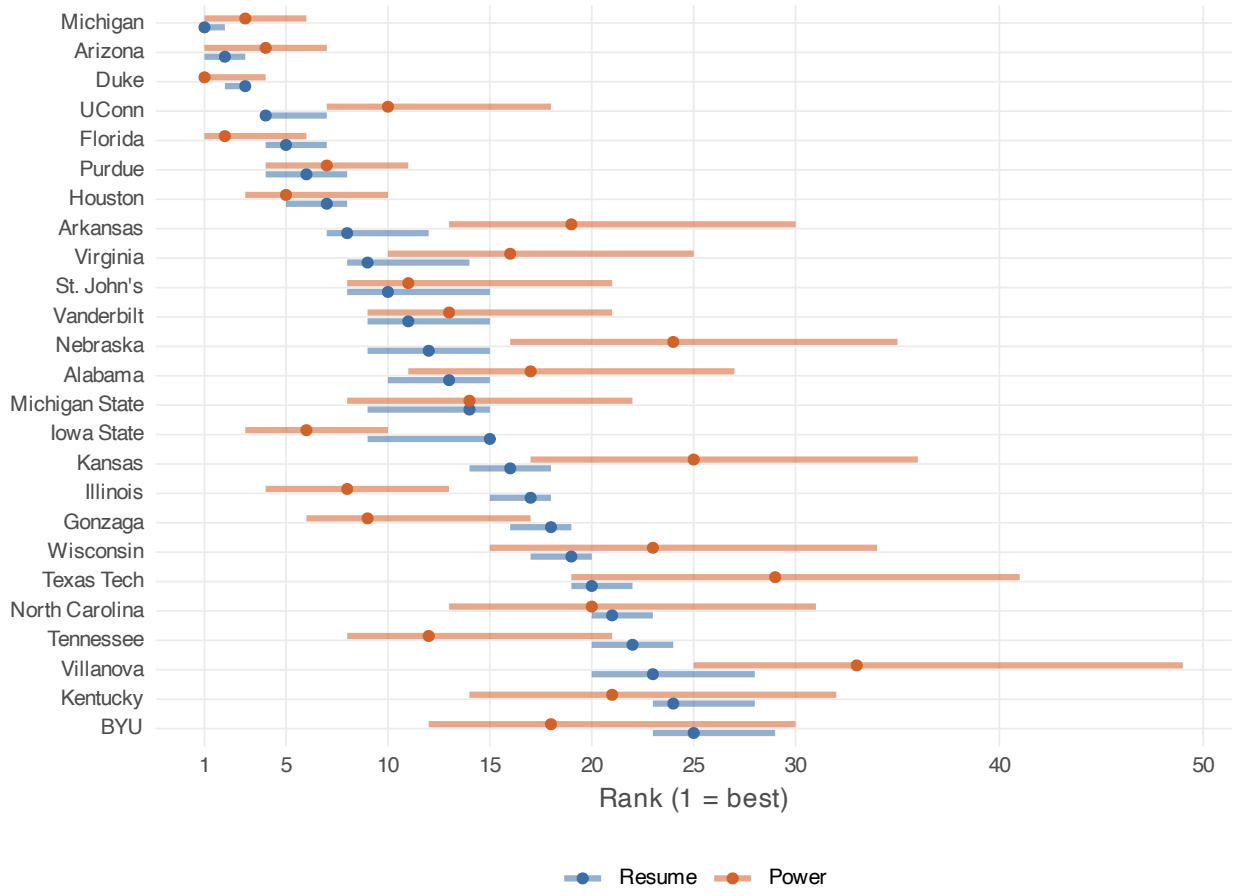


Figure 9: Posterior rank intervals at Selection Monday of 2025–26 for the top 25 of the resume. Points are the published ranks; bars the central 80% of the rank across posterior draws. Teams are ordered by resume rank.

Figure 10 plots the two products against each other for every Division I team at Selection Monday. The relation is monotone but loose. The right panel shows what the scatter around the curve is: the vertical distance of each team from the curve against its *luck*, actual wins minus the wins its own rating would predict against its schedule (the sum of its win probabilities in each game under the posterior). The two are correlated at 0.81: teams above the curve have won more games than their strength predicts, usually close ones, and teams below it have lost them.

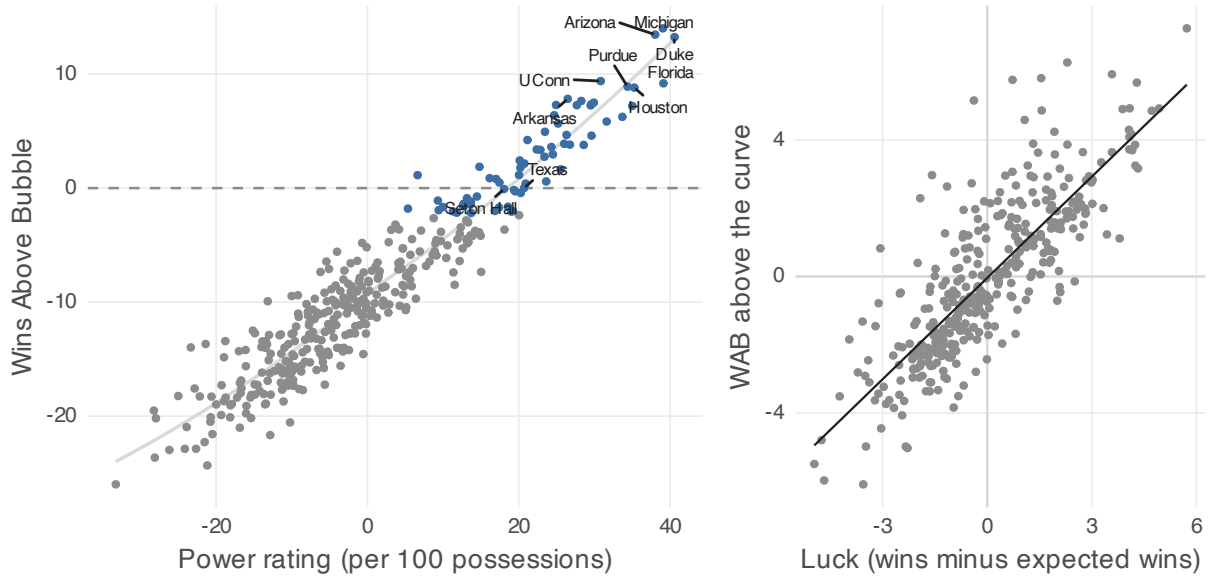


Figure 10: Resume against power at Selection Monday of 2025–26, all Division I teams. Left: the dashed line is zero Wins Above Bubble, teams above it did better than the bubble team would have with their schedule; the grey curve is a LOESS fit; labelled, the resume top 8, the last team in and the first team out. Right: each team's distance above or below that curve against its luck, actual minus expected wins.

Figure 11 follows the power rating of the resume top six through the season. Preseason ratings come from the preseason prediction alone and are wide: the median 80% interval is 22.9 points per 100 on opening night, 7.0 at Selection Monday and 7.0 after the tournament. With the random walk a rating can fall as well as rise late in the season, and the bands do not narrow to zero: the walk keeps adding uncertainty of ϕ per week while each game removes some.

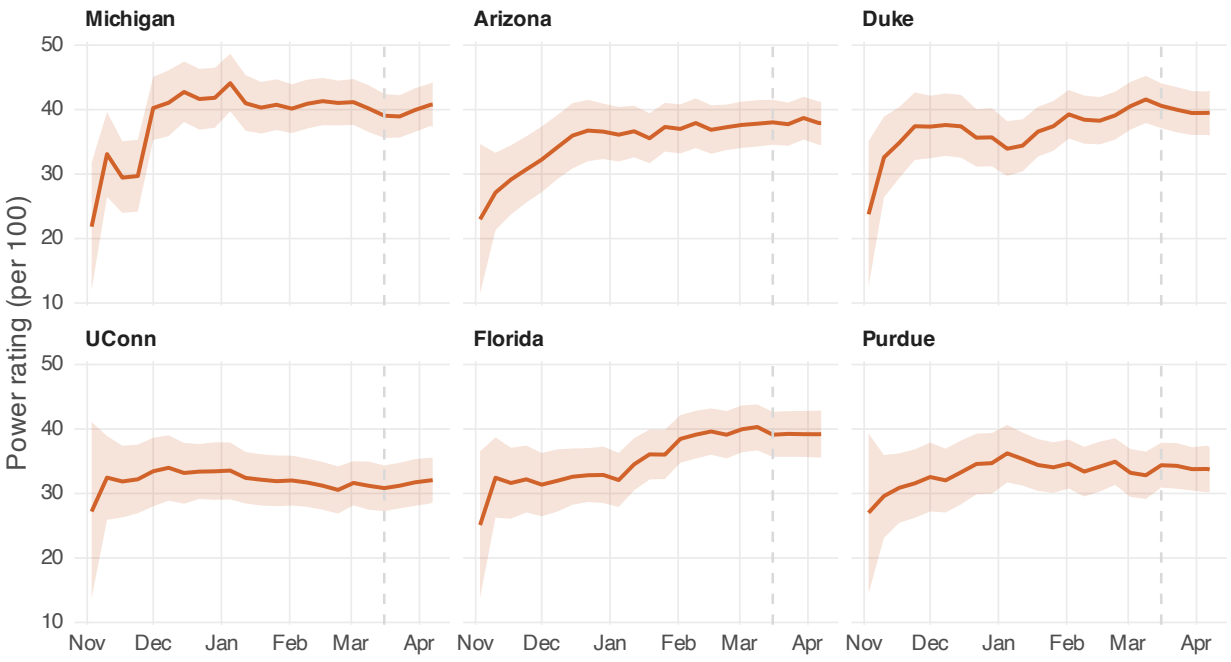


Figure 11: Power rating through 2025–26 for the Selection Monday resume top six, with 80% posterior intervals; the first point is opening night and the last the final posterior after the tournament.

Figure 12 splits the final 2025–26 power ratings into offence and defence (Section 3.3). The best offence by this measure is Purdue's, 130.9 points per 100 possessions against an average defence, and the best defence Michigan's, 88.3 allowed. Across Division I good offences and good defences tend to go together (correlation of Off with $-$Def 0.86), and teams with the same power rating can get there in very different ways: every point on one diagonal of the figure has the same Off $-$ Def.

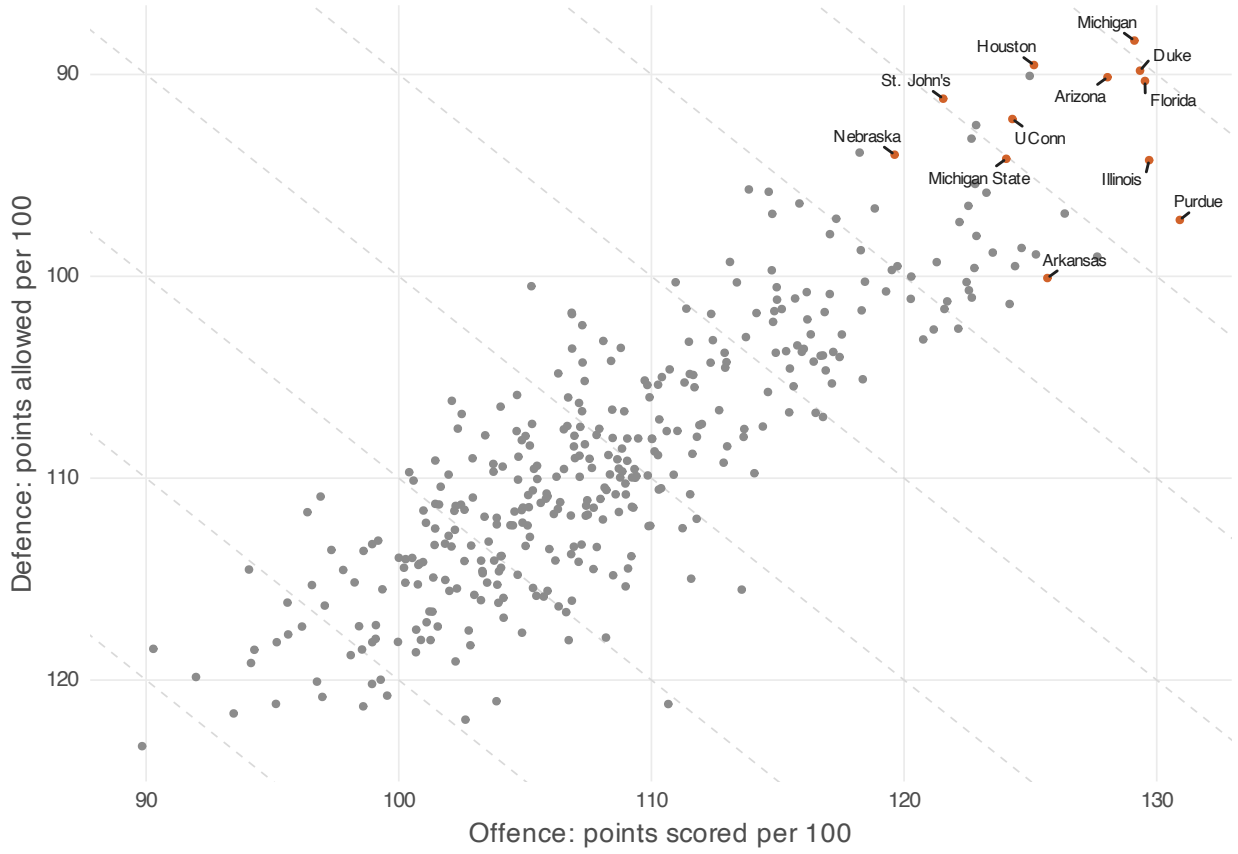


Figure 12: Offence and defence at the end of 2025–26, all Division I teams: points scored and allowed per 100 possessions against an average team on a neutral floor. The defence axis is reversed, so better teams are up and to the right; dashed diagonals are lines of equal power rating, 10 points apart. Labelled, the final resume top 12.

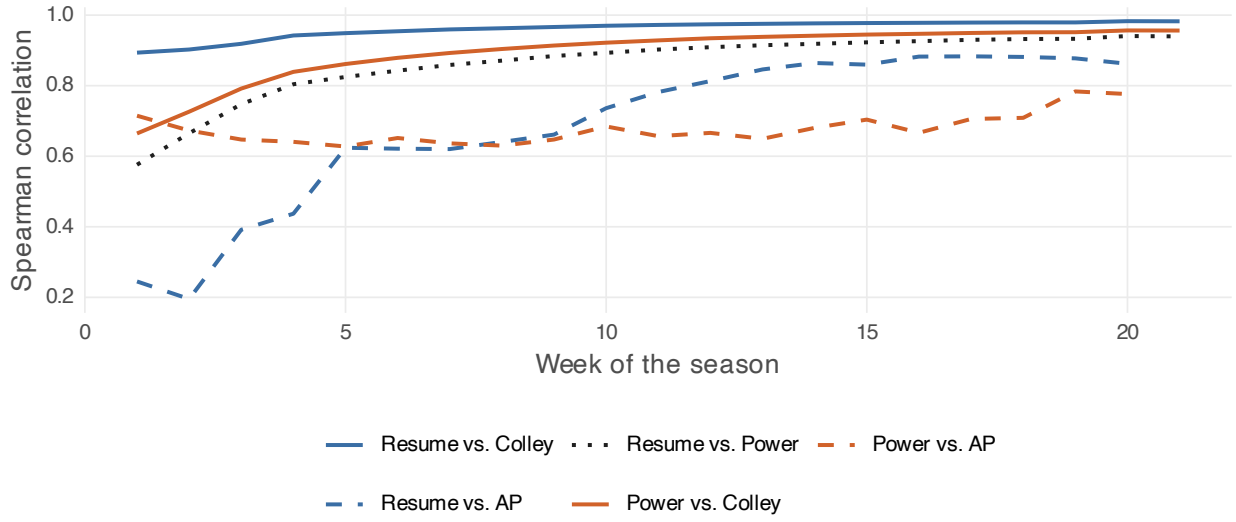


Figure 13: Spearman rank correlation between rankings of Division I teams, by week of the season, averaged over complete seasons. Correlations with the AP poll use the teams it ranks.

Figure 13 measures agreement between the products and the references by Spearman rank correlation among Division I teams, week by week, averaged over the complete seasons. The resume tracks the record-only Colley ranking closely (mean correlation 0.92 in weeks 2–4 and 0.98 in weeks 15–18) — both are functions of the record — and differs from it in how opponents and sites are valued. The power rating is less correlated with Colley (0.79 early, 0.95 late), as it should be, because it uses margin and efficiency. Agreement with the AP poll, which ranks only 25 teams, is computed on those 25.

Example. The products disagree most for teams that are strong but lost games. At Selection Monday of 2025–26, Baylor had power rank 41 (20.0 per 100) but resume rank 69, at 15–16. Its own posterior gave it 18.0 expected wins against this schedule, so its luck was -3.0; 16 losses, 3 of them by five points or fewer, cost it 5.92 wins of WAB, while its 15 wins earned 3.54. The power rating reads those losses as close games that a strong team loses sometimes; the resume counts them as losses.

9.1 The 2025–26 Field

The first question a reader will ask of a resume is whether it picks the teams the committee picked. Given the actual automatic bids, the 37 at-large places of 2025–26 filled by WAB at Selection Monday would have taken 36 of the committee's 37 at-large teams. Chosen by the committee but not by WAB: SMU (resume 48, seed 11). Chosen by WAB but not by the committee: Auburn (resume 42, power 28). Figure 14 shows how early the season simulation knew: the probability of a place in the field that each snapshot gave every team, against whether the team was selected.

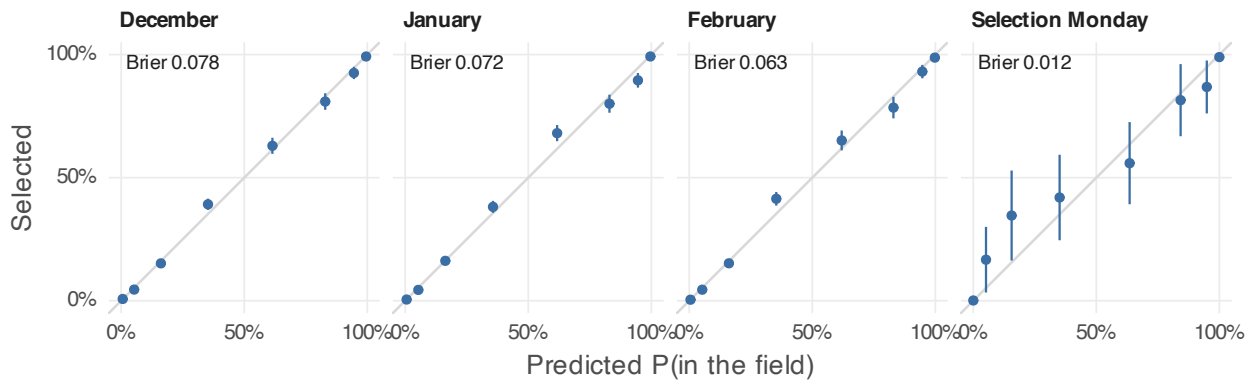


Figure 14: Calibration of the season simulation's probability of a place in the NCAA tournament field, every team at every snapshot of the complete seasons with a tournament, by month of the snapshot; the last panel is Selection Monday, when only the at-large choices remain uncertain. Points are bins of predicted probability with a 95% binomial interval; the label gives the Brier score.

9.2 The Resume Against the Committee

A retrodictive ranking has no future outcome to score it against, so the evidence for the resume has to be external. Table 21 compares the rankings with the committee's choices in every complete season with a tournament: the number of the committee's at-large teams that each ranking's top at-large candidates include (given the actual automatic bids), the rank correlation with the seed lines over the field, and the correlation of our WAB with Torvik's on Selection Monday. It also reports retrodictive accuracy: the share of the season's decided Division I games whose winner the final ranking places higher.

Over 11 tournaments, WAB's at-large candidates include 368 of the committee's 399 at-large teams (92%), strength of record's 366 and the power rating's 325. The disagreements are a handful of teams each year, at the edge of the field, where the committee weighs things a resume measure does not — quadrant records, road records, the eye test, injuries — and where the posterior itself is uncertain: the last few teams in and the first few out have overlapping WAB intervals (Figure 9). The committee selects on results, as the resume does, more than on strength. The seeds, which rank the whole field, follow WAB (median rank correlation 0.96) more closely than the power rating (0.93). Our WAB and Torvik's, built with different ratings and a differently defined bubble, agree almost perfectly (median 0.997). Retrodictive accuracy at the end of the season is 75.2% for the resume, 74.5% for SOR, 75.6% for Colley and 74.0% for power: a ranking built from the record explains the record better than one built to predict.

10 Discussion

The design principle of this paper is that a ranking should say which question it answers, and that different questions can share a model without sharing a formula. The power rating and the resume

Table 21: The resume against the committee, by season, at Selection Monday: at-large teams the committee chose that each ranking's top at-large candidates include, out of the number of at-large places; Spearman correlation of WAB and of the power rating with the seed (sign reversed); and Spearman correlation of our WAB with Torvik's, all teams.

Season	At large	By WAB	By SOR	By power	Seed ρ, WAB	Seed ρ, power	ρ with Torvik's WAB
2014–15	36	33	33	29	0.96	0.91	0.998
2015–16	36	31	31	29	0.95	0.89	0.998
2016–17	36	34	34	29	0.96	0.92	0.997
2017–18	36	31	31	28	0.97	0.95	0.996
2018–19	36	33	33	29	0.97	0.96	0.997
2020–21	37	36	36	29	0.96	0.93	0.997
2021–22	36	33	32	28	0.97	0.91	0.998
2022–23	36	33	33	29	0.95	0.93	0.998
2023–24	36	33	32	29	0.95	0.95	0.998
2024–25	37	35	35	33	0.98	0.92	0.997
2025–26	37	36	36	33	0.98	0.97	0.997

come from one posterior and are never mixed: the resume borrows the model's estimate of how hard each game was, and nothing else. That separation has a cost — two numbers where the public expects one — but each number can then be checked against something observable: the power rating against future games (Section 8), and the resume, less directly, against the committee's selections and the season's own results (Section 9.2). Basketball is kinder to the power model than football: thirty games a season, seventy possessions a game, and a play-by-play rich enough to separate what a team controls from how its shots happened to fall. The model comes within 0.06 points of the closing line on average, ahead of a widely used public rating built from the same games.

Limitations. Several choices deserve more work.

1. *The end of close games.* The model underpredicts the share of games tied after regulation (Section 7.1), and no location-scale family removes the misfit (Section 7.4). The end of a close game is played to the score: fouling, three-point attempts to tie, holding for the last shot. A likelihood for the regulation margin that respects the integer score and puts mass at zero — a rounded or interval-censored version of the present one, or a model for the two scores themselves [45, 46] — would fit the centre of the distribution, at the price of a likelihood that is no longer conjugate to the efficiency margin's. It would matter for the spread more than for the win probability.
2. *One step size for every team and week.* The random walk of Equation 4 moves every team by Gaussian steps of the same size, so a team that loses its best player in February is pulled down only gradually, and a slow decline and a sudden one look alike. Heavier-tailed steps, or steps informed by the injury report and by minutes lost, would let a rating jump; one season identifies even one step size only weakly (Section 3.2), so such extensions need informative structure rather than more parameters alone.
3. *Asymmetric garbage time.* The efficiency margin excludes garbage time, but the scoreboard

margin cannot: a team that leads by 25 and rests its starters concedes late points that enter y_g but not e_g. The model can absorb that only in λ and κ, which are therefore averages over close and lopsided games.

4. *Priors from the whole history.* The pooled priors use every complete season (Section 3.4), and the leave-one-season-out comparisons of Section 8.5 bound what that borrows. The non-pooled priors are weakly informative and set by judgement, and the game-level ones are irrelevant; the preseason loadings β and β_2 are not, and could be pooled like the hierarchy's scales.
5. *A separate tempo model.* Possessions and totals come from a ridge regression outside the Bayesian fit (Section 3.3), so their uncertainty is not propagated into the spreads and the offence–defence split, and the margin and the total are not modelled jointly. Folding tempo into the posterior would make the spread's uncertainty complete.
6. *Games are exchangeable.* Rest, travel, injuries, altitude and the size of the crowd are all absent, and home court is one number per season for every arena (Figure 2); the gap to the closing line, which persists through the season (Section 8.2), is a measure of how much these and roster knowledge matter.
7. *The committee is not modelled.* The season simulation's at-large selection is by projected WAB, a stand-in for a committee that reads team sheets, not a model of it. Its probabilities of a place in the field are well calibrated (Figure 14) because the committee's choices and the resume largely agree, and they will be wrong in the same places where the two disagree.

11 Appendix: Supplementary Posterior Checks

The snapshots of Section 6 are checked by gates and summaries only, because keeping every draw of 277 fits is not needed for publication. This appendix examines two posteriors in full, following the checklist of Kahle et al. [47] and Miyakawa and Kahle [48] for applied Bayesian computation — sampler behaviour; convergence; posterior geometry; the justification of informative priors and the priors they induce on derived quantities; sensitivity of the posterior to changes in the prior; predictive adequacy; and influence — together with the visual workflow of Gabry et al. [35]. The two posteriors bracket the range the pipeline fits:

- **2025–26, first Monday**: 311 Division I games, about 1.7 per team. The data barely constrain a team's strength, the hierarchy and the preseason prediction do most of the work, and this is where the hierarchical geometry is hardest for the sampler.
- **2025–26, final**: all 5,752 Division I games of the season, the tournaments included.

Both are refitted with the production settings of Section 6.1 (4 chains, 1,000 post-warm-up draws each, 4,000 in all) by the `supplement` target.

11.1 Sampler Behaviour Across All Snapshots

Before the two detailed fits, Figure 15 shows the gate statistics of every snapshot.

Table 22: Sampler diagnostics by chain. E-BFMI below 0.2 would indicate that the momentum resampling cannot move the chain across energy levels fast enough; the maximum tree depth is 10.

Fit	Chain	Step size	Leapfrog steps	Max depth	At max depth	Divergent	E-BFM
2025–26, first Monday	1	0.198	31.0	5	0	0	0.9
	2	0.201	30.1	5	0	0	0.9
	3	0.208	21.1	5	0	0	0.9
	4	0.190	31.0	5	0	0	1.0
2025–26, final	1	0.115	31.0	5	0	0	0.9
	2	0.114	31.0	5	0	0	1.0
	3	0.113	31.0	5	0	0	0.9
	4	0.108	31.0	5	0	0	1.0

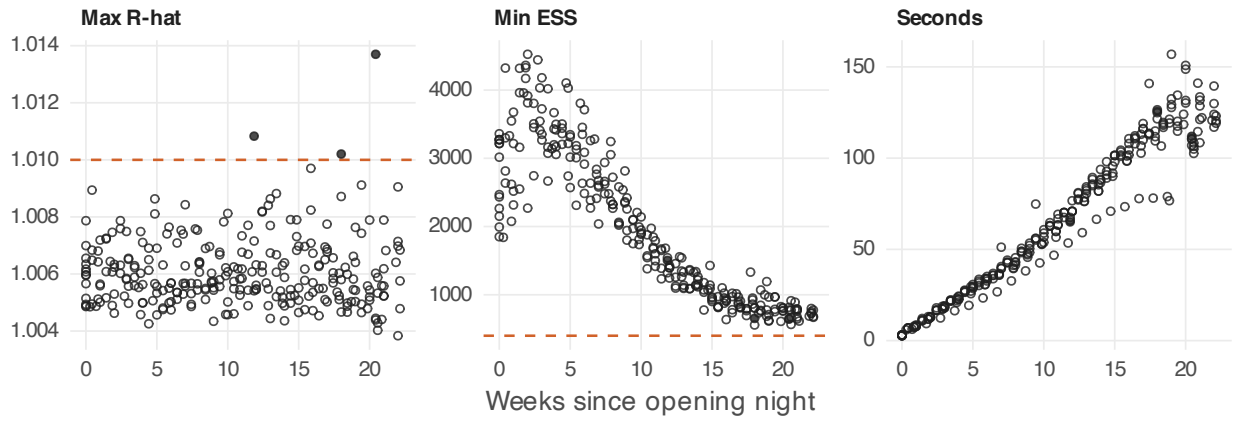


Figure 15: Gate statistics of all snapshots by week of the season, every season overlaid. Left: the largest rank-normalised $\widehat{R}$ over all monitored parameters, with the gate at 1.01. Centre: the smallest bulk effective sample size, with the gate at 400. Right: wall-clock time of the fit, four chains in parallel. Filled points failed a gate.

11.2 Sampler Diagnostics

Table 22 reports the Hamiltonian Monte Carlo diagnostics of each chain [26]: the adapted step size, the mean number of leapfrog steps per iteration, how often the trajectory hit the maximum tree depth, divergent transitions, and the energy Bayesian fraction of missing information, E-BFMI [49].

Neither fit has a single divergent transition, no iteration reaches the maximum tree depth, and the smallest E-BFMI is 0.93, far above 0.2. Figure 16 compares the marginal energy distribution with the distribution of energy changes between iterations; overlapping histograms are what a healthy fit looks like.

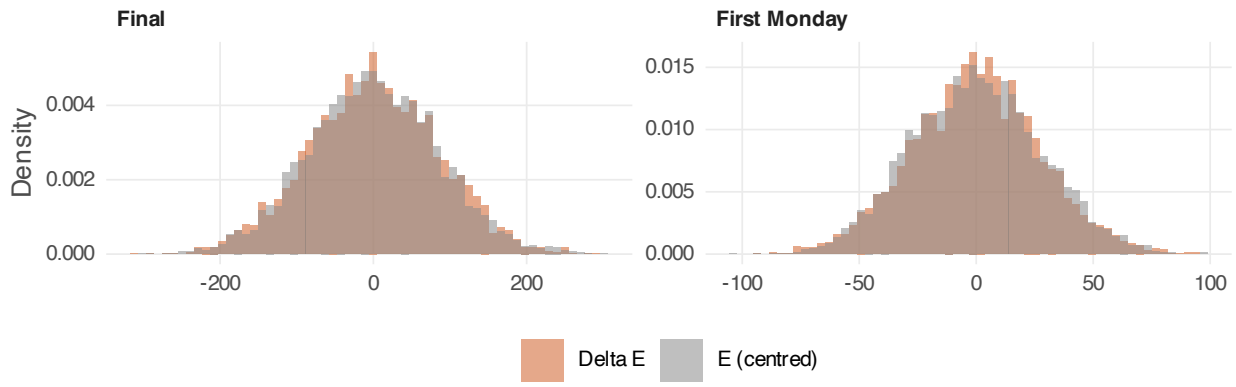


Figure 16: Energy diagnostic. Grey: marginal energy E, centred; orange: the change in energy between successive iterations, ΔE. Pooled over chains.

11.3 Convergence

Table 23 gives posterior summaries and convergence diagnostics of the structural parameters: the rank-normalised split-$\widehat{R}$ [32, 50], bulk and tail effective sample sizes [51], and the Monte Carlo standard error of the posterior mean.

Every structural $\widehat{R}$ is at most 1.004, and the smallest bulk effective sample size is 670, for ϕ in the final fit. Monte Carlo standard errors are at most 3.9% of the posterior standard deviations. The team strengths are too many to tabulate; Figure 17 shows the distribution of their $\widehat{R}$ and of their bulk effective sample size per draw.

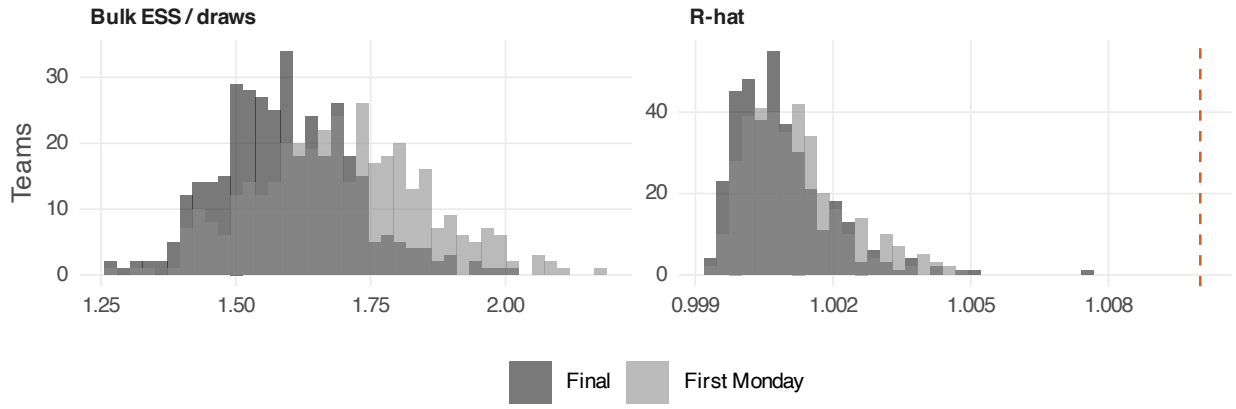


Figure 17: Convergence of all team strengths r_i. Left: rank-normalised $\widehat{R}$, with the gate at 1.01. Right: bulk effective sample size per draw; values above one are possible for anti-correlated chains, which NUTS often produces for well-identified parameters.

Trace plots (Figure 18) show the chains iteration by iteration; rank plots (Figure 19) rank the draws of all chains together, and if the chains sample the same distribution each chain's ranks are uniform

Table 23: Posterior summaries and convergence diagnostics for the structural parameters. *MCSE* is the Monte Carlo standard error of the posterior mean.

Fit	Parameter	Mean	SD	5%	95%	$\widehat{R}$	ESS bulk	ESS tail	MCSE
2025–26, first Monday	η	4.65	0.40	4.00	5.31	1.002	7,856	2,656	0.005
	β	1.04	0.08	0.91	1.16	1.001	3,676	2,215	0.001
	β_2	0.07	0.08	-0.06	0.20	1.000	4,525	2,891	0.001
	λ	0.95	0.03	0.90	1.00	1.000	6,283	2,951	0.000
	κ	0.58	0.05	0.49	0.66	1.000	4,884	3,302	0.001
	σ_e	14.77	0.71	13.64	15.98	1.001	3,902	3,390	0.011
	σ_m	13.29	0.57	12.39	14.26	1.001	6,234	3,357	0.007
	ω	1.52	0.20	1.19	1.85	1.002	5,955	2,699	0.003
	τ	4.15	0.34	3.60	4.70	1.002	5,284	3,011	0.005
	ϕ	0.56	0.10	0.40	0.72	1.004	5,570	2,177	0.001
2025–26, final	η	4.09	0.19	3.78	4.39	1.000	4,209	3,242	0.003
	β	0.98	0.05	0.90	1.06	1.003	931	1,341	0.002
	β_2	0.13	0.04	0.06	0.21	1.002	877	1,469	0.001
	λ	0.93	0.01	0.91	0.95	1.001	5,014	2,823	0.000
	κ	0.59	0.01	0.57	0.61	1.002	4,596	2,427	0.000
	σ_e	13.94	0.14	13.71	14.17	1.000	4,625	2,941	0.002
	σ_m	12.98	0.12	12.78	13.19	1.002	5,182	2,913	0.002
	ω	1.41	0.19	1.09	1.72	1.000	2,381	2,311	0.004
	τ	4.39	0.24	3.99	4.78	1.000	1,825	2,489	0.006
	ϕ	0.56	0.09	0.42	0.70	1.001	670	1,352	0.003

[32]. Autocorrelation (Figure 20) should damp quickly with the lag, and the Gelman–Rubin shrink factor (Figure 21) should fall to 1 as the chains lengthen and stay there [50].

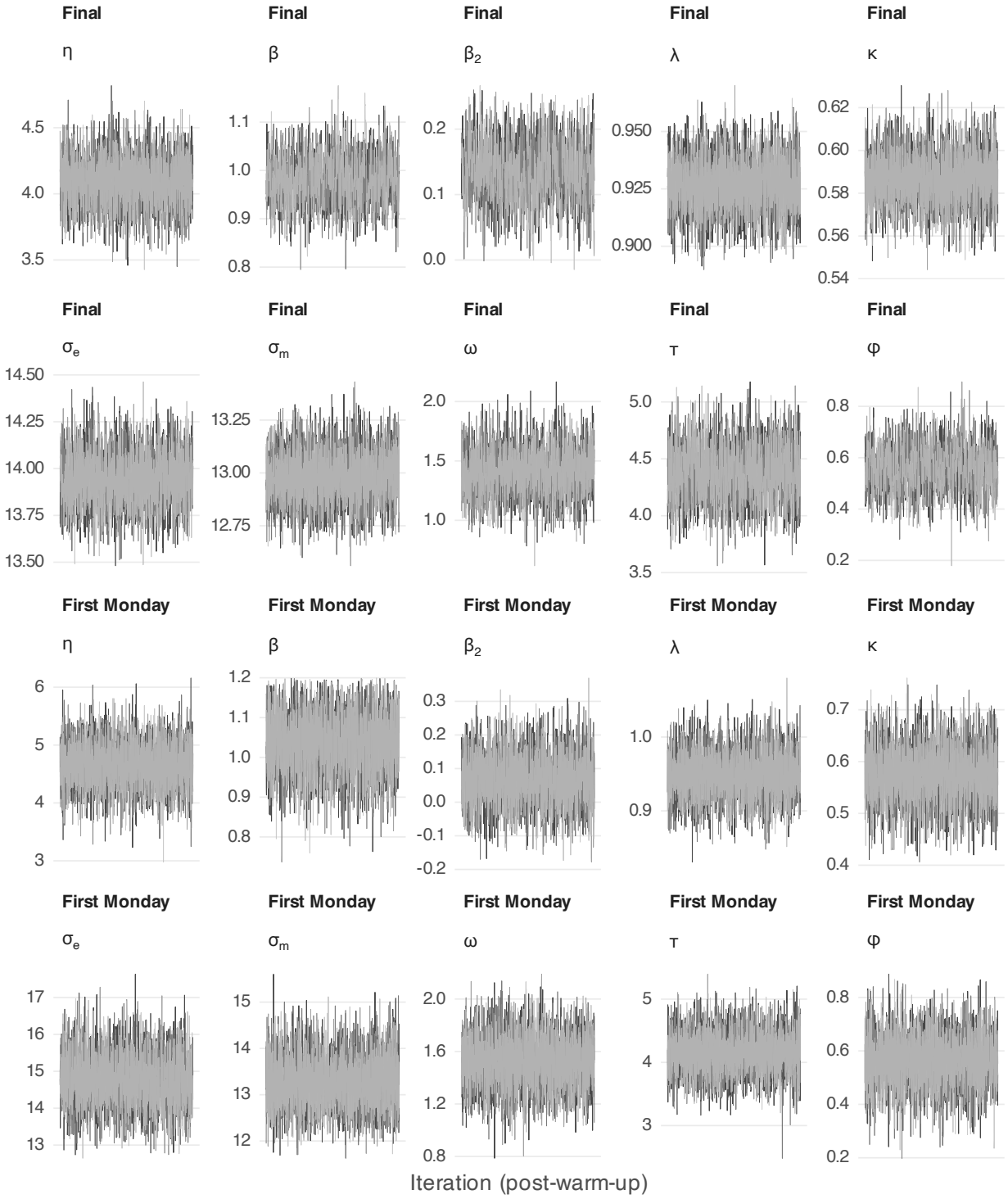


Figure 18: Trace plots of the structural parameters, 2025–26, final (top block) and 2025–26, first Monday. One shade of grey per chain.

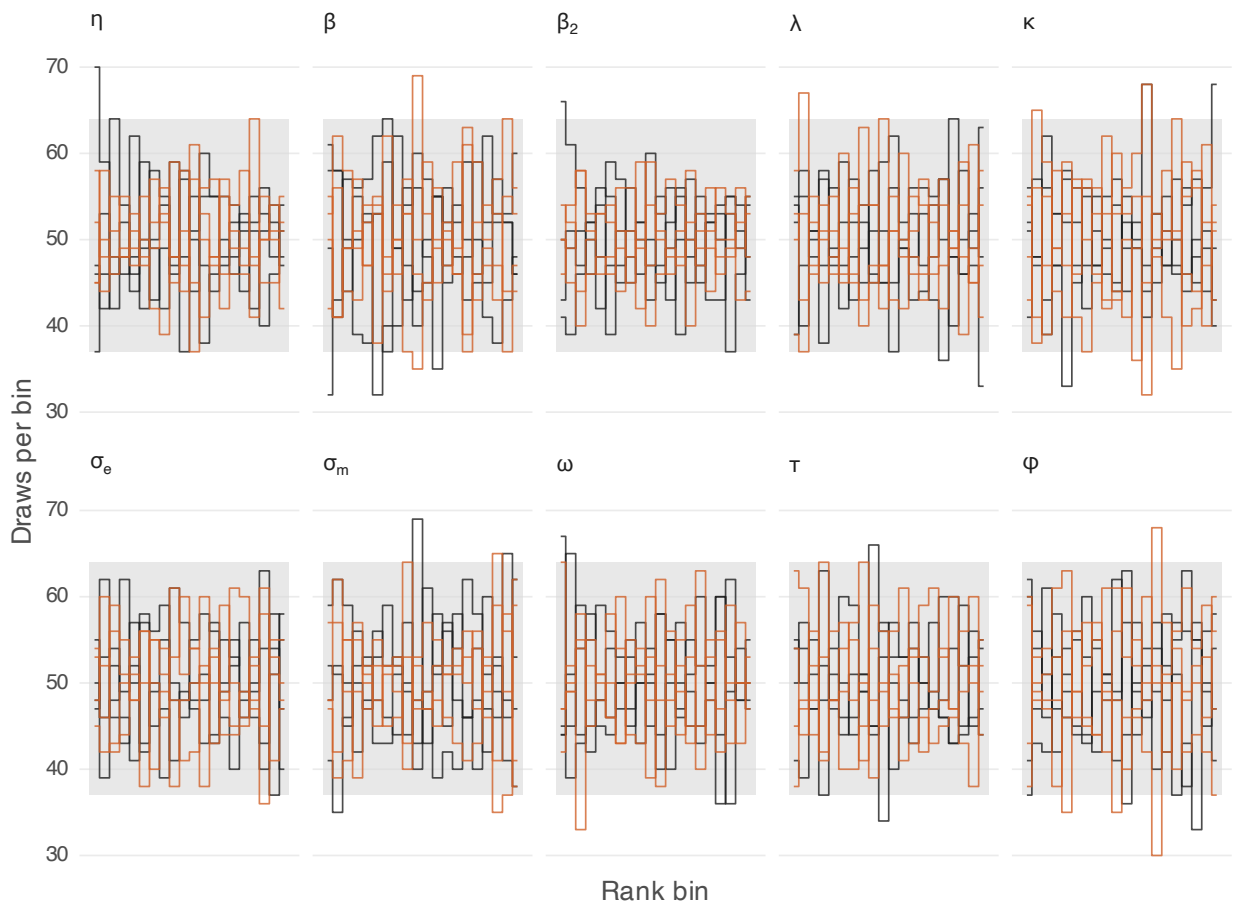


Figure 19: Rank plots of the structural parameters for both fits. For each parameter the draws of all chains are ranked jointly and each chain's ranks are binned into 20 bins; the band is the central 95% range of a bin count under uniformity. Lines are chains; 2025–26, first Monday in orange, 2025–26, final in grey.

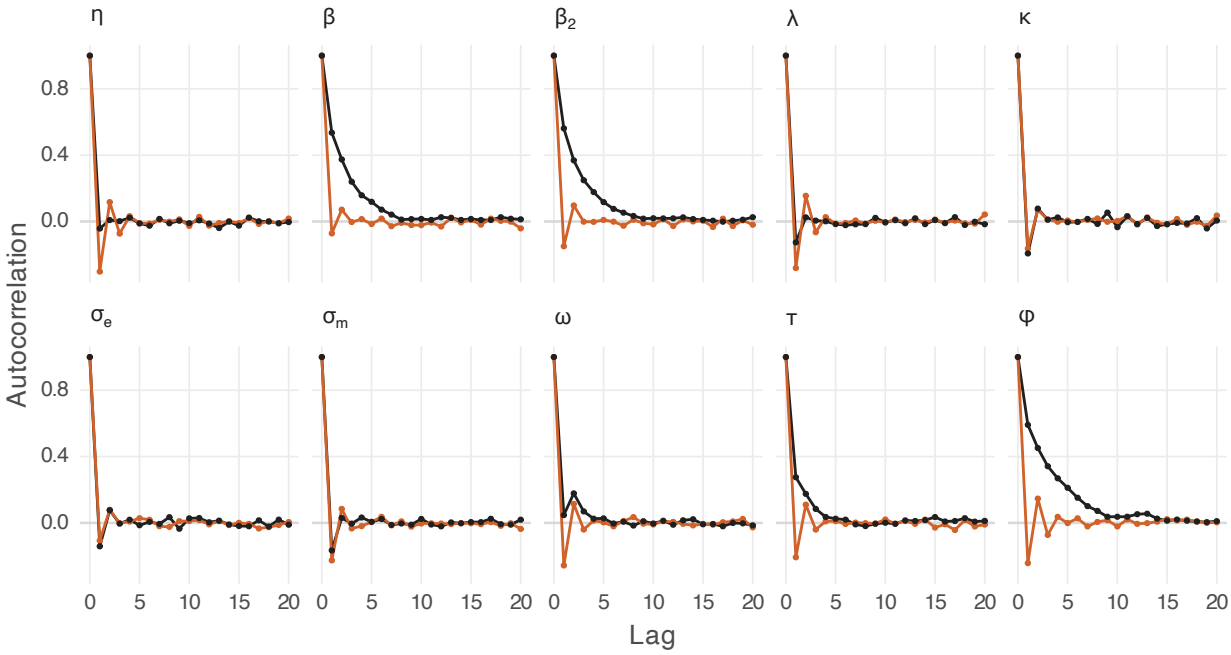


Figure 20: Autocorrelation of the structural parameters up to lag 20, averaged over chains. Orange: first Monday; grey: final.

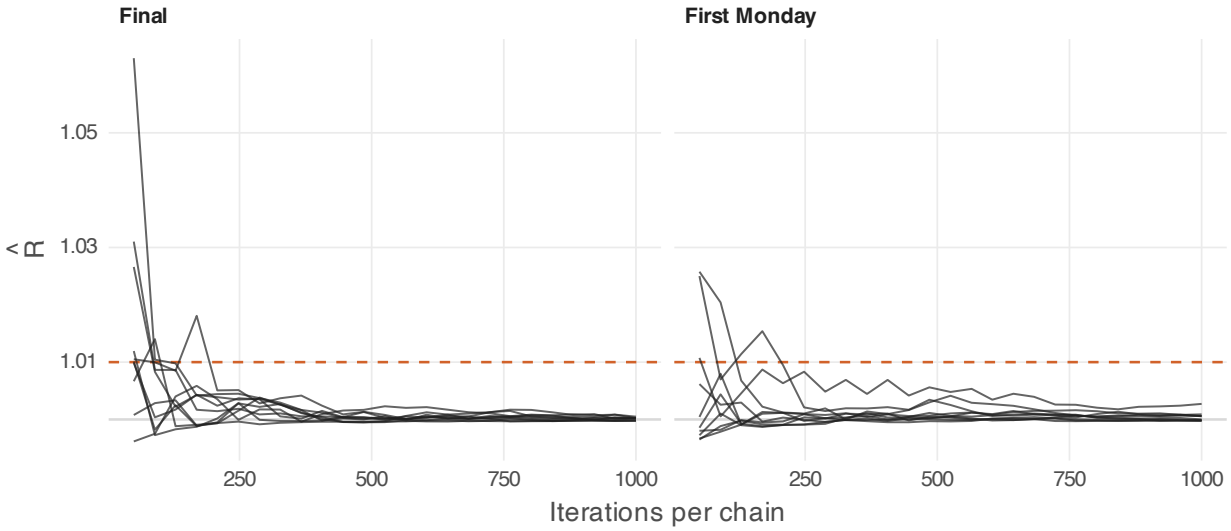


Figure 21: Gelman–Rubin shrink factor (rank-normalised split-$\widehat{R}$) of each structural parameter computed on the first n post-warm-up iterations of every chain, as n grows. The dashed line is the gate at 1.01.

The autocorrelation of 17 of the 20 parameter–fit pairs drops below 0.1 within three lags; the slowest is ϕ in the final fit, which needs 8 lags. The shrink factors of every parameter are below the gate after 208 iterations per chain in both fits and stay there.

11.4 Posterior Geometry

Hierarchical models have funnels: when a scale parameter such as ω or τ is small, the effects it governs must be close to their mean, and the posterior narrows into a region the sampler can struggle to enter. The non-centred parameterisation of Equation 3 removes the funnel from the sampled coordinates when the data are weak [25]. Figure 22 plots each scale against a standardised effect it governs, in the sampled coordinates; divergent transitions, if any, would be drawn in orange.

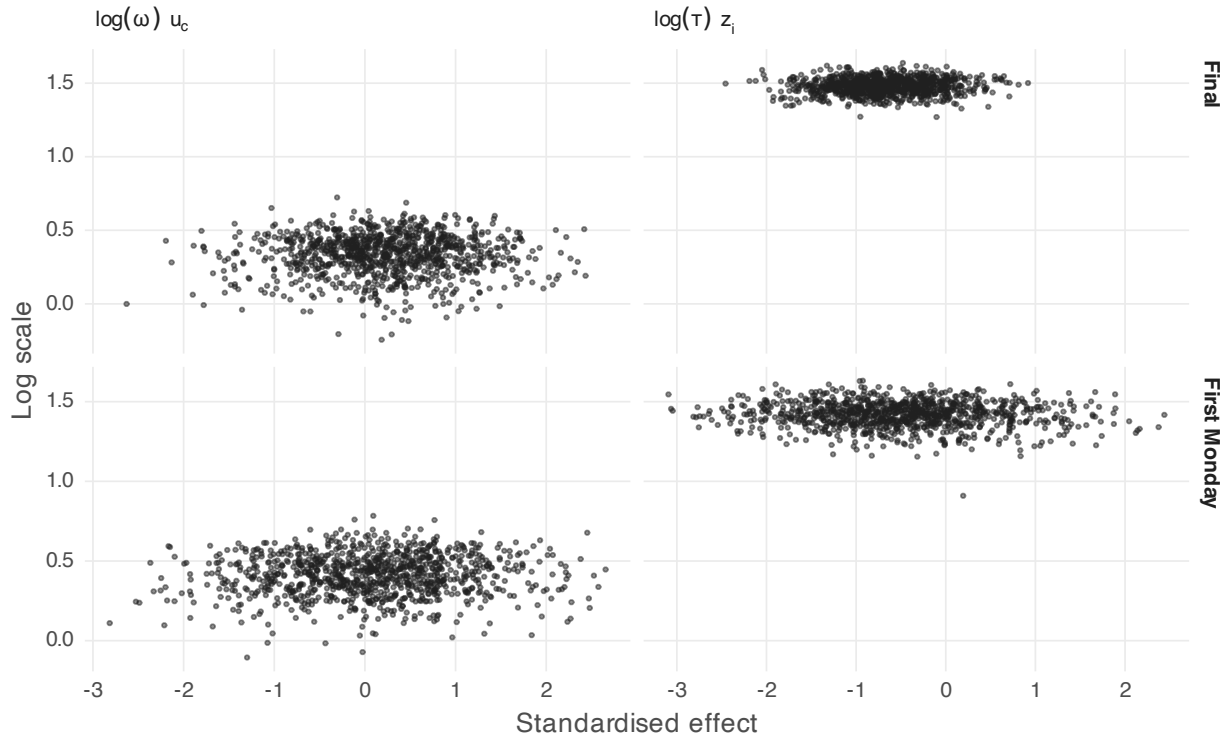


Figure 22: Posterior geometry in the sampled coordinates, 1,000 draws per panel: the conference effect u_c of the first conference against $\log \omega$, and the team effect z_i of the team with the fewest games against $\log \tau$. Divergent transitions would be orange.

Neither fit shows a funnel in the sampled coordinates: the standardised effects have the same spread at every value of the scale. In the final fit the team effect is concentrated, because a full season pins the team down; on the first Monday it spreads more widely. The random walk has a funnel of the same kind, between ϕ and the standardised steps of Equation 4, which is not drawn; the evidence that the sampler enters its neck is the absence of divergent transitions together with the informative prior that keeps ϕ away from zero.

11.5 Prior and Posterior

Figure 23 overlays the prior density of each structural parameter on its posterior in both fits, and Figure 24 condenses the comparison into two numbers per parameter [52]: the posterior

contraction $1 - \mathrm{Var}_{\mathrm{post}} / \mathrm{Var}_{\mathrm{prior}}$, near one when the data dominate the prior, and the posterior z-score $(\mathbb{E}_{\mathrm{post}} - \mathbb{E}_{\mathrm{prior}})/\mathrm{sd}_{\mathrm{post}}$.

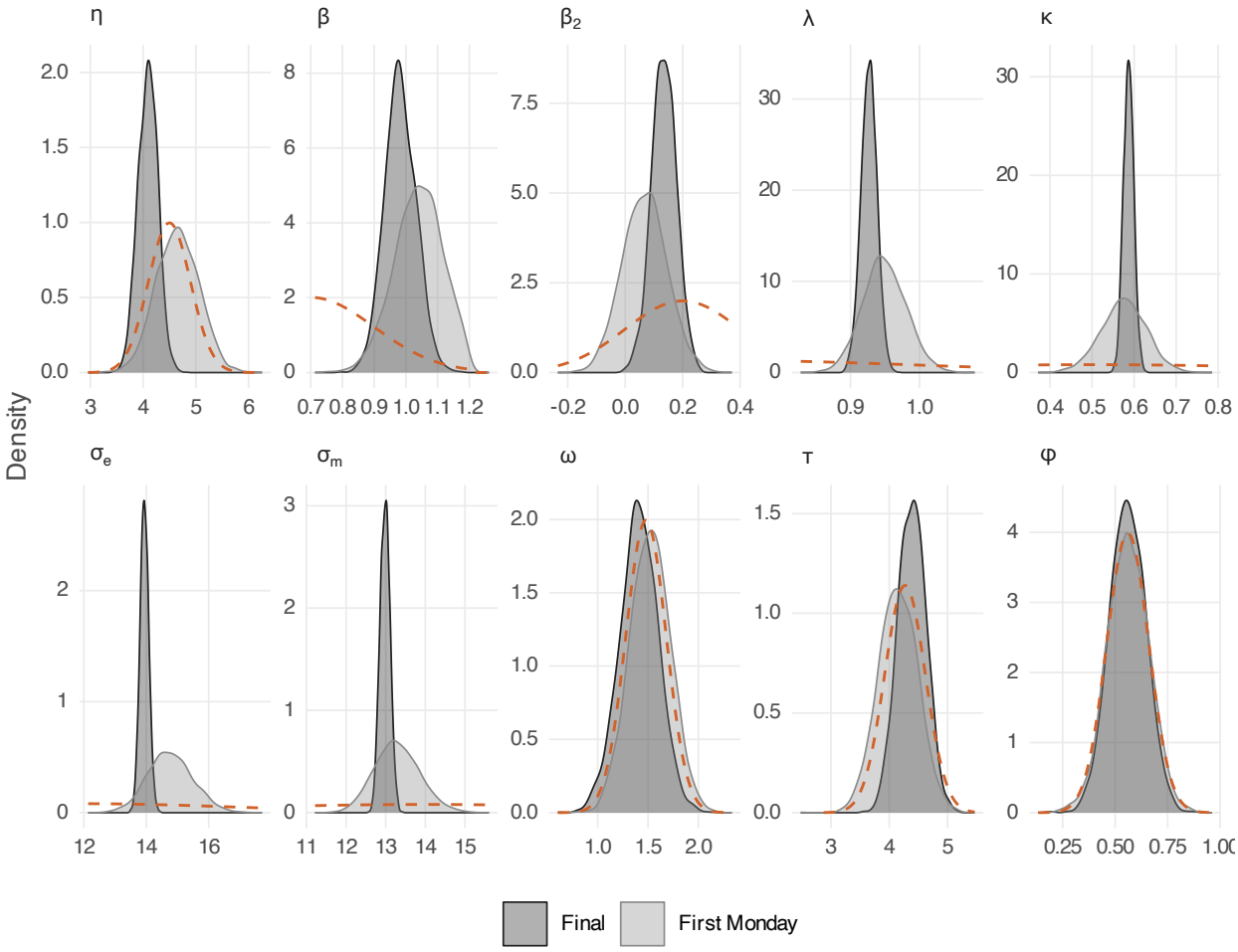


Figure 23: Prior (dashed) and posterior densities of the structural parameters. Priors are normal, truncated to the parameter's support; η has the prior of the season.

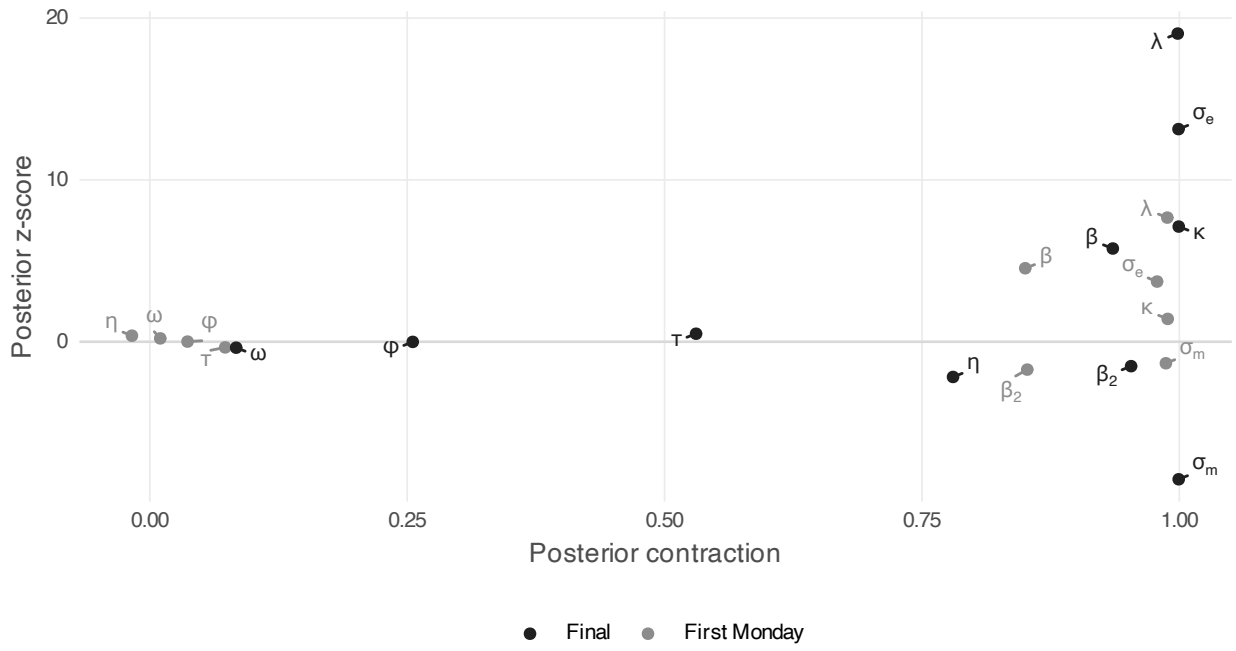


Figure 24: Posterior contraction against posterior z-score for the structural parameters, for both fits. Prior moments are those of the truncated priors.

On the first Monday, 4 of 10 structural parameters contract by less than half — η, τ, ω and ϕ — so their priors carry the fit, as intended in a week with two games per team. By the end of the season every parameter except ω and ϕ contracts by more than half: the pooled priors are about as informative as a season. Posteriors more than two of their own standard deviations from the prior mean are β (first Monday, $z = 4.5$); λ (first Monday, $z = 7.7$); σ_e (first Monday, $z = 3.7$); η (final, $z = -2.2$); σ_m (final, $z = -8.5$); β (final, $z = 5.8$); λ (final, $z = 19.0$); σ_e (final, $z = 13.1$); κ (final, $z = 7.1$): for these the prior is centred away from where the data put them, but is too wide to pull them back.

11.6 Justification of the Informative Priors

Informative priors need a justification that does not come from the data they are applied to. Ours come from the history of the sport. The `prior_refits` target refits the final posterior of every complete season, each with about 5,500 games, under deliberately vague priors — $\eta \sim \mathcal{N}(0, 10^2)$, $\beta \sim \mathcal{N}(0.5, 1)$ on $(0, 1.2)$, $\beta_2 \sim \mathcal{N}(0, 1)$, half-normal priors with scale 10 on ω and τ, scale 2 on λ and κ (normal for κ) and scale 50 on σ_e and σ_m, and $\phi \sim \mathcal{N}^{+}(0.5, 0.5^2)$, wide but centred away from zero, because under a prior with its mode at zero the sampler stalls in the random walk's funnel (Section 3.2) — so that a season's estimate owes nothing to the published priors; the chains are twice as long as the published ones. Figure 25 plots the resulting per-season posteriors against the priors in use, and Table 24 gives the numbers.

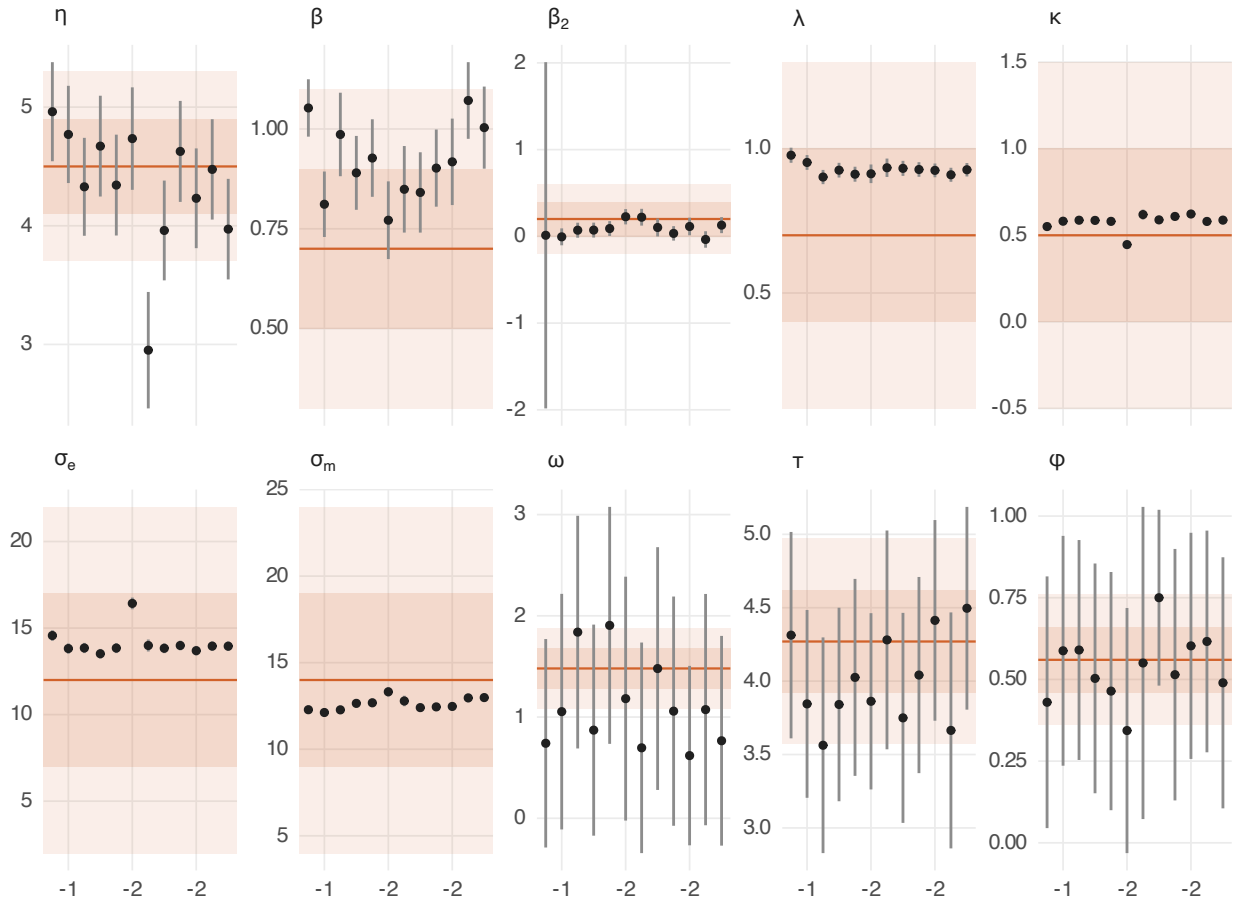


Figure 25: Per-season full-season posteriors under vague priors (points: mean; bars: ± 2 posterior sd) against the prior in use (line: prior mean; bands: ± 1 and ± 2 prior sd).

Two points qualify this justification. First, pooling every complete season is empirical Bayes at the level of the sport, and saying that a season's estimate lies inside a prior partly built from it would be circular. The last column of Table 24 avoids that: for each season it pools the *other* seasons only and asks whether the season's own estimate lies within two standard deviations of that leave-one-season-out prior. For the four pooled parameters it does in every season, so a new season should be expected to fall inside its prior. Second, the prior standard deviation removes estimation noise from the between-season spread (Equation 5): for parameters whose between-season spread is no larger than estimation noise alone would produce, the prior is as narrow as the uncertainty of a mean over a dozen seasons. That is the right width if the parameter is the same every season, and the leave-one-season-out column checks it.

11.7 Induced Priors

A prior on parameters induces a prior on everything computed from them, and a set of individually reasonable priors can combine into an unreasonable one. The quantities that matter here are not ω or τ but the spread of team strengths, the probability that the home team wins an even game,

Table 24: Empirical basis of the priors: the pooled mean of the per-season full-season posterior means under vague priors, their between-season standard deviation (DerSimonian–Laird), the typical within-season posterior sd (root mean square), the predictive sd of Equation 5, and the prior in use. *LOSO* is the share of seasons whose own estimate lies within two standard deviations of the prior pooled from the other seasons, counting the estimate's own posterior sd. Only the first four priors are pooled; the rest are weakly informative.

Parameter	Pooled mean	Between-season sd	Posterior sd	Predictive sd	Prior	LOSO
ω	1.05	0.00	0.55	0.16	$\mathcal{N}(1.48, 0.20^2)$	100%
τ	4.01	0.00	0.35	0.10	$\mathcal{N}(4.27, 0.35^2)$	100%
ϕ	0.56	0.00	0.18	0.05	$\mathcal{N}(0.56, 0.10^2)$	100%
η	4.46	0.25	0.21	0.27	$\mathcal{N}(4.50, 0.40^2)$	100%
β	0.92	0.09	0.05	0.09	$\mathcal{N}(0.70, 0.20^2)$	100%
β_2	0.09	0.06	0.29	0.07	$\mathcal{N}(0.20, 0.20^2)$	92%
λ	0.93	0.02	0.01	0.02	$\mathcal{N}(0.70, 0.30^2)$	92%
σ_e	14.12	0.67	0.15	0.70	$\mathcal{N}(12.00, 5.00^2)$	92%
κ	0.58	0.05	0.01	0.05	$\mathcal{N}(0.50, 0.50^2)$	92%
σ_m	12.61	0.32	0.13	0.34	$\mathcal{N}(14.00, 5.00^2)$	92%
m_{nd}	-47.17	5.71	1.17	5.95	$\mathcal{N}(-25.00, 10.00^2)$	100%
σ_{nd}	17.34	1.62	1.01	1.71	$\mathcal{N}(10.00, 5.00^2)$	92%

and the distribution of margins the model expects before seeing any game. Table 25 simulates them from the priors alone, on the 2025–26 schedule with its preseason predictions, each game played at the random-walk states of its period and its margin drawn from the model.

The induced priors are on the scale of the sport: before any game, the model expects the best team to be about 20 points per 100 better than the 45th (90% prior interval 12–29), an even game to be won at home 61% of the time, and margins with a standard deviation of about 15 points. All three observed statistics lie inside the central 95% of their prior predictive distribution. The prior on $\bar{\sigma}$, which sets every win probability and therefore every π_g in the resume, is induced by three separate weakly informative priors and is wide (90% interval 8.2–26.3); the data narrow it to a few tenths in any snapshot after the first week.

11.8 Posterior Predictive Checks

Section 7.1 compared summary statistics with their replicates. Here the whole distributions are compared (Figure 26), then the predictive distribution of each game without that game, by the leave-one-out probability integral transform.

Table 25: Induced priors, by simulation from the priors alone (4,000 draws) on the 2025–26 schedule and preseason predictions. The last column gives the value observed in 2025–26 where one exists.

Quantity	5%	Median	95%	Observed
Margin scale $\bar{\sigma}$ (per 100, 70 possessions)	8.19	16.34	26.32	
Share of margin variance from efficiency, R^2	0.00	0.14	0.74	
Weekly random-walk step ϕ	0.40	0.56	0.73	
SD of team strengths (per 100)	7.57	11.93	16.73	
Best minus 45th-best team (per 100)	12.35	19.54	28.61	
P(home win), evenly matched teams	0.57	0.61	0.71	
P(best team beats an average one), neutral	0.87	0.98	1.00	
SD of regulation margin (points)	10.03	14.59	20.55	15.01
Share of games within 3 points	0.13	0.19	0.27	0.19
Home regulation win rate, home floor	0.58	0.62	0.67	0.61

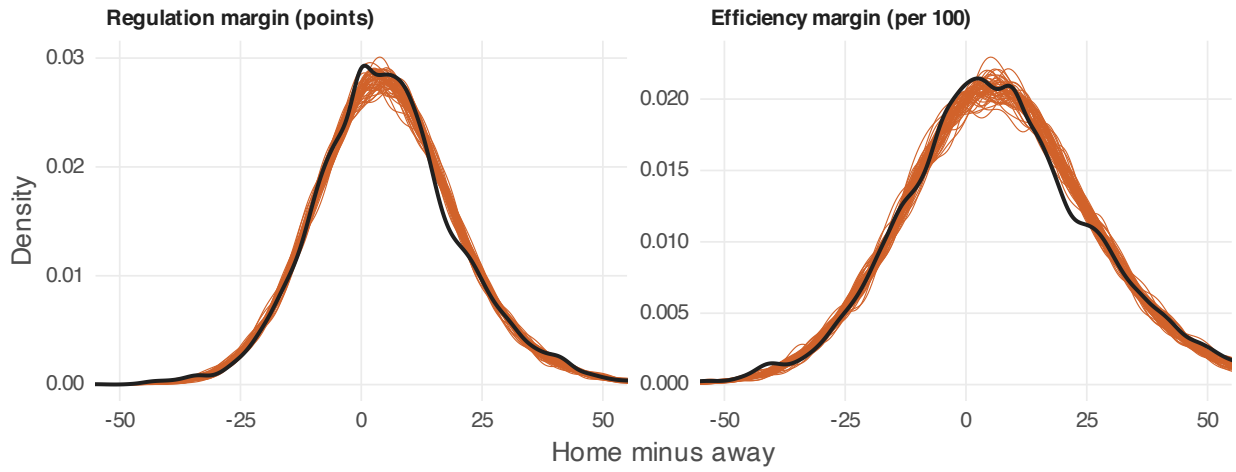


Figure 26: Observed distribution (black) against 50 replicated data sets (orange) for the 2025–26, final fit. Left: regulation margin in points, replicated from the marginal law of Equation 2 and rounded. Right: efficiency margin per 100 live possessions, for games that have one.

The density overlay shows the misfit of Section 7.1 directly: the observed margin has a spike at zero, the overtime games, that no replicate has. The leave-one-out probability integral transform (LOO-PIT) checks calibration game by game: for each game, $\Pr(y_g^{\text{rep}} \leq y_g \mid \text{all other games})$, estimated with the Pareto-smoothed importance weights of the LOO computation. If the model is calibrated these values are uniform; a U shape indicates predictive distributions that are too narrow, a hump that they are too wide [35, 53]. Figure 27 shows them for both signals and both fits.

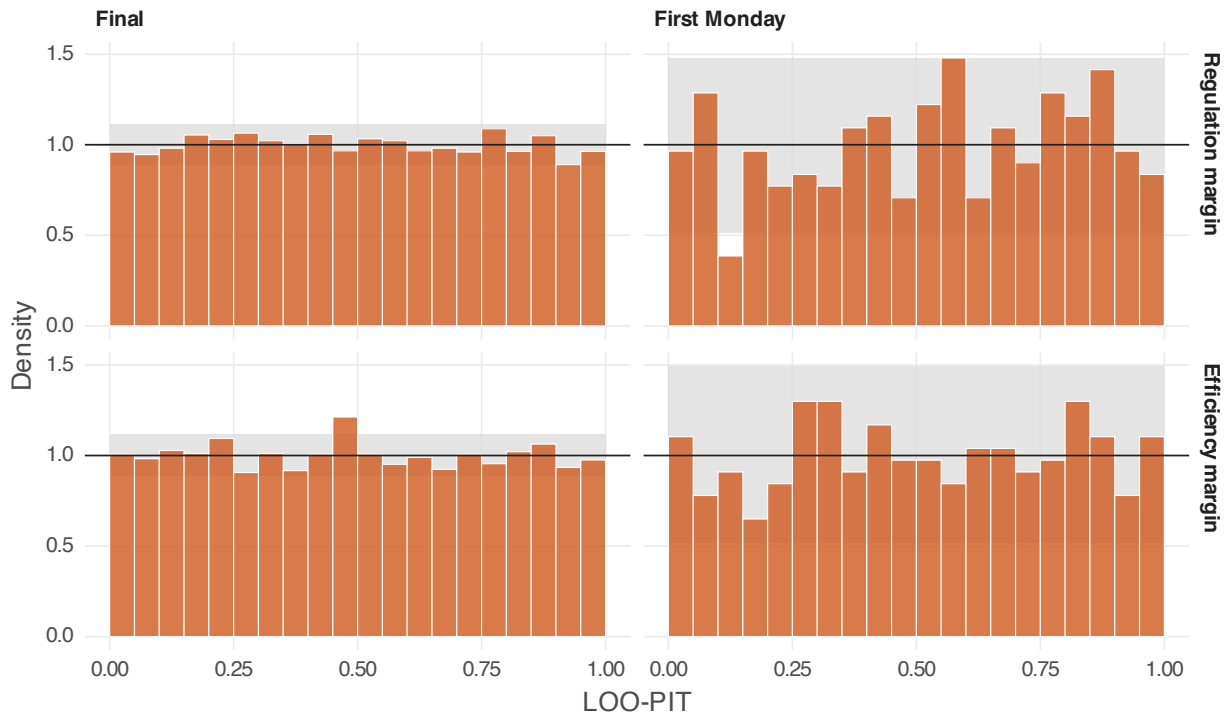


Figure 27: LOO-PIT histograms, 20 bins; the band is the central 95% range of a bin count under uniformity. The margin's PIT uses its marginal law; the spike of regulation ties sits at the PIT value of zero margin, which depends on the game.

Of the 80 histogram bins across the four panels, 2 fall outside their 95% band, against about 4 expected by chance. Figure 28 plots the observed margin minus its posterior mean against that mean; a systematic trend would mean that the model's spreads are biased for lopsided or for close matchups — for example, if strong favourites won by less than predicted.

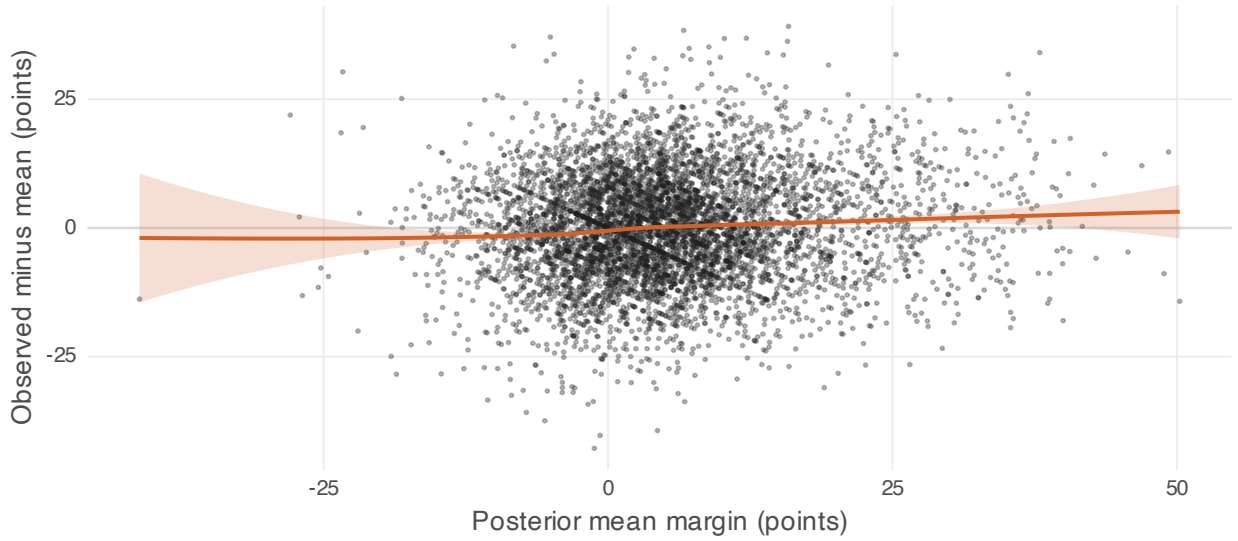


Figure 28: Residual regulation margin (points) against the posterior mean margin, 2025–26, final. The line is a LOESS smooth with a 95% band.

11.9 Influence and PSIS Diagnostics

The Pareto $\hat{k}$ of each game estimates the tail shape of its importance ratios in PSIS-LOO; values above 0.7 mean the leave-one-out approximation is unreliable for that game, and in practice that the game is highly influential [38]. Figure 29 plots every game's $\hat{k}$ in both fits.

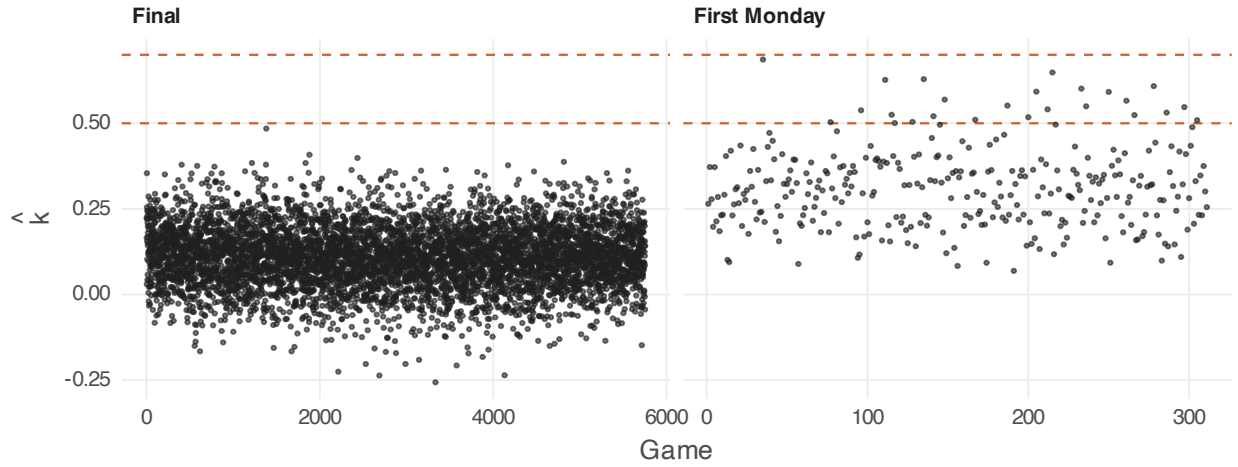


Figure 29: Pareto $\hat{k}$ for each game, in schedule order. Dashed lines at 0.5 and 0.7.

0 of 5,752 games exceed 0.7 in the final fit and 0 of 311 on the first Monday (25 above 0.5). Early in the season a single game is a large share of the evidence about two teams, so removing it moves their posteriors more, but with two games per team and a strong preseason prediction no game is the only evidence about anyone.

11.10 Power-scaling Sensitivity

Table 12 summarised power-scaling sensitivity by a single distance per parameter. Figure 30 shows what lies behind it for the parameters flagged most often: the posterior density as the prior (top row) or the likelihood (bottom row) is raised to powers between 0.8 and 1.25 [36]. A density that shifts with the prior power is prior-sensitive; one that shifts with the likelihood power is informed by the data.

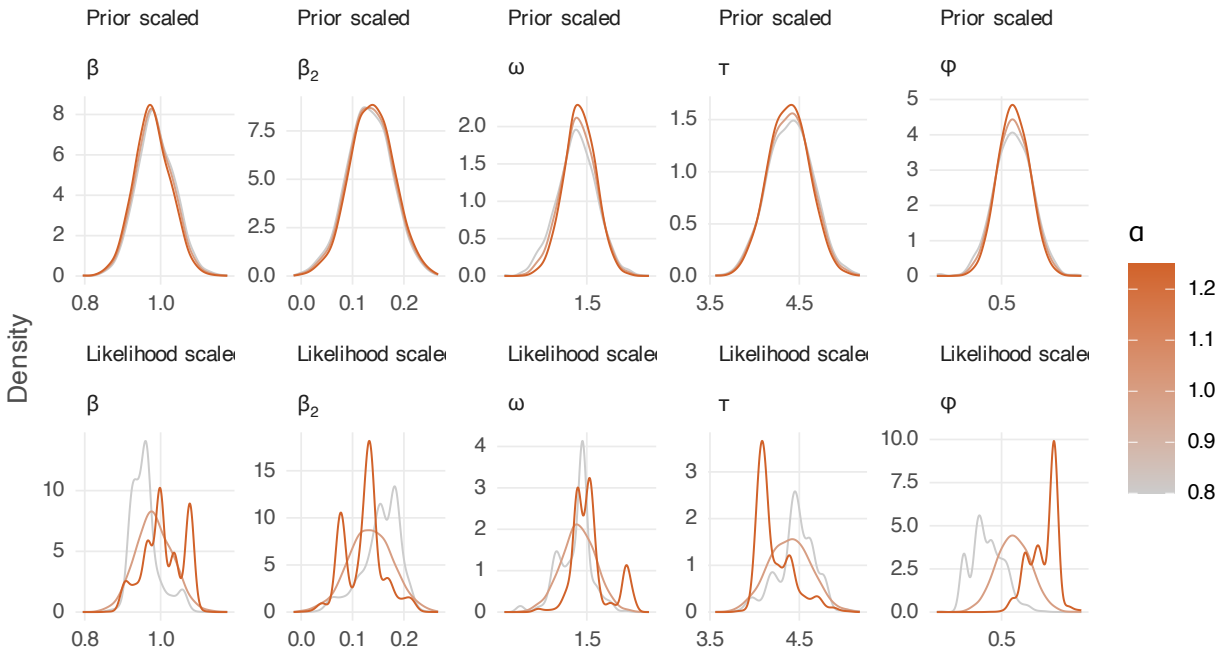


Figure 30: Power-scaled posterior densities, 2025–26, final. Darker lines are larger powers α; the base posterior is $\alpha = 1$.

11.11 Sensitivity to the Prior: Perturbation Refits

Power-scaling is a local check. The requirement that matters in practice is global — that a reasonable analyst with a different prior would publish essentially the same ratings. We therefore refit both posteriors under four joint alternatives for every structural prior: all prior standard deviations doubled, all multiplied by five, and all prior means moved up, or down, by one prior standard deviation. Joint changes can cancel, so each prior group is also widened fivefold on its own. Table 26 compares what is published: the power ratings and the resume.

At the end of the season the prior is immaterial to what is published: under every joint alternative the power ratings correlate with the baseline at $\rho \geq 0.9999$, no team's rating moves by more than 0.40 points per 100, and no team's WAB by more than 0.12. On the first Monday the ratings move more, as they should when the priors carry the fit: by up to 2.0 points per 100 (mean 0.56), with rank correlation at least 0.998. One prior group at a time, the largest effect in both fits comes from widening the prior on ω, τ (0.29 points at most at the end of the season, 1.72 on the first Monday). 4 of the 22 refits fail a convergence gate, all of them refits of the final posterior, and

Table 26: Effect of the prior on the published products, relative to the baseline prior, all Division I teams. Power: Spearman correlation of the ratings, and the mean and largest absolute change in a team's rating (per 100 possessions). Resume: Spearman correlation of WAB and the largest absolute change in a team's WAB. *Gates*: whether the refit passed them.

Fit	Prior	ρ power	Mean $\lvert\Delta\rvert$	Max $\lvert\Delta\rvert$	ρ WAB	Max $\lvert\Delta\text{WAB}\rvert$	Gates
first Monday	SD $\times$ 2	0.9995	0.24	0.90	0.9998	0.04	pass
	SD $\times$ 5	0.9975	0.56	1.97	0.9991	0.08	pass
	Means $+1$ SD	0.9997	0.18	0.63	0.9999	0.04	pass
	Means -1 SD	0.9998	0.17	0.67	0.9999	0.04	pass
	ω, τ sd $\times$ 5	0.9980	0.49	1.72	0.9996	0.04	pass
	ϕ sd $\times$ 5	0.9999	0.07	0.25	1.0000	0.01	pass
	η sd $\times$ 5	0.9997	0.43	1.36	0.9989	0.10	pass
	β, β_2 sd $\times$ 5	0.9994	0.27	1.26	0.9998	0.04	pass
	σ_m sd $\times$ 5	0.9999	0.07	0.22	1.0000	0.01	pass
	$\lambda, \sigma_e, \kappa$ sd $\times$ 5	0.9999	0.07	0.26	1.0000	0.01	pass
	$m_{\text{nd}}, \sigma_{\text{nd}}$ sd $\times$ 5	0.9999	0.11	0.39	1.0000	0.01	pass
final	SD $\times$ 2	1.0000	0.06	0.23	1.0000	0.06	fail
	SD $\times$ 5	0.9999	0.11	0.35	1.0000	0.12	fail
	Means $+1$ SD	0.9999	0.10	0.40	1.0000	0.06	pass
	Means -1 SD	0.9999	0.10	0.33	1.0000	0.06	pass
	ω, τ sd $\times$ 5	0.9999	0.08	0.29	1.0000	0.05	pass
	ϕ sd $\times$ 5	1.0000	0.06	0.25	1.0000	0.03	fail
	η sd $\times$ 5	1.0000	0.05	0.19	1.0000	0.06	pass
	β, β_2 sd $\times$ 5	1.0000	0.04	0.22	1.0000	0.03	pass
	σ_m sd $\times$ 5	1.0000	0.04	0.15	1.0000	0.02	pass
	$\lambda, \sigma_e, \kappa$ sd $\times$ 5	1.0000	0.04	0.14	1.0000	0.02	pass
	$m_{\text{nd}}, \sigma_{\text{nd}}$ sd $\times$ 5	1.0000	0.04	0.16	1.0000	0.02	fail

their changes are included in the ranges above; with chains this long a failed gate means slower mixing, not a different posterior. The pooled priors therefore stabilise the sampler in data-poor weeks (Section 3.4) without determining what is published once the season is under way.

11.12 Simulation-based Calibration

The checks so far ask whether the model fits the data. Simulation-based calibration asks whether the *computation* recovers the model: if parameters are drawn from the prior, data simulated from them, and the posterior computed, then the rank of each true value among the posterior draws is uniform when the sampler and the code are correct [54, 55]. We simulate 200 seasons on the real structure of the 2025–26 snapshot of December 1 — its 365 teams, conferences, preseason predictions, 1,245 games with their weights and missing efficiency margins, and the non-Division I games — each with its own hyperparameters, team effects, random walk, efficiency margins and margins drawn as the model specifies, refit each with two chains of 500 draws, and rank the true values among 99 thinned draws. Besides the structural parameters the test quantities include what is published: every team's centred rating at the snapshot date, its power rank, and its WAB against the 45th-best team (the true value computed from the true strengths and the simulated results).

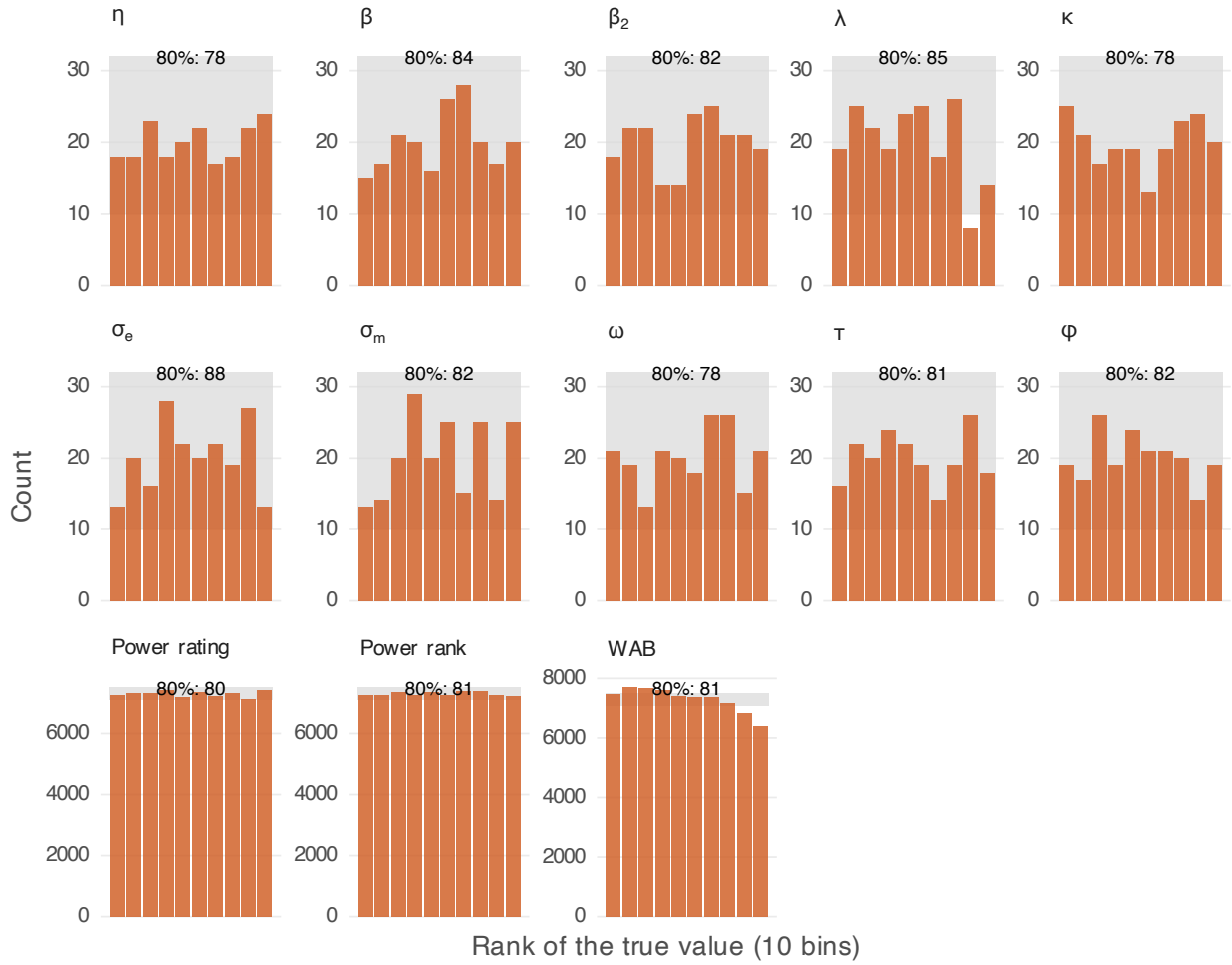


Figure 31: Simulation-based calibration, 200 simulated seasons. Histograms of the rank of the true value among 99 posterior draws, in 10 bins; the band is the central 99% range of a bin count under uniformity (for the team-level quantities, pooled over all teams, so the band is narrow). The label gives the coverage of the central 80% posterior interval.

The structural parameters are recovered with coverage of the 80% interval between 78% and 88%, and for the published quantities it is power rank 81%, power rating 80%, WAB 81%. A chi-square test of uniformity of the rank histograms rejects at the 1% level for WAB; with 365 team-level values per simulated season the team-level tests are very sensitive, and the rank statistics of different teams in one simulation are not independent, so the size of the departure, visible in the figure, matters more than its significance. No simulated fit had a divergent transition. Ranks are discrete, so ties between a true rank and the posterior draws are broken at random.

12 Appendix: The Stan Program

The program below is `stan/power.stan`, reproduced verbatim at render time. Its header comment records the options that were tested; the published configuration uses the luck-adjusted efficiency

margin, the normal margin family, the two-season carry-over and one home-court advantage.

```
// ============================================================================
// CBB power model (pilot). Each D1 team has a net strength r in points per 100 possessions
// against an average D1 team on a neutral floor; it moves week to week as a random walk.
//
//   mu = r[home] - r[away] + hca * home_adv          (expected margin per 100 possessions)
//
// Two per-game signals share r (as in the CFB model):
//  eff   : shot-quality margin per 100 live possessions    eff ~ Normal(lambda mu, sigma_eff / sqrt(w_e))
//   margin : regulation margin per 100 regulation possessions (OT -> 0), conditioned on the eff
//            residual so the two are not counted as independent evidence:
//            margin ~ Student-t(nu, mu + kappa (eff - lambda mu), sigma_m / sqrt(w_m))
// Games without eff use the margin's marginal law (scale widened by kappa^2 sigma_eff^2).
// w = possessions / 70: per-100 rates from short or slow games are noisier.
//
// Preseason level: r0 = mu_conf[conf] + b_prev * prev_dev + sqrt(sigma_team^2 + (b_prev prev_sd)^2) z.
//   conf_carry = 0: mu_conf ~ Normal(0, sigma_conf); prev_dev = last season's final rating
//                   (0 when unknown). Conference strength is in both terms, so mu_conf and
//                   b_prev trade off.
//  conf_carry = 1: mu_conf ~ Normal(b_conf * prev_conf, sigma_conf), prev_conf = mean last-season
//                  rating of the conference's current members; prev_dev = a team's rating minus
//                  its conference's prev_conf. The two carry-overs then act on separate levels.
// conf_centred = 1 samples mu_conf directly (centred), better once each conference has many games.
// Non-D1 games (N_nd): the D1 team's margin against one pooled non-D1 level r_nd, with extra
// spread sigma_nd for the mix of D2/D3/NAIA opponents; margin signal only.
// Dynamics: one state per period (state_days, in R) with a game; step sd tau * sqrt(periods
// elapsed); dynamic = 0 freezes ratings.
// Options tested by backtest (pilot/out/12_variants.md; R's model_config() sets the defaults:
// use_prev2 on, normal margins): use_prev2 adds b_prev2 * the rating two seasons back to r0; team_hca gives each t
// margin_family 1 = Student-t, 2 = normal, 3 = contaminated normal (share p_out with scale k_out
// times sigma); save_loglik stores each D1 game's log density (for LOO). Every prior's
// parameters are data (model_config() holds the published values); lprior is their joint log
// density, for power-scaling sensitivity.
// ============================================================================

functions {
  // Pointwise margin log density for the chosen family
 vector margin_pointwise(vector y, vector mu, vector sigma, int family, real nu, real p_out, real k_out) {
    int n = num_elements(y);
    vector[n] lp;
    for (i in 1:n) {
      if (family == 1) lp[i] = student_t_lpdf(y[i] | nu, mu[i], sigma[i]);
      else if (family == 2) lp[i] = normal_lpdf(y[i] | mu[i], sigma[i]);
    else lp[i] = log_mix(p_out, normal_lpdf(y[i] | mu[i], k_out * sigma[i]), normal_lpdf(y[i] | mu[i], sigma[i]))
    }
    return lp;
  }
  // Summed margin log density (vectorised for the t and the normal)
 real margin_total(vector y, vector mu, vector sigma, int family, real nu, real p_out, real k_out) {
    if (family == 1) return student_t_lpdf(y | nu, mu, sigma);
    if (family == 2) return normal_lpdf(y | mu, sigma);
    return sum(margin_pointwise(y, mu, sigma, family, nu, p_out, k_out));
  }
```

```
  // Margin mean and sd per game: conditioned on the eff residual where a game has eff, the
  // margin's marginal law (scale widened by kappa sigma_eff) where it does not
  matrix margin_moments(vector mu, vector eff, array[] int idx_eff, array[] int idx_noeff, vector sw_m, vector sw_
                        real lambda, real kappa, real sigma_m, real sigma_eff) {
    int n = num_elements(mu);
    matrix[n, 2] m;
    m[:, 1] = mu;
    m[idx_eff, 1] = mu[idx_eff] + kappa * (eff[idx_eff] - lambda * mu[idx_eff]);
    m[idx_eff, 2] = sigma_m ./ sw_m[idx_eff];
   m[idx_noeff, 2] = sqrt(square(sigma_m ./ sw_m[idx_noeff]) + square(kappa * sigma_eff ./ sw_e[idx_noeff]));
    return m;
  }
}

data {
  int<lower=1> N_teams;
  int<lower=1> N_conf;
  array[N_teams] int<lower=1, upper=N_conf> team_conf;
  vector[N_teams] prev_dev;
  vector<lower=0>[N_teams] prev_sd;
  vector[N_conf] prev_conf;
  int<lower=0, upper=1> conf_carry;

  int<lower=0> N_states;
  array[N_states] int<lower=1, upper=N_teams> state_team;
  array[N_states] int<lower=1, upper=N_states> state_first;
  vector<lower=0>[N_states] state_gap;
  array[N_teams] int<lower=0, upper=N_states> now_state;
  vector<lower=0>[N_teams] now_gap;

  int<lower=0> N_games;
  array[N_games] int<lower=1, upper=N_states> home_state;
  array[N_games] int<lower=1, upper=N_states> away_state;
 vector<lower=-1, upper=1>[N_games] home_adv;  // +1 home team on its floor, -1 on the away team's, 0 neutral
  vector[N_games] margin;
  vector<lower=0>[N_games] w_m;
  array[N_games] int<lower=0, upper=1> has_eff;
  vector[N_games] eff;
  vector<lower=0>[N_games] w_e;

  int<lower=0> N_nd;
  array[N_nd] int<lower=1, upper=N_states> nd_state;
  vector<lower=-1, upper=1>[N_nd] nd_venue;     // +1 D1 team at home, -1 away, 0 neutral
  vector[N_nd] nd_margin;                       // D1 - non-D1, per 100 regulation possessions
  vector<lower=0>[N_nd] nd_w;

  int<lower=0, upper=1> use_eff;
  int<lower=0, upper=1> dynamic;
  int<lower=0, upper=1> conf_centred;
  // (mean, sd) of the normal priors on the hierarchy scales and the weekly drift
  vector[2] sigma_conf_prior;
  vector[2] sigma_team_prior;
  vector[2] tau_prior;
  vector[2] hca_prior;
```

```
  // (mean, sd) of the normal priors on the other structural parameters; nu ~ Gamma(nu_prior),
  // p_out ~ Beta(p_out_prior)
  vector[2] b_prev_prior;
  vector[2] b_conf_prior;
  vector[2] b_prev2_prior;
  vector[2] sigma_m_prior;
  vector[2] nu_prior;
  vector[2] p_out_prior;
  vector[2] k_out_prior;
  vector[2] sigma_hca_prior;
  vector[2] lambda_prior;
  vector[2] sigma_eff_prior;
  vector[2] kappa_prior;
  vector[2] r_nd_prior;
  vector[2] sigma_nd_prior;

  int<lower=0, upper=1> use_prev2;
  vector[N_teams] prev2_dev;                    // rating two seasons back (0 when unknown)
  int<lower=0, upper=1> team_hca;
  int<lower=1, upper=3> margin_family;
  int<lower=0, upper=1> save_loglik;
}

transformed data {
  int N_eff = use_eff ? sum(has_eff) : 0;
  array[N_eff] int idx_eff;
  array[N_games - N_eff] int idx_noeff;
  {
    int i = 1;
    int j = 1;
    for (k in 1:N_games) {
      if (use_eff && has_eff[k] == 1) { idx_eff[i] = k; i += 1; }
      else { idx_noeff[j] = k; j += 1; }
    }
  }
  vector[N_states] sqrt_gap = sqrt(state_gap);
  vector[N_games] sw_m = sqrt(w_m);
  vector[N_games] sw_e = sqrt(w_e);
  // the team whose floor it is (home_adv = 0: unused)
  array[N_games] int host;
 for (k in 1:N_games) host[k] = home_adv[k] < 0 ? state_team[away_state[k]] : state_team[home_state[k]];
}

parameters {
  real<lower=0> sigma_conf;
  vector[conf_centred ? 0 : N_conf] conf_raw;
  vector[conf_centred ? N_conf : 0] mu_conf_c;
  real<lower=0> sigma_team;
  real<lower=0, upper=1.2> b_prev;
  vector[conf_carry] b_conf_raw;
  vector[N_teams] z;
  vector<lower=0>[dynamic] tau_raw;
  vector[dynamic ? N_states : 0] eps;
```

```
  real hca;
  real<lower=0> sigma_m;
  vector<lower=2>[margin_family == 1] nu_raw;
  vector<lower=0, upper=0.5>[margin_family == 3] p_out_raw;
  vector<lower=1>[margin_family == 3] k_out_raw;
  vector[use_prev2] b_prev2_raw;
  vector<lower=0>[team_hca] sigma_hca_raw;
  vector[team_hca ? N_teams : 0] hca_z;
  vector<lower=0>[use_eff] lambda_raw;
  vector<lower=0>[use_eff] sigma_eff_raw;
  vector[use_eff] kappa_raw;
  vector[N_nd > 0] r_nd_raw;
  vector<lower=0>[N_nd > 0] sigma_nd_raw;
}

transformed parameters {
  real b_conf = conf_carry ? b_conf_raw[1] : 0;
  vector[N_conf] mu_conf;
  if (conf_centred) mu_conf = mu_conf_c;
  else mu_conf = b_conf * prev_conf + sigma_conf * conf_raw;
  real b_prev2 = use_prev2 ? b_prev2_raw[1] : 0;
  vector[N_teams] r0 = mu_conf[team_conf] + b_prev * prev_dev + b_prev2 * prev2_dev
                       + sqrt(square(sigma_team) + square(b_prev * prev_sd)) .* z;
 // normal and contaminated-normal margins: nu stands in as a large df for the win-probability code
  real nu = margin_family == 1 ? nu_raw[1] : 1e4;
  real p_out = margin_family == 3 ? p_out_raw[1] : 0;
  real k_out = margin_family == 3 ? k_out_raw[1] : 1;
  real sigma_hca = team_hca ? sigma_hca_raw[1] : 0;
  vector[N_teams] hca_team = rep_vector(hca, N_teams);
  if (team_hca) hca_team += sigma_hca * hca_z;
  real tau = dynamic ? tau_raw[1] : 0;
  real lambda = use_eff ? lambda_raw[1] : 0;
  real sigma_eff = use_eff ? sigma_eff_raw[1] : 0;
  real kappa = use_eff ? kappa_raw[1] : 0;
  vector[N_states] r_state = r0[state_team];
  if (dynamic && N_states > 0) {
    vector[N_states] step = sqrt_gap .* eps;
    vector[N_states] cs = cumulative_sum(step);
    r_state += tau * (cs - cs[state_first] + step[state_first]);
  }
}

model {
  sigma_conf ~ normal(sigma_conf_prior[1], sigma_conf_prior[2]);
  sigma_team ~ normal(sigma_team_prior[1], sigma_team_prior[2]);
  b_prev ~ normal(b_prev_prior[1], b_prev_prior[2]);
  b_conf_raw ~ normal(b_conf_prior[1], b_conf_prior[2]);
  hca ~ normal(hca_prior[1], hca_prior[2]);
  sigma_m ~ normal(sigma_m_prior[1], sigma_m_prior[2]);
  nu_raw ~ gamma(nu_prior[1], nu_prior[2]);
  p_out_raw ~ beta(p_out_prior[1], p_out_prior[2]);
  k_out_raw ~ normal(k_out_prior[1], k_out_prior[2]);
  b_prev2_raw ~ normal(b_prev2_prior[1], b_prev2_prior[2]);
  sigma_hca_raw ~ normal(sigma_hca_prior[1], sigma_hca_prior[2]);
```

```
  hca_z ~ std_normal();
  tau_raw ~ normal(tau_prior[1], tau_prior[2]);
  lambda_raw ~ normal(lambda_prior[1], lambda_prior[2]);
  sigma_eff_raw ~ normal(sigma_eff_prior[1], sigma_eff_prior[2]);
  kappa_raw ~ normal(kappa_prior[1], kappa_prior[2]);
  conf_raw ~ std_normal();
  if (conf_centred) mu_conf_c ~ normal(b_conf * prev_conf, sigma_conf);
  r_nd_raw ~ normal(r_nd_prior[1], r_nd_prior[2]);
  sigma_nd_raw ~ normal(sigma_nd_prior[1], sigma_nd_prior[2]);
  z ~ std_normal();
  eps ~ std_normal();

  if (N_games > 0) {
   vector[N_games] mu = r_state[home_state] - r_state[away_state] + hca_team[host] .* home_adv;
    if (use_eff) {
      vector[N_eff] mu_e = mu[idx_eff];
      target += normal_lpdf(eff[idx_eff] | lambda * mu_e, sigma_eff ./ sw_e[idx_eff]);
    }
   matrix[N_games, 2] mm = margin_moments(mu, eff, idx_eff, idx_noeff, sw_m, sw_e, lambda, kappa, sigma_m, sigma_
    target += margin_total(margin, mm[:, 1], mm[:, 2], margin_family, nu, p_out, k_out);
  }
  if (N_nd > 0) {
    target += margin_total(nd_margin, r_state[nd_state] - r_nd_raw[1] + hca * nd_venue,
                  sqrt((square(sigma_m) + square(kappa * sigma_eff)) ./ nd_w + square(sigma_nd_raw[1])),
                           margin_family, nu, p_out, k_out);
  }
}

generated quantities {
  // Rating at the snapshot date: latest state plus the drift since
  vector[N_teams] r;
  // Marginal sd of a game's margin per 100 at w = 1 (for win probabilities)
  real sigma_game = sqrt(square(sigma_m) + square(kappa * sigma_eff));
  // Each D1 game's log density: log_lik the game's whole contribution (efficiency, then the
  // margin given it), log_lik_margin the margin's marginal law alone (comparable across models
  // with and without the efficiency signal)
  vector[save_loglik ? N_games : 0] log_lik;
  vector[save_loglik ? N_games : 0] log_lik_margin;
  if (save_loglik && N_games > 0) {
   vector[N_games] mu = r_state[home_state] - r_state[away_state] + hca_team[host] .* home_adv;
   matrix[N_games, 2] mm = margin_moments(mu, eff, idx_eff, idx_noeff, sw_m, sw_e, lambda, kappa, sigma_m, sigma_
    log_lik = margin_pointwise(margin, mm[:, 1], mm[:, 2], margin_family, nu, p_out, k_out);
    for (k in idx_eff) log_lik[k] += normal_lpdf(eff[k] | lambda * mu[k], sigma_eff / sw_e[k]);
    array[0] int none;
   matrix[N_games, 2] mo = margin_moments(mu, eff, none, linspaced_int_array(N_games, 1, N_games), sw_m, sw_e,
                                          lambda, kappa, sigma_m, sigma_eff);
   log_lik_margin = margin_pointwise(margin, mo[:, 1], mo[:, 2], margin_family, nu, p_out, k_out);
  }
  // Joint log density of the structural priors (priorsense power-scales it); the hierarchical
  // standard normals (z, eps, conf_raw, hca_z) are part of the model, not the prior
  real lprior = normal_lpdf(sigma_conf | sigma_conf_prior[1], sigma_conf_prior[2])
                + normal_lpdf(sigma_team | sigma_team_prior[1], sigma_team_prior[2])
                + normal_lpdf(b_prev | b_prev_prior[1], b_prev_prior[2])
                + normal_lpdf(b_conf_raw | b_conf_prior[1], b_conf_prior[2])
```

```
                + normal_lpdf(b_prev2_raw | b_prev2_prior[1], b_prev2_prior[2])
                + normal_lpdf(hca | hca_prior[1], hca_prior[2])
                + normal_lpdf(sigma_m | sigma_m_prior[1], sigma_m_prior[2])
                + gamma_lpdf(nu_raw | nu_prior[1], nu_prior[2])
                + beta_lpdf(p_out_raw | p_out_prior[1], p_out_prior[2])
                + normal_lpdf(k_out_raw | k_out_prior[1], k_out_prior[2])
                + normal_lpdf(sigma_hca_raw | sigma_hca_prior[1], sigma_hca_prior[2])
                + normal_lpdf(tau_raw | tau_prior[1], tau_prior[2])
                + normal_lpdf(lambda_raw | lambda_prior[1], lambda_prior[2])
                + normal_lpdf(sigma_eff_raw | sigma_eff_prior[1], sigma_eff_prior[2])
                + normal_lpdf(kappa_raw | kappa_prior[1], kappa_prior[2])
                + normal_lpdf(r_nd_raw | r_nd_prior[1], r_nd_prior[2])
                + normal_lpdf(sigma_nd_raw | sigma_nd_prior[1], sigma_nd_prior[2]);
  for (i in 1:N_teams) {
    r[i] = now_state[i] == 0 ? r0[i] : r_state[now_state[i]];
    if (dynamic) r[i] += tau * sqrt(now_gap[i]) * normal_rng(0, 1);
  }
}
```

13 Appendix: Reproducibility

The published pipeline is run with `targets::tar_make()` from the project root, after setting an API key for CollegeBasketballData.com; the first full build fits all 278 posteriors in one to two hours on sixteen cores, and later runs fit only new snapshots. The analyses of this paper and the paper itself are targets defined in `_targets.R` and `R/08_analysis.R`, built with `CBB_PAPER=1 Rscript -e 'targets::tar_make()'` (several more hours) and rerun whenever their inputs change. Every number and figure is read from the store and nothing is refitted at render time.

Randomness. Every fit uses a fixed seed (2026 for the published fits and most analyses, $2026 + i$ for the i-th simulation-based calibration season and for each ablation snapshot, and a seed per step for the replicated data and bootstraps), so a rebuild on the same data and software reproduces every number.

Software. R package versions are recorded in `renv.lock` at the project root; Table 27 lists the main ones and CmdStan.

Data. The providers revise past data occasionally (late score corrections, reclassified games, new betting lines), so a rebuild months later need not reproduce this paper exactly. Every raw file is cached under `data/raw/` and never re-fetched once a season is final, which is the snapshot the paper was built from; Table 28 records, for each season, what the models saw and a hash of it. This build read the data on 2026-11-03.

References

[1] D. Harville, "The use of linear-model methodology to rate high school or college football teams," *Journal of the American Statistical Association*, vol. 72, no. 358, pp. 278–289, 1977. DOI: 10.1080/01621459.1977.10480991

Table 27: Software versions used for this render.

Component	Version
R	4.6.2
CmdStan	2.37.0
targets	1.12.0
tarchetypes	0.14.1
cmdstanr	0.9.0.9000
posterior	1.7.0
loo	2.10.1
priorsense	1.4.0
dplyr	1.2.1
ggplot2	4.0.3
igraph	2.3.3

Table 28: The data behind this build, by season: Division I teams, completed Division I and non-Division I games, Division I games with an efficiency margin and with a closing line, snapshots, and an xxHash64 hash of the season's games table as the models read it.

Season	Teams	D-I games	Non-D-I	With eff.	With line	Snapshots	Hash
2014–15	351	5,492	435	4,914	3,988	23	`be8061a6e9b1babd`
2015–16	351	5,506	433	5,026	3,991	23	`fd46dbffcea45bf3`
2016–17	351	5,485	441	5,096	3,973	23	`3c7462d39767507a`
2017–18	351	5,500	467	5,196	4,033	23	`45a5373c87e2cb2a`
2018–19	353	5,595	452	5,123	5,535	24	`ed36a28d307b9a56`
2019–20	353	5,324	433	5,006	5,318	20	`4fd4a462856a368a`
2020–21	347	3,943	340	3,732	3,929	21	`1c23633daad468a3`
2021–22	358	5,475	490	5,361	5,454	23	`08974f9460b8ee0b`
2022–23	363	5,714	506	5,633	5,714	23	`416d2a09b632234c`
2023–24	362	5,723	517	5,642	5,301	24	`e4a9f1d0ff3cda8b`
2024–25	364	5,759	526	5,649	5,432	24	`89f47488ca2d3227`
2025–26	365	5,752	547	5,740	5,748	24	`ba1a5f98bc0cb464`

[2] K. Massey, "Statistical models applied to the rating of sports teams," Honors thesis, Bluefield College, 1997.
[3] R. A. Bradley and M. E. Terry, "Rank analysis of incomplete block designs: I. The method of paired comparisons," *Biometrika*, vol. 39, no. 3/4, pp. 324–345, 1952. DOI: 10.2307/2334029
[4] W. N. Colley, "Colley's bias free college football ranking method: The Colley matrix explained," Princeton University, Tech. Rep., 2002. [Online]. Available: https://www.colleyrankings.com/matrate.pdf
[5] A. N. Langville and C. D. Meyer, *Who's #1? The Science of Rating and Ranking*. Princeton, NJ: Princeton University Press, 2012.
[6] K. Pomeroy, *Ratings explanation*, https://kenpom.com/blog/ratings-explanation/, 2016.
[7] B. Torvik, *T-rank: College basketball ratings, player statistics and time machine*, https://barttorvik.com/, 2026.
[8] D. Oliver, *Basketball on Paper: Rules and Tools for Performance Analysis*. Washington, DC: Potomac Books, 2004.
[9] J. Kubatko, D. Oliver, K. Pelton, and D. T. Rosenbaum, "A starting point for analyzing basketball statistics," *Journal of Quantitative Analysis in Sports*, vol. 3, no. 3, 2007.
[10] NCAA, *NCAA evaluation tool (NET) and team sheets, division i men's basketball*, NCAA.org, 2024.
[11] D. Harville, "Predictions for National Football League games via linear-model methodology," *Journal of the American Statistical Association*, vol. 75, no. 371, pp. 516–524, 1980. DOI: 10.1080/01621459.1980.10477504
[12] H. Stern, "On the probability of winning a football game," *The American Statistician*, vol. 45, no. 3, pp. 179–183, 1991. DOI: 10.1080/00031305.1991.10475798
[13] D. A. Harville and M. H. Smith, "The home-court advantage: How large is it, and does it vary from team to team?" *The American Statistician*, vol. 48, no. 1, pp. 22–28, 1994.
[14] P. Kvam and J. S. Sokol, "A logistic regression/Markov chain model for NCAA basketball," *Naval Research Logistics*, vol. 53, no. 8, pp. 788–803, 2006.
[15] M. J. Lopez and G. J. Matthews, "Building an NCAA men's basketball predictive model and quantifying its success," *Journal of Quantitative Analysis in Sports*, vol. 11, no. 1, pp. 5–12, 2015.
[16] B. J. Coleman and A. K. Lynch, "Identifying the NCAA tournament "dance card"," *Interfaces*, vol. 31, no. 3, pp. 76–86, 2001.
[17] M. E. Glickman and H. S. Stern, "A state-space model for National Football League scores," *Journal of the American Statistical Association*, vol. 93, no. 441, pp. 25–35, 1998. DOI: 10.1080/01621459.1998.10474084
[18] M. E. Glickman, "Parameter estimation in large dynamic paired comparison experiments," *Journal of the Royal Statistical Society: Series C (Applied Statistics)*, vol. 48, no. 3, pp. 377–394, 1999. DOI: 10.1111/1467-9876.00159
[19] L. Knorr-Held, "Dynamic rating of sports teams," *Journal of the Royal Statistical Society: Series D (The Statistician)*, vol. 49, no. 2, pp. 261–276, 2000. DOI: 10.1111/1467-9884.00235
[20] M. J. Lopez, G. J. Matthews, and B. S. Baumer, "How often does the best team win? A unified approach to understanding randomness in North American sport," *The Annals of Applied Statistics*, vol. 12, no. 4, pp. 2483–2516, 2018. DOI: 10.1214/18-AOAS1165
[21] A. Gelman et al., *Bayesian workflow*, 2020. arXiv: 2011.01808 [stat.ME].

[22] SportsDataverse, *Sportsdataverse-data: ESPN men's college basketball schedules, box scores and play-by-play*, https://github.com/sportsdataverse/sportsdataverse-data, 2026.

[23] CollegeBasketballData.com, *CollegeBasketballData.com API*, https://collegebasketballdata.com/, 2026.

[24] D. B. Rubin, "Inference and missing data," *Biometrika*, vol. 63, no. 3, pp. 581–592, 1976. DOI: 10.1093/biomet/63.3.581

[25] O. Papaspiliopoulos, G. O. Roberts, and M. Sköld, "A general framework for the parametrization of hierarchical models," *Statistical Science*, vol. 22, no. 1, pp. 59–73, 2007. DOI: 10.1214/088342307000000014

[26] M. Betancourt, *A conceptual introduction to Hamiltonian Monte Carlo*, 2017. arXiv: 1701.02434 [stat.ME].

[27] R. J. Carroll, D. Ruppert, L. A. Stefanski, and C. M. Crainiceanu, *Measurement Error in Nonlinear Models: A Modern Perspective*, 2nd ed. Chapman and Hall/CRC, 2006. DOI: 10.1201/9781420010138

[28] A. Gelman, J. B. Carlin, H. S. Stern, D. B. Dunson, A. Vehtari, and D. B. Rubin, *Bayesian Data Analysis*, 3rd ed. Boca Raton, FL: Chapman and Hall/CRC, 2013.

[29] Y. Hong, "On computing the distribution function for the Poisson binomial distribution," *Computational Statistics & Data Analysis*, vol. 59, pp. 41–51, 2013. DOI: 10.1016/j.csda.2012.10.006

[30] M. D. Hoffman and A. Gelman, "The No-U-Turn sampler: Adaptively setting path lengths in Hamiltonian Monte Carlo," *Journal of Machine Learning Research*, vol. 15, no. 47, pp. 1593–1623, 2014.

[31] B. Carpenter et al., "Stan: A probabilistic programming language," *Journal of Statistical Software*, vol. 76, no. 1, pp. 1–32, 2017. DOI: 10.18637/jss.v076.i01

[32] A. Vehtari, A. Gelman, D. Simpson, B. Carpenter, and P.-C. Bürkner, "Rank-normalization, folding, and localization: An improved $\widehat{R}$ for assessing convergence of MCMC," *Bayesian Analysis*, vol. 16, no. 2, pp. 667–718, 2021. DOI: 10.1214/20-BA1221

[33] W. M. Landau, "The targets R package: A dynamic Make-like function-oriented pipeline toolkit for reproducibility and high-performance computing," *Journal of Open Source Software*, vol. 6, no. 57, p. 2959, 2021. DOI: 10.21105/joss.02959

[34] R Core Team, *R: A language and environment for statistical computing*, R Foundation for Statistical Computing, Vienna, Austria, 2026. [Online]. Available: https://www.R-project.org/

[35] J. Gabry, D. Simpson, A. Vehtari, M. Betancourt, and A. Gelman, "Visualization in Bayesian workflow," *Journal of the Royal Statistical Society: Series A (Statistics in Society)*, vol. 182, no. 2, pp. 389–402, 2019. DOI: 10.1111/rssa.12378

[36] N. Kallioinen, T. Paananen, P.-C. Bürkner, and A. Vehtari, "Detecting and diagnosing prior and likelihood sensitivity with power-scaling," *Statistics and Computing*, vol. 34, no. 1, p. 57, 2024. DOI: 10.1007/s11222-023-10366-5

[37] A. Vehtari, A. Gelman, and J. Gabry, "Practical Bayesian model evaluation using leave-one-out cross-validation and WAIC," *Statistics and Computing*, vol. 27, no. 5, pp. 1413–1432, 2017. DOI: 10.1007/s11222-016-9696-4

[38] A. Vehtari, D. Simpson, A. Gelman, Y. Yao, and J. Gabry, "Pareto smoothed importance sampling," *Journal of Machine Learning Research*, vol. 25, no. 72, pp. 1–58, 2024.

[39] L. J. Tashman, "Out-of-sample tests of forecasting accuracy: An analysis and review," *International Journal of Forecasting*, vol. 16, no. 4, pp. 437–450, 2000. DOI: 10.1016/S0169-2070(00)00065-0

[40] G. W. Brier, "Verification of forecasts expressed in terms of probability," *Monthly Weather Review*, vol. 78, no. 1, pp. 1–3, 1950. DOI: 10.1175/1520-0493(1950)078<0001:VOFEIT>2.0.CO;2

[41] T. Gneiting and A. E. Raftery, "Strictly proper scoring rules, prediction, and estimation," *Journal of the American Statistical Association*, vol. 102, no. 477, pp. 359–378, 2007. DOI: 10.1198/016214506000001437

[42] S. D. Levitt, "Why are gambling markets organised so differently from financial markets?" *The Economic Journal*, vol. 114, no. 495, pp. 223–246, 2004. DOI: 10.1111/j.1468-0297.2004.00207.x

[43] F. X. Diebold and R. S. Mariano, "Comparing predictive accuracy," *Journal of Business & Economic Statistics*, vol. 13, no. 3, pp. 253–263, 1995. DOI: 10.1080/07350015.1995.10524599

[44] D. J. Spiegelhalter, "Probabilistic prediction in patient management and clinical trials," *Statistics in Medicine*, vol. 5, no. 5, pp. 421–433, 1986. DOI: 10.1002/sim.4780050506

[45] D. Karlis and I. Ntzoufras, "Analysis of sports data by using bivariate Poisson models," *Journal of the Royal Statistical Society: Series D (The Statistician)*, vol. 52, no. 3, pp. 381–393, 2003. DOI: 10.1111/1467-9884.00366

[46] G. Baio and M. Blangiardo, "Bayesian hierarchical model for the prediction of football results," *Journal of Applied Statistics*, vol. 37, no. 2, pp. 253–264, 2010. DOI: 10.1080/02664760802684177

[47] D. Kahle, J. Seaman, and J. Stamey, "An overview of Bayesian computation," in *Case Studies in Bayesian Methods for Biopharmaceutical CMC*, ser. Chapman & Hall/CRC Biostatistics Series, P. Faya and T. Pourmohamad, Eds., Boca Raton, FL: Chapman and Hall/CRC, 2022, pp. 5–24.

[48] E. Miyakawa and D. Kahle, "A practical comparison of Bayesian computing platforms in R," in *New Frontiers in Data Science (SDSC 2025)*, ser. Communications in Computer and Information Science, H. Han and J. Stamey, Eds., vol. 2556, Cham: Springer, 2025, pp. 58–74. DOI: 10.1007/978-3-031-99879-9_4

[49] M. Betancourt, *Diagnosing suboptimal cotangent disintegrations in Hamiltonian Monte Carlo*, 2016. arXiv: 1604.00695 [stat.ME].

[50] A. Gelman and D. B. Rubin, "Inference from iterative simulation using multiple sequences," *Statistical Science*, vol. 7, no. 4, pp. 457–472, 1992. DOI: 10.1214/ss/1177011136

[51] C. J. Geyer, "Practical Markov chain Monte Carlo," *Statistical Science*, vol. 7, no. 4, pp. 473–483, 1992. DOI: 10.1214/ss/1177011137

[52] D. J. Schad, M. Betancourt, and S. Vasishth, "Toward a principled Bayesian workflow in cognitive science," *Psychological Methods*, vol. 26, no. 1, pp. 103–126, 2021. DOI: 10.1037/met0000275

[53] T. Säilynoja, P.-C. Bürkner, and A. Vehtari, "Graphical test for discrete uniformity and its applications in goodness-of-fit evaluation and multiple sample comparison," *Statistics and Computing*, vol. 32, no. 2, p. 32, 2022. DOI: 10.1007/s11222-022-10090-6

[54] S. Talts, M. Betancourt, D. Simpson, A. Vehtari, and A. Gelman, *Validating Bayesian inference algorithms with simulation-based calibration*, 2018. arXiv: 1804.06788.

[55] M. Modrák et al., "Simulation-based calibration checking for Bayesian computation: The choice of test quantities shapes sensitivity," *Bayesian Analysis*, 2023, Advance publication. arXiv: 2211.02383.